Higher-rank Suzuki geometries

Complete source treatment and proofs for L110

This volume treats the fixed twenty-page higher-rank article. The opening manuscript and claim ledger give the current disposition. Exact earlier proofs retain their historical scope statements; the separate scalar geometry supplement proves the stronger conclusion where indicated. Two additional necessity papers repair the conditional-extension gap identified in the retained review of the first assembly. The final papers are reused common theory, with classical premises explicitly attributed. No new original, principal-refutation count, Lean formalization or compression measurement is claimed.

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Subfield descent and the higher-rank Suzuki constructions

Human Compression project

Abstract

We account for the mathematical claims in the fixed twenty-page edition of *The residually weakly primitive pre-geometries of the Suzuki simple groups*. Its exhaustive theorem is false. We retain the earlier counterexample and give proofs or explicit corrections for its remaining claims. The surviving construction families follow from subfield descent, cyclic coset coordinates and scalar irreducibility. Several other prescribed intersection lattices are impossible, and a scalar codimension test rules out the mixed construction. This is a complete treatment of the fixed source's assertions, not a corrected classification of all Suzuki geometries.

1 The source and the meaning of completion

The source is D. Leemans, Note di Matematica 20(1) (2000/2001), 1–20, received 26 July 1999 and accepted 21 September 1999. We use the retained published edition, whose physical and printed page numbers agree. The exact source file and page references are in the claim ledger. The companion volume includes all the proof texts used below, and its acquisition record binds their earlier independent reviews.

For $G_i \leq G$ and a finite type set I, write $G_J = \bigcap_{j\in J} G_j$ and $G_\varnothing = G$. The source's condition is

$$\text{for every } J \subsetneq I \text{ there is } i \notin J \text{ with } G_{J\cup\{i\}} \text{ maximal proper in } G_J. \tag{1}$$

The source uses left cosets and left translation. Its residues for flags inside the distinguished chamber are the algebraic intersection systems. Its local two-transitivity condition applies to those algebraic rank-one residues. Actual thinness counts all chambers incident with an actual panel. The distinction is necessary for the corrections imported from the rank-three treatment. On p. 5 the source defines groups of Suzuki type to be simple and explicitly excludes Sz(2). We apply that convention to the Suzuki labels in its residue lists. In particular the common Suzuki bottom in R_{10}, R_{11}, R_{13} is simple; the parameter is at least eight.

We use flag transitivity in the source's stated strong sense, including transitivity on every flag type. In the positive results here we prove that every flag has a common representative in G, which implies both flag transitivity and the geometry axiom. No inference from chamber transitivity alone is needed.

The source explicitly states on p. 3 that its principal theorem concerns pregeometries, without assuming flag transitivity or the geometry axiom. On p. 15 it requires the *whole intended intersection lattice* of each construction, as inferred from the corresponding rank-four case. A list of abstract parabolic isomorphism types is therefore weaker than the construction being asserted. We keep these two meanings separate. A refuted exhaustive theorem is disposed of by a counterexample; it does not create an obligation to prove a stronger repaired classification.

2 Common theory and the rank-three input

The classical premises are the four-dimensional Suzuki model, its maximal-subgroup and centralizer structure, simple subfield maximality, Lang–Steinberg surjectivity and Schur–Zassenhaus. None is presented as new. The source itself cites Lüneburg for the subgroup

structure. The written common theory in the volume supplies the needed matrix intersections, torus arithmetic, normalizer restrictions and affine uniqueness arguments. The source's own exhaustive rank-three list is never used as a premise.

Lemma 1, that every rank-three coset pregeometry is a geometry, has the short common-representative proof in the acquired preliminary paper. Lemma 2 is the odd-exponent gcd identity for $2^{2m}+1$. Lemma 3 is the proper-overgroup statement for an embedded full split dihedral in a maximal simple subfield; the ambient group itself must be excluded from its literal word "subgroups." Lemma 4 follows from the proved exclusion of the other maximal subgroup types in rank at least four. The consequences of (1) on pp. 4–5, including strict intersections and a maximal chain obtained by ordering labels, are proved in the common prime-degree argument.

Theorems 1 and 2 are recovered: the three necessary prime-degree patterns are proved without an enumeration assumption, each allowed pattern has an explicit realization, and a prime field degree with composite $q-1$ admits no rank at least three satisfying (1). The thirteen labels in Theorem 3 have the following exact relation to the rank-three treatment.

L110 labels	Rank-three source labels	Disposition
R_1, R_2, R_3	Theorem 5.3, patterns 1–3	necessary and realizable
R_4, R_5, R_6	Theorem 5.5, patterns 1–3	prescribed meets impossible
R_7	Theorem 5.5, pattern 4	canonical realization
R_8	Theorem 5.5, pattern 5	prescribed meet impossible
R_9, R_{10}, R_{11}	Theorem 5.5, patterns 6–8	canonical realizations
R_{12}	Theorem 5.5, pattern 9	bare meets exist; (1) fails
R_{13}	Theorem 5.5, pattern 10	canonical realization

The exhaustive assertion in Theorem 3 is false: the acquired $\mathrm{Sz}(8), \mathrm{Sz}(32), C_{533}:C_4$ family in $\mathrm{Sz}(32768)$ has intersection orders $20, 52, 164, 4$, whereas the applicable printed pattern requires the two simple parabolics to meet in order eight. Its geometry, firmness, residual connectedness and flag transitivity are proved. The exact corrections, including the scalar obstruction to R_{12}, are imported with their proofs rather than rederived by a new search.

Theorem 4 is explicitly a quotation of the separate dihedral classification, reference [20]. That original is outside the fixed accessible intake. We retain the quotation as an attributed premise, without claiming recovery of the cited original. The needed odd-dihedral maximality consequences also have elementary proofs in the common theory and the prime-residue extension paper; the higher constructions do not rely on an unchecked machine enumeration of dihedral geometries.

3 The rank-four extension argument

The new nonextension paper proves that none of R_1, R_2, R_3 extends to a rank-four system with its simple parabolic maximal in the ambient Suzuki group. For R_1, the affine subgroup forces its containing soluble parabolic into a point stabilizer, which cannot contain the required odd dihedral. For R_2, the common-reflection argument aligns the two prime-degree simple subfields and makes their meet $\mathrm{Sz}(2)$, of order twenty, not the prescribed dihedral of order ten. For R_3, the common full torus forces an affine meet, and the unique simple subfield containing that embedded affine group makes two parabolics equal. Equal orders alone would not justify that last step. Thus the source's claimed surviving R_2 candidate and Figure 1 are refuted, while the nonextension conclusions for R_1, R_3 are recovered. Section 3.2 follows from the recovered prime-degree obstruction.

In Section 3.3, the R_4, R_5, R_6, R_8 extension branches already start from impossible full residue lattices. Their higher-rank continuations cannot repair the obstruction. The R_{12} branch

fails (1) by the scalar codimension test. The other five branches require a separate necessity argument: existence of the canonical families does not prove that every extension of a prescribed residue has their lattice.

The split extension paper proves this necessity for R_7 and R_9. A field-generating cyclic complement identifies the embedded subfield inside every simple overgroup of the prescribed dihedral or central affine subgroup. Its intersection with the maximal parabolic then excludes such a simple overgroup. The remaining dihedral parabolic lies in the ambient split normalizer; the affine parabolic lies in the appropriate point stabilizer. Scalar irreducibility and the absence of intermediate overgroups force the full central radical and the same cyclic complement in both soluble parabolics. The last cyclic and dihedral maximality conditions force two distinct primes and every remaining intersection. This proves the conditional extension statement also in the repeated R_7 branch of Section 3.3.2. On p. 13 the stray G_4 means G_3. The printed divisibility for the new prime must be $n' \mid (s'-1)/(t-1)$; the quotient $(q-1)/(s'-1)$ is not the correct condition. These textual corrections do not weaken the necessity claim.

For R_{10} and R_{11}, use $H = C_m : C_c$ with the full torus complement, of order two in the split case and four in the nonsplit case. Its normalizer is H itself. Conjugacy inside each subfield group therefore makes the containing subfield of each degree unique. Two such groups containing H meet in the compatible gcd field; in the nonsplit argument the full rotation order exceeds five. We do not assume that the bottom torus of order five determines Sz(8). A divisor argument supplies the remaining information: an intersection system of r cyclic extensions with a common complement has r prime-index steps, and strictness supplies r different prime valuation witnesses. Its top rotation order divided by its bottom order is thus a product of r distinct primes, with one omitted at each parabolic. This remains true when a selected prime divides the bottom order.

The torus necessity paper also derives the input field degrees from (1). A composite relative degree would insert an intermediate compatible torus order, contradicting the prime quotient unless the simple bottom were maximal, which is impossible at that composite degree. Applying the same argument in the ambient residue gives the two distinct relative primes and their product. The source's R_{11} row had not stated those degree identities explicitly. With these premises proved, the fixed maximal simple parabolic determines the other two simple parabolics, and the full-complement and divisor arguments determine the remaining torus parabolic and all its intersections. The R_{10} paragraph's later interchange of G_2, G_3 is corrected to agree with its opening lattice. For R_{13} the common simple bottom and uniqueness align all field groups; maximal prime degrees then give every intersection.

These are conditional necessities for the exact source residues. They do not make the source's global branch list exhaustive. The first assembly conflated these two questions; its independent review held completion on that gap. That assembly and hold are preserved. The two new necessity papers supply the missing arguments in this edition, while the global exhaustive conclusions remain refuted.

4 Every numbered construction

Put $S_d = \mathrm{Sz}(2^d)$. In the positive families choose distinct odd primes a_i, an odd $e > 1$, $t = 2^e$, $n = e\prod_i a_i$, and standard field parabolics S_{n/a_i}. Selecting a set J of field labels leaves S_{n_J} with $n_J = e\prod_{i\notin J} a_i$. The following table states the complete dispositions; the accompanying papers give all subgroup and residue proofs.

No.	Disposition of the source recipe
1	Its intended full lattice contains the forbidden shared-reflection prime-subfield meet. It is impossible; the additional affine parabolic cannot repair it.

No.	Disposition of the source recipe
2–4	The intended lattices contain the impossible R_4, R_5, R_6 residues. Separately, their literal full intersections of orders 10, 4, 4 cannot be subgroups of the displayed odd dihedral: its rotation order has no factor 5 and its two-part is 2. Deleting the last index would repair that notation, but would not repair the intended lattice.
5	After restoring the missing factor $t-1$ in the dihedral order and correcting the conflicting indices and the undefined product index, the canonical split family satisfies (1) for every allowed parameter. The two cases have e prime, including the repeated-prime case $a_i = e$. If several distinguished primes give valid choices, different omitted simple-subfield orders distinguish the resulting stated-action families.
6	The intended R_8 lattice is impossible. This does not exclude the bare parabolic types: the retained example with relative degrees $3, 5, 7$ and torus primes $13, 41, 113$ realizes those types with a different lower lattice, as a thick, residually connected, flag-transitive geometry.
7	Relabel the repeated subfield/dihedral index. The common scalar field lemma proves all intersections and (1) for every allowed parameter.
8–9	Relabel the repeated final subfield/torus index. Every allowed parameter has a canonical realization satisfying (1), thickness, residual connectedness and flag transitivity. Selected primes may divide the base torus order; the valuation proof includes this case.
10	Relabeling the repeated final index does not repair the mathematics. The bare intended intersections exist. For every embedding with the prescribed mixed parabolic and pair intersections, the remaining intersections are all nonmaximal in that parabolic, so (1) fails.
11	Every literal allowed parameter has a canonical thick, residually connected, flag-transitive realization satisfying (1). The exact criterion is $e > 1$, which also permits all distinct prime divisors to be selected when a repeated prime factor remains. That boundary is absent from the source rank bound.

For Constructions 5 and 7 put $M = (t-1)\prod_i \ell_i$, where the ℓ_i are distinct primes dividing $(t^{a_i}-1)/(t-1)$. The two additional parabolics are D_{2M} and $Z_{2^n} : C_M$. The identity

$$\gcd(M, 2^{n_J} - 1) = (t-1)\prod_{i \notin J} \ell_i$$

gives all intersections. That cyclic scalar subgroup generates the full remaining coefficient field, so its affine complement is maximal. The four possibilities for selecting neither, either, or both soluble types then prove (1) for every subset.

For Constructions 8 and 9 the additional group is $K = C_{\gamma\prod_i \ell_i} : C_c$, with $c = 2$ or 4. Compatible tori give $K \cap S_{n_J} = C_{\gamma\prod_{i\notin J}\ell_i} : C_c$. Lang–Steinberg gives a common representative for pairwise incident standard subfield cosets. The Chinese remainder theorem makes $K \longrightarrow \prod_i K/(K \cap S_{n/a_i})$ surjective as a map of coset sets. This proves the common-representative property for the whole family. No nonnormal coset set is treated as a quotient group.

For Construction 10, conjugate the mixed parabolic to $Q = Z_q U_t H_t$. A prescribed pair intersection H surjects onto Q/Z_q, and its overgroups correspond exactly to the $\mathbb{F}_t$-subspaces containing $H \cap Z_q$. Its scalar codimension is $\prod_i a_i - \prod_i a_i/a_j > 1$. Thus every such H has a proper intermediate overgroup in Q. For Construction 11, the field intersection formula and descent give all the positive claims. If $e = 1$, a final $S_1 = 5 : 4$ lies strictly inside a torus normalizer of some S_{a_i}, so (1) fails.

5 The principal correction and final remarks

The earlier accepted principal correction uses

$$n = 3^2 \cdot 5 \cdot 7 \cdot 11 = 3465, \qquad (G_1, G_2, G_3, G_4) = (S_{1155}, S_{693}, S_{495}, S_{315}).$$

Their common intersection is $S_3 = \mathrm{Sz}(8)$ and every available step is a maximal simple-subfield inclusion. All four parabolics are simple, excluding Constructions 1–10, while the literal Construction 11 allows rank at most three for four distinct prime divisors. This refutes Theorem 5. The descent supplement also proves thickness, residual connectedness and flag transitivity. The correction remains an omission from the literal list; it is repaired by replacing that one rank bound with $e > 1$, without thereby obtaining a complete classification.

The source's introductory assertions about local two-transitivity are quotations of the rank-three paper. That paper's complete corrected treatment supplies the exact witnesses: two nonthin rank-three geometries in $\mathrm{Sz}(8)$ and $\mathrm{Sz}(32)$ satisfying the literal algebraic local condition, and a scalar-subspace family of arbitrary rank under the same algebraic convention. The separate new coset-coordinate join now proves that this last family consists of thick, residually connected, flag-transitive geometries with actual local two-transitivity. Thus the quoted rank restriction is false even for actual geometries. Its rank-three members also occur at composite field degrees. The earlier conservative pregeometry proof remains unchanged; this stronger conclusion and its two-page proof are separately identified.

The diagram paper proves the conditional star statements for the surviving prescribed families: the centre is the torus type in 8/9 and the dihedral type in corrected 5/7; pure fields have complete diagrams. The pentagon figure for Construction 1 presupposes an impossible lattice. The future flag-transitivity question is answered for the canonical families 8, 9 and 11 and the retained bare-type example for 6. Flag transitivity of corrected 5/7 and of arbitrary relative positions remains unproved here; it was not an assertion of the source's classification theorem.

The new contribution in this assembly consists of explicit nonextension arguments, the conditional necessity proofs for all viable rank-four branches, the full parameter applications of the common descent and scalar theory, the correction ledger, the diagram proof, and the explicit join establishing actual geometries for the scalar family. Earlier project proofs and finite certificates are retained with attribution. The already checked Lean fragment concerns only the abstract coset-coordinate implication; no Suzuki classification or whole manuscript is claimed formalized. Completing this source adds no new principal-refutation count, no ordinary recovery, no new conjecture count and no compression measurement.

A missing boundary in the Suzuki subfield construction

Human Compression project

Abstract

The number of distinct prime divisors of a field degree does not determine whether its smallest selected subfield is trivial. A repeated prime factor allows all distinct primes to be selected while leaving a simple Suzuki group at the bottom. This yields a rank-four residually weakly primitive pregeometry omitted from the eleven literal constructions in the stated classification. We give the complete subgroup argument, reusing the earlier project proof of this boundary. We do not assert a corrected exhaustive classification or a new solution of an open problem.

1 The subgroup condition

Let G be a group, let I be a finite type set, and let $G_i \le G$ for $i \in I$. The coset pregeometry has elements gG_i, of type i; two elements are incident when their cosets intersect. Put

$$G_J = \bigcap_{j \in J} G_j \quad (J \neq \varnothing), \qquad G_\varnothing = G.$$

The source calls this system residually weakly primitive when, for every $J \subsetneq I$, at least one $i \in I \setminus J$ makes $G_{J\cup\{i\}}$ a maximal proper subgroup of G_J. This is the definition on printed page 4 of the source. Its principal classification concerns pregeometries with precisely this subgroup condition; flag transitivity and the geometry axiom are not additional hypotheses of that classification.

Write $S_d = \mathrm{Sz}(2^d)$ for odd d. We use standard simultaneous subfield embeddings into the faithful four-dimensional Suzuki model. Two classical facts are retained: S_d is nonabelian simple for $d \ge 3$, and S_d is maximal in S_e if e/d is prime and $d > 1$. These are among the Suzuki structural inputs used by the source itself (its Table 1, printed page 6).

We also record why subfield intersections are exact in these embeddings. In characteristic two the standard root and diagonal matrices are

$$U(a,b) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ a^{\theta+1}+b & a^\theta & 1 & 0 \\ a^{\theta+2}+ab+b^\theta & b & a & 1 \end{pmatrix}, \qquad h(t) = \mathrm{diag}(t^{\theta+1}, t, t^{-1}, t^{-\theta-1}),$$

where $\theta = 2^{(e+1)/2}$ in $\mathbb{F}_{2^e}$; let w be the antidiagonal identity matrix. The standard Bruhat forms are $U(a,b)h(t)$ and $U(a,b)h(t)wU(c,f)$. This matrix description is retained classical Suzuki theory. For odd $d \mid e$, the exponent θ restricts to $2^{(d+1)/2}$ on $\mathbb{F}_{2^d}$.

Lemma 1. *For odd divisors a, b of an odd integer n, the standard embedded subgroups satisfy*

$$S_a \cap S_b = S_{\gcd(a,b)}.$$

Proof. First, the elements of S_n whose matrix entries lie in $\mathbb{F}_{2^d}$ are exactly S_d. In the first Bruhat form, the entry g_{22} gives t, the ratio g_{21}/g_{11} gives a, and g_{42}/t gives b. Thus all parameters lie in the smaller field when the entries do. In the second form,

$$g_{14} = t^{\theta+1}, \quad c = g_{13}/g_{14}, \quad f = g_{12}/g_{14}, \quad a = g_{24}/g_{14}, \quad b = g_{34}/g_{14} + a^{\theta+1}.$$

The power $\theta+1$ is coprime to 2^n-1: an odd common divisor would give both $2^{n+1}=1$ and $2^n=1$ modulo that divisor. Consequently t is a fixed integer power of g_{14}, and hence also lies in the smaller field. The two forms are distinguished by whether g_{14} is zero. This proves the asserted equality with the matrices over $\mathbb{F}_{2^d}$. Finally $\mathbb{F}_{2^a}\cap\mathbb{F}_{2^b}=\mathbb{F}_{2^{\gcd(a,b)}}$, which gives the result. □

2 The omitted pregeometry

Let $p_1,\dots,p_k$ be distinct prime divisors of an odd integer n, and put

$$d=\frac{n}{p_1\cdots p_k},\qquad G=S_n,\qquad G_i=S_{n/p_i}.$$

All embeddings are the standard ones just fixed.

Theorem 2. *If $d>1$, the resulting rank-k coset pregeometry is residually weakly primitive. In fact every available one-step inclusion in its intersection lattice is maximal, and all its 2^k intersections are distinct.*

Proof. Repeated use of Lemma 1 gives, for every $J\subseteq I$,

$$G_J=S_{n_J},\qquad n_J=\frac{n}{\prod_{j\in J}p_j}.$$

For $i\notin J$ the ratio $n_J/n_{J\cup\{i\}}$ is the prime p_i. The smaller exponent is at least $d>1$. Classical subfield maximality therefore applies to every such inclusion. Unique factorization makes the integers n_J distinct for distinct subsets J. The order $2^{2r}(2^{2r}+1)(2^r-1)$ of S_r is strictly increasing with r, so the corresponding subgroups are distinct as well. These statements give every condition in the source's definition. □

For an explicit example take

$$n=3^2\cdot5\cdot7\cdot11=3465,\qquad(p_1,p_2,p_3,p_4)=(3,5,7,11).$$

The four maximal parabolics are

$$S_{1155},\quad S_{693},\quad S_{495},\quad S_{315},$$

and their common intersection is $S_3=\mathrm{Sz}(8)$. Theorem 2 proves that this is a rank-four residually weakly primitive pregeometry.

Theorem 3. *The displayed pregeometry is absent from Constructions 1–11 in the fixed published edition of* The residually weakly primitive pre-geometries of the Suzuki simple groups. *It therefore contradicts the literal exhaustive assertion in its Theorem 5.*

Proof. Constructions 1–10 all include a soluble maximal parabolic: a dihedral group, an affine soluble group, a cyclic group extended by a group of order four, or the specified mixed root group. Every maximal parabolic in our example is nonabelian simple. Thus a permutation of types and an ambient group automorphism cannot identify our system with any of those constructions.

Construction 11 writes $n=p_1^{e_1}\cdots p_m^{e_m}$ with the p_i pairwise distinct, and permits ranks $\alpha\in\{2,\dots,m-1\}$. Here $m=4$, so its largest permitted rank is three. Our rank is four. Theorem 5 states that all residually weakly primitive pregeometries of rank at least four are those in Constructions 1–11, with no additional operation of taking truncations. The example is therefore outside that literal list. □

3 Attribution and limits

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20. The relevant definition is on printed pages 4–5, the construction conventions on page 15, and Construction 11 and Theorem 5 on page 18. The source was received on 26 July 1999 and accepted on 21 September 1999.

The argument is a reworking of Sections 2–3 of the project's earlier Suzuki subfield manuscript, whose separate review already checked this exact omitted boundary against the source. The old proof establishes additional geometry and flag-transitivity statements using descent. Those stronger conclusions are not needed here and are not acquired as premises of this note. The elementary matrix intersection argument comes from the earlier project treatment of compatible Suzuki tori. Classical Suzuki structure and prime-degree subfield maximality remain explicit retained inputs.

The correction concerns the literal rank range. The condition

$$n/(p_1 \cdots p_k) > 1$$

includes the example; suitable extensions of the construction can therefore repair this omission. We make no claim that such a repair exhausts all embeddings, and do not classify the remaining ten constructions. Neither a new mathematical discovery nor historical priority is claimed by this reworking. It contributes no additional conjecture solution, Lean theorem or compression factor. The supplementary control checks the matrix parameter identities symbolically, and the sixteen exponents and thirty-two prime-degree inclusions by integer arithmetic. It does not enumerate S_{3465}.

Prime subfield residues do not extend to rank four

Human Compression project

Abstract

We complete the three extension arguments in Section 3.1 of the fixed higher-rank Suzuki article. None of the prime split-torus rank-three residues extends to a rank-four family satisfying the source's subgroup axiom. The proof uses maximal subgroups of an odd dihedral group and uniqueness of an embedded affine subgroup inside a fixed-size subfield. In particular, equal abstract orders are never used to identify two subfield groups. The surviving intermediate candidate in the source's second case is excluded by the already proved alignment of split tori.

1 Classical and previously proved inputs

Write $q = 2^n$, with n odd, and $G = \mathrm{Sz}(q)$. For a family (G_i) put $G_J = \bigcap_{j\in J} G_j$. We use exactly the source condition that for every proper J, some unused type gives a maximal proper inclusion $G_{J\cup\{i\}} < G_J$. The acquired strict-intersection lemma implies that adding *any* unused type strictly decreases the intersection. An induced family in any G_J satisfies the same axiom. No flag-transitivity hypothesis is imposed.

The classical Suzuki subgroup classification, torus partition and ovoid model are retained as explicit premises. A nonidentity element in a split torus has that torus as its centralizer. A full split torus has normalizer $D_q = C_{q-1} : C_2$. No point stabilizer contains an odd dihedral group with nontrivial rotation subgroup: in its abelian odd quotient, an odd rotation has nontrivial image, while an inverting involution has trivial image, which is impossible. A proper nonsolvable subgroup is a simple subfield group. A solvable subgroup is contained in a point stabilizer or one of the torus normalizers. These are the subgroup-classification consequences already used in the accepted common notes.

We also reuse the following proved consequences.

1. Every parabolic in an odd-dihedral family satisfying the subgroup axiom is maximal in that dihedral group.

2. The full split normalizer of a standard $\mathrm{Sz}(s)$ is self-normalizing in G, and determines exactly one embedded subfield of a prescribed containing degree. The same-size affine subgroup $A_s = Z_s : C_{s-1}$ determines exactly one copy of $\mathrm{Sz}(s)$.

3. The proper overgroups of a full split torus in $\mathrm{Sz}(t)$ are that torus, its split normalizer, the two affine groups at its fixed points, and the two point stabilizers. A point stabilizer cannot be a parabolic of a family of rank at least two satisfying the subgroup axiom.

4. If $t = 2^f$ has prime degree f but composite $t - 1$, there is no family of rank at least three satisfying the subgroup axiom.

5. Two full split dihedrals of subfield groups inside a common split normalizer, sharing an involution, align the two subfields. Their intersection is the standard group over the intersection field.

These are respectively the accepted cyclic/dihedral lemma, embedded normalizer and affine uniqueness lemmas, full-torus overgroup lemma, prime-degree obstruction, and shared-reflection alignment lemma. Their proofs are imported explicitly; the source's exhaustive geometry list is not a premise.

2 One prime-degree maximal subfield

Suppose $G_0 = \mathrm{Sz}(s)$ is maximal in G, where $s = 2^d \geq 8$ and $p = s - 1$ is prime. Then d is an odd prime, and $n = dr$ for an odd prime r. The two primes d, r may coincide. Fix its full split dihedral $\Delta_s = D_{2p}$ and an affine subgroup $A_s = Z_s : C_p$.

Lemma 1. *A proper subgroup containing Δ_s and not contained in G_0 is contained in the unique ambient split normalizer D_q containing Δ_s. A proper subgroup containing A_s and not contained in G_0 is contained in the unique point stabilizer fixed by Z_s.*

Proof. An odd element of order p determines a split torus in G. A simple subfield $\mathrm{Sz}(2^f)$ containing it must have $p \mid 2^f - 1$: its nonsplit orders divide $(2^f)^2 + 1$, which divides $q^2 + 1$, and $q-1$ is coprime to q^2+1. Since $p = 2^d - 1$ is prime, 2 has order d modulo p; it cannot have order one, and d is prime. Thus $d \mid f$. The only proper divisor f of dr that is divisible by d is $f = d$. The acquired embedded uniqueness for Δ_s or A_s therefore makes such a containing subfield equal to G_0.

A subgroup containing Δ_s cannot lie in a point stabilizer, by the odd-dihedral observation, or in a nonsplit normalizer, by torus coprimality. Its only remaining soluble position is in the unique D_q fixed by its rotation subgroup. A subgroup containing A_s cannot lie in any torus normalizer: it contains the elementary abelian Z_s of order $s \geq 8$, whereas their Sylow two-subgroups have order two or four. Its point stabilizer is unique, since each nonidentity element of Z_s fixes precisely one ovoid point. The subgroup classification now proves both assertions. □

Lemma 2. *Suppose a rank-four family has $G_{01} = \Delta_s$, and $G_{013} \cong C_2$. Then*

$$G_1 = D_{2p\ell}, \qquad G_{13} = D_{2\ell}, \tag{1}$$

where $\ell \neq p$ is an odd prime dividing $(q-1)/p$. If $G_{012} \cong C_2$, then also $G_{12} = D_{2\ell}$ and $G_{123} = C_\ell$. If $G_{012} = C_p$, then G_{12} is either $C_{p\ell}$ or a dihedral subgroup of order $2p$ distinct from G_{01}.

Proof. Strict intersections give $G_1 \not\leq G_0$. Lemma 1 places it inside D_q. It contains Δ_s, so is dihedral. The induced rank-three family in G_1 has all parabolics maximal. Maximality of Δ_s gives $G_1 = D_{2p\ell}$ with ℓ prime. If $\ell = p$, every maximal subgroup of G_1 contains its order-p rotation subgroup, contradicting $G_{013} \cong C_2$. For distinct p, ℓ, its maximal subgroups are the rotation group $C_{p\ell}$ and dihedrals of orders $2p$ and 2ℓ. Only the last type can intersect G_{01} in order two. A dihedral of order 2ℓ and G_{01} have a common reflection: their reflection congruences in the cyclic rotation group have coprime moduli p, ℓ. Thus the asserted intersection also has that order. Two distinct dihedrals of order 2ℓ in G_1 share exactly their order-ℓ rotation group, proving the last intersection claim. The case of order-p intersection leaves precisely the other two maximal-subgroup types. Equality of any two displayed parabolics is excluded by strict intersections. □

Lemma 3. *Under (1), suppose G_{03} is not contained in G_{01}. Then $G_3 = \mathrm{Sz}(t)$, where $t = 2^r$, $r \neq d$, and $t - 1 = \ell$ is prime.*

Proof. The group G_3 contains $G_{13} = D_{2\ell}$. This excludes a point stabilizer and the nonsplit normalizers. If G_3 were soluble, it would therefore lie in D_q, the unique split normalizer containing this dihedral. But $G_0 \cap D_q = \Delta_s$: its odd rotation subgroup is the full split torus of G_0. This would put G_{03} inside G_{01}, a contradiction.

Thus G_3 is a proper simple subfield group. Its possible field exponents are d and r, since $n = dr$. The order-ℓ rotation is split in this subfield by the same coprimality argument as above. The first size is impossible, since $\ell \neq p = s - 1$. Consequently $G_3 = \mathrm{Sz}(2^r)$ and $r \neq d$. Its induced rank-three family satisfies the subgroup axiom. The prime-degree obstruction implies that $2^r - 1$ is prime. Since it is divisible by ℓ, it equals ℓ. □

3 The three residue types

Theorem 4. *No rank-four family satisfying the source axiom has a maximal parabolic $G_0 = \mathrm{Sz}(s)$ with $s - 1$ prime. In particular none of residues R_1, R_2, R_3 in Section 3.1 of the fixed L110 source extends as proposed.*

Proof. The acquired necessary-pattern theorem gives exactly these three possibilities in the induced rank-three family of G_0. In all cases we can label $G_{01} = \Delta_s$ and $G_{013} = C_2$.

For R_1, we have $G_{02} = A_s$ and $G_{012} = C_2$. Lemma 2 gives $G_{12} = D_{2\ell}$. By strict intersections $G_2 \not\leq G_0$, so Lemma 1 puts G_2 in a point stabilizer. But it contains the odd dihedral G_{12}, a contradiction. This proves the first nonextension directly, without any appeal to an ovoid-circle intersection assumption.

For R_2, $G_{02} = A_s$, $G_{03} = D_{2a}$ for an odd prime a, and $G_{012} = C_p$. Again G_2 lies in its unique point stabilizer, so it cannot contain a dihedral group of order $2p$. Lemma 2 therefore gives $G_{12} = C_{p\ell}$. The subgroup G_{03} is not contained in G_{01} by strict intersections, so Lemma 3 gives $G_3 = \mathrm{Sz}(t)$ with $t - 1 = \ell$. The full split dihedrals G_{01} and G_{13} of G_0, G_3 lie in G_1 and share $G_{013} = C_2$. Alignment forces

$$G_0 \cap G_3 \cong \mathrm{Sz}(2^{\gcd(d,r)}) = \mathrm{Sz}(2),$$

which has order twenty. This contradicts $G_{03} = D_{2a}$ for prime a. The source's final $a = 5$ candidate is therefore impossible as well; the extra affine parabolic cannot repair this obstruction.

For R_3, $G_{02} = D_{2a}$ and $G_{03} = D_{2b}$, where the permitted half-orders are primes and at most one is two. All three intersections $G_{012}, G_{013}, G_{023}$ have order two, and the full intersection is trivial. Lemma 2 gives

$$G_{12}, G_{13} \cong D_{2\ell}, \qquad G_{123} = C_\ell.$$

Apply Lemma 3 also with types 2 and 3 interchanged. Both G_2, G_3 are copies of the same $\mathrm{Sz}(t)$, where $t-1 = \ell$. Equal sizes alone do not identify these subgroups. Instead their common subgroup G_{23} contains the full split torus C_ℓ. Within G_2, the acquired full-torus overgroup list leaves C_ℓ, its normalizer $D_{2\ell}$, the two affine groups, and the two point stabilizers. The first choice would equal G_{123}; the second would equal G_{12}, since the full torus has a unique normalizer. Both violate strict intersections. A point stabilizer is excluded as a parabolic of the induced family in G_2. Thus G_{23} is an affine group $A_t = Z_t : C_{t-1}$. The embedded affine uniqueness theorem now forces its two containing copies G_2, G_3 of $\mathrm{Sz}(t)$ to coincide, the final contradiction. □

Attribution and scope. The source is D. Leemans, *The residually weakly primitive pregeometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20, especially printed pp. 10–12. The source's case division and nonextension conclusions for R_1, R_3 are retained; the proofs here make their overgroup and uniqueness steps explicit. The R_2

obstruction reuses the earlier shared-reflection proof already applied to the full prescribed lattice of Construction 1. The general Suzuki structure is classical. No assertion that the other composite-degree residue lists are exhaustive, no whole-original completion credit, and no additional search, Lean result or complexity measurement is made by this component.

Obstructions to five prescribed Suzuki constructions

Human Compression project

Abstract

The higher-rank constructions in the fixed Suzuki pregeometry article prescribe a full intersection pattern, not just abstract parabolic types. We separate an elementary defect in three printed intersection formulas from the nonexistence of the intended patterns. Two previously established normalizer arguments exclude those patterns and the related first and sixth constructions. The arguments are reproduced here so that the source disposition does not depend on a label in a working ledger.

Write $q = 2^n$, with n odd, and work in the classical Suzuki matrix model. For odd $d \mid n$, let S_d be the standard subgroup obtained by restricting parameters to $\mathbb{F}_{2^d}$. For $d > 1$ this is the simple group $\mathrm{Sz}(2^d)$; $S_1 = \mathrm{Sz}(2)$ has order 20 and is isomorphic to $C_5 : C_4$. Let H_q be the split torus and w its standard involutory normalizer. Put $\Delta_d = H_{2^d}\langle w\rangle$, of order $2(2^d - 1)$, where H_{2^d} denotes the unique subgroup of H_q of order $2^d - 1$.

We retain the classical conjugacy of subfield groups, conjugacy of their maximal split dihedrals, the split-torus centralizer and normalizer, and the involution and Sylow-two structure. These are explicit earlier inputs, not consequences of a finite sample. In the standard coordinates the unique Bruhat parameters give

$$S_a \cap S_b = S_{\gcd(a,b)}. \tag{1}$$

Indeed a matrix belongs to both groups precisely when its uniquely recovered parameters belong to both fields. This also proves the statement when the common field is $\mathbb{F}_2$.

1 Two rigidity arguments

Lemma 1. *For $d > 1$, the normalizer of Δ_d in $\mathrm{Sz}(q)$ is Δ_d. Exactly one conjugate of S_d contains Δ_d.*

Proof. The odd cyclic subgroup of Δ_d is characteristic, and its centralizer is H_q. Every normalizer of Δ_d therefore belongs to $H_q\langle w\rangle$. For $h \in H_q$, conjugation preserves Δ_d if and only if $h^2 \in H_{2^d}$. The quotient H_q/H_{2^d} has odd order, so this is equivalent to $h \in H_{2^d}$. This proves the normalizer claim. If a conjugate of S_d contains Δ_d, pull the latter back into S_d. It is a maximal split dihedral there and can be moved to the standard one by internal conjugacy. After this adjustment the defining conjugator normalizes Δ_d, hence belongs to $\Delta_d \le S_d$. The conjugate is S_d. □

Theorem 2. *Suppose $a, b > 1$ are odd divisors of n. Let S, T be conjugates of S_a, S_b. Suppose one split dihedral subgroup contains maximal split dihedrals of S and T with a common involution. Then S, T are simultaneously conjugate to S_a, S_b. In particular, $S \cap T \cong \mathrm{Sz}(2^{\gcd(a,b)})$.*

Proof. Conjugate the ambient split-torus normalizer to $H_q\langle w\rangle$. Conjugating further by H_q moves the common reflection to w, since squaring is bijective in this odd cyclic torus. Its rotation subgroups of orders $2^a - 1$ and $2^b - 1$ are unique. The two prescribed dihedrals are therefore Δ_a, Δ_b. Lemma 1 forces the two subfield groups to be standard. Now apply (1). □

For the second argument use the standard Sylow-two subgroup U_q and its center $Z_q = \{u(0,t) : t \in \mathbb{F}_q\}$. Its nonidentity central elements are precisely its involutions. Fix $z = u(0,1)$. The retained involution-centralizer theorem says $C_G(z) = U_q$.

Lemma 3. *If a conjugate K of S_d contains z, then the identity together with the involutions in $C_K(z)$ is exactly Z_{2^d}. Consequently, conjugates of S_a, S_b with $\gcd(a,b) = 1$ cannot have a Klein four subgroup in their intersection.*

Proof. The involutions of S_d form one conjugacy class. Adjusting a defining conjugator internally therefore writes $K = S_d^v$ with $v \in C_G(z) = U_q$. Thus $C_K(z) = U_{2^d}^v$. Its identity and involutions form $Z_{2^d}^v = Z_{2^d}$, because U_q centralizes Z_q. If two subfield groups share an involution, first conjugate it to z. Their common commuting involutions lie in $Z_{2^a} \cap Z_{2^b} = Z_{2^{\gcd(a,b)}}$. For coprime degrees this group has order two. A Klein four group would supply three nonidentity commuting involutions, a contradiction. □

2 The rank-three patterns and their extensions

The fixed article L110 reproduces as R4, R5 and R6 three patterns in $\mathrm{Sz}(2^{ab})$, where a, b are distinct odd primes. They require two subfield groups $S \cong \mathrm{Sz}(2^a)$ and $T \cong \mathrm{Sz}(2^b)$ and a split dihedral D with

$$|D| = 2(2^a - 1)(2^b - 1), \quad S \cap D \cong D_{2(2^a-1)}, \quad T \cap D \cong D_{2(2^b-1)}, \quad S \cap T \cap D \cong C_2.$$

They prescribe $S \cap T$ to be, respectively, D_{10}, C_2^2, or C_4. Theorem 2 instead gives $S \cap T \cong S_1$, of order 20. None of these three full patterns exists. The primality conditions on $2^a - 1$ and $2^b - 1$ are not needed for this obstruction.

The pattern R8 has the same coprime subfield degrees and prescribes $S \cap T \cong C_4 \times C_2$. This group contains a Klein four group, so Lemma 3 excludes it. No primality condition on the chosen nonsplit-torus factors is needed for that conclusion.

The implication for a prescribed extension is elementary but must be stated with its hypotheses. If a larger family of actual subgroups has a residue with one of these full intersection patterns, intersecting the family with the residue stabilizer would realize the forbidden pattern. Hence such a larger family cannot exist. This reasoning concerns the full prescribed lattice; a bare list of abstract subgroup types does not specify the required intersections.

L110 Section 3.3.1, printed pp.12–13, derives Constructions 2, 3, 4 and 6 from residues R4, R5, R6 and R8, respectively. The introduction to the constructions on p.15 explicitly requires the full sublattice obtained from the corresponding rank-four pattern. In higher rank, fixing all but the appropriate two subfield types in these prescribed patterns leaves the corresponding rank-three pattern in a subfield stabilizer. The obstruction therefore applies to every intended full lattice in these four constructions, not just to a convenient coordinate placement.

Construction 1 requires a separate check. The rank-four lattice at the end of Section 3.1, printed p.11, has two subfield groups of distinct prime degrees and the split dihedral of order $2(2^a - 1)(2^b - 1)$. Their intersections with that dihedral are their maximal split dihedrals, with a common involution. The two subfield groups are asserted to intersect in D_{10} (the case $n = 5$ in that paragraph). This is exactly the contradiction in Theorem 2; the additional affine parabolic cannot repair it. Construction 1 extends this prescribed rank-four lattice. For more than two subfield types, fixing the other subfield types leaves the same rank-four residue. Thus none of its intended full lattices exists either. Figure 1 is explicitly conditional on existence and flag transitivity; the impossible lattice supplies no realized diagram.

3 The literal formulas are a separate issue

Let the odd field degree be the squarefree product $p_1 \cdots p_m$ and put $n_i = 2^{p_i} - 1$, as in Constructions 2–4. Their printed intersection runs from $j = 0$ through m, so includes $G_m \cong D_{2\prod_i n_i}$. The printed values are D_{10}, $D_4 = C_2^2$, and C_4, respectively. All three are impossible even before any subfield argument. Since each p_i is odd, $5 \nmid 2^{p_i} - 1$; hence the ambient dihedral order is not divisible by 5. Its two-part is exactly 2, so it has no subgroup of order 4. Lagrange's theorem excludes the three printed intersections.

The following source remark instead says that these constructions differ only in the intersection of the Suzuki subgroups. Deleting the last index from the displayed intersection repairs that particular order defect, but does not remove the independent rigidity obstruction to the intended full lattices. We therefore record both conclusions rather than treating an index correction as a proof of existence.

Attribution and scope. The source is D. Leemans, *The residually weakly primitive pregeometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20. The subgroup and normalizer premises are classical. The rigidity proofs reproduce the project's earlier Suzuki realization and higher-subfield notes; the contribution here is their explicit application to the first four and sixth higher-rank constructions and the distinction between literal and intended formulas. This is not a classification of all higher-rank Suzuki geometries, all relative positions, or the remaining constructions. It gives no new principal-refutation, open-conjecture, Lean or compression credit.

Scalar fields and split torus families

Human Compression project

Abstract

One elementary divisibility identity determines all intersections in a family of Suzuki coset pregeometries. The same identity proves that the cyclic scalar group in each affine residue acts irreducibly. This gives a uniform proof of the intended families in Constructions 5 and 7 of the fixed higher-rank source, after stating its necessary indexing and order corrections explicitly. Only residual weak primitivity is asserted here.

1 The arithmetic of the scalar subgroup

Let $e \geq 3$ be odd, let $k \geq 1$, let $a_1, \ldots, a_k$ be distinct odd primes, and put

$$t = 2^e, \quad A = \prod_{i=1}^{k} a_i, \quad q = t^A.$$

Choose distinct primes ℓ_i such that

$$\ell_i \mid \frac{t^{a_i} - 1}{t - 1}, \qquad M = (t-1) \prod_{i=1}^{k} \ell_i.$$

The primes ℓ_i need not be coprime to $t - 1$. For $J \subseteq \{1, \ldots, k\}$ set

$$D_J = \prod_{i \notin J} a_i, \quad q_J = t^{D_J}, \quad M_J = (t-1) \prod_{i \notin J} \ell_i.$$

Lemma 1. *For every J one has $\gcd(M, q_J - 1) = M_J$. In particular $M \mid q - 1$. The multiplicative subgroup of order M_J in $\mathbb{F}_{q_J}^*$ generates $\mathbb{F}_{q_J}$ as a field over $\mathbb{F}_2$.*

Proof. If $\ell_i \nmid t - 1$, the order of t modulo ℓ_i is a_i: it divides that prime and is not one. Hence $\ell_i \mid t^{D_J} - 1$ exactly when $i \notin J$. Its exponent in M is one, since the chosen ℓ_i are distinct.

If $\ell_i \mid t - 1$, reduction of $1 + t + \cdots + t^{a_i - 1}$ modulo ℓ_i shows that $\ell_i = a_i$. For an odd prime $\ell \mid t - 1$, successive binomial expansions give

$$v_\ell(t^D - 1) = v_\ell(t - 1) + v_\ell(D).$$

For completeness, raising a number congruent to one modulo ℓ to a prime power step r preserves the valuation when $r \neq \ell$: the first binomial term has coefficient r, and all later terms have larger valuation. When $r = \ell$, the first term gains exactly one power of ℓ and all later terms have strictly larger valuation, since ℓ is odd. Iterating proves the displayed formula. Since D_J is squarefree, the additional factor ℓ_i in M occurs in $q_J - 1$ precisely when $a_i \mid D_J$. All other prime powers of $t - 1$ occur on both sides. This proves the gcd identity.

Suppose that the subgroup of order M_J is contained in a subfield $\mathbb{F}_{2^f} \subseteq \mathbb{F}_{q_J}$. Its subgroup of order $2^e - 1$ gives $2^e - 1 \mid 2^f - 1$, so $e \mid f$ by the usual gcd identity for powers of two. Write $f = eD$, where $D \mid D_J$. Applying the identity just proved to $t^D - 1$ shows that each a_i with $i \notin J$ divides D; otherwise its additional prime factor in M_J would be missing. Thus $D = D_J$, and the subfield is the whole field. □

2 All subgroup intersections and the maximality condition

We use the classical Suzuki matrix model, its standard subfield groups, and maximality of prime-relative-degree inclusions of simple Suzuki groups as attributed inputs. Write $S_d = \mathrm{Sz}(2^d)$ for a standard subfield group, Z_{2^d} for the involution centre of its standard Sylow two-subgroup, and H_r for the subgroup of order r in the ambient split torus. The standard Weyl involution w inverts that torus. Unique matrix coordinates give

$$S_b \cap S_c = S_{\gcd(b,c)}.$$

Conjugation by the split torus on Z_q is multiplication by a power coprime to $q-1$. These coordinate and action facts have complete derivations, with the classical ambient model explicit, in the acquired Suzuki preliminary and normalizer notes. We do not assume the source's classification of all residually weakly primitive pregeometries.

In $G = S_{eA}$ consider the following $k+2$ parabolics, with separate labels for the last two:

$$P_i = S_{eA/a_i}\ (1 \le i \le k), \qquad P_D = H_M \rtimes \langle w \rangle, \qquad P_E = Z_q \rtimes H_M.$$

For a set of selected subfield labels J, their intersection is S_{eD_J}. Intersecting further with either or both of the last two parabolics gives exactly

$$\begin{aligned} S_{eD_J} \cap P_D &= H_{M_J} \rtimes \langle w \rangle, \\ S_{eD_J} \cap P_E &= Z_{q_J} \rtimes H_{M_J}, \\ S_{eD_J} \cap P_D \cap P_E &= H_{M_J}. \end{aligned}$$

Indeed the diagonal parameter must belong to $\mathbb{F}_{q_J}$, so Lemma 1 gives its precise cyclic order. The central unipotent parameter is restricted to that same field. The reflection coset of the split normalizer is outside the point stabilizer, leaving only the torus in the third intersection. This also proves the formulas when J is empty.

Theorem 2. *The displayed parabolics define a residually weakly primitive coset pregeometry of rank $k+2$. Its full intersection is H_{t-1}.*

Proof. The condition is that every proper subset of the labels admits a remaining parabolic whose intersection is maximal proper in the current intersection. We check all four possibilities concerning the labels D, E.

If neither has been selected and a subfield label remains, adjoining any such label gives a prime-relative-degree inclusion $S_{eD_J/a_i} < S_{eD_J}$. Both are simple, since their exponents are at least $e \ge 3$, so this is maximal. If no subfield label remains, the current intersection is S_e, and adjoining D gives its full split torus normalizer, a classical maximal subgroup.

If D but not E has been selected, adjoining E passes from $H_{M_J} \rtimes \langle w \rangle$ to its cyclic subgroup of index two.

If E but not D has been selected, adjoining D passes from $Z_{q_J} \rtimes H_{M_J}$ to H_{M_J}. This is maximal because the scalar action is irreducible over $\mathbb{F}_2$. To verify this last assertion, an additive subspace invariant under the cyclic scalar group is invariant under the field it generates. Lemma 1 identifies that field with $\mathbb{F}_{q_J}$, whose one-dimensional vector space has only the zero and full subspaces. Every intermediate subgroup containing H_{M_J} is the product of that complement and its intersection with Z_{q_J}; hence no proper nontrivial intermediate subgroup exists.

Finally, if both D, E have been selected, a subfield label i remains because the chosen subset is proper. Adjoining it passes from H_{M_J} to H_{M_J/ℓ_i}, of prime index ℓ_i. This exhausts the possibilities and proves the condition. The intersection formulas give the stated full intersection. □

3 The two source families and their literal corrections

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20. We use its fixed twenty-page edition. The introduction to the constructions, on p. 15, instructs the reader to retain the intended whole intersection lattice from the rank-four discussion. We therefore distinguish this intended lattice from literal indexing or order errors in the displayed recipes.

For Construction 7 put $k = \alpha - 2$, and take its selected relative prime degrees as $a_1, \ldots, a_k$. Its definition of t gives our $e = \log_2 t$. Its range $\alpha \leq \sum_j e_j$ leaves at least two prime factors, with multiplicity, in e, so $e \geq 3$. Its cyclic parameters are our ℓ_i. The two displayed soluble group orders are exactly those in Theorem 2. The source labels the selected Suzuki groups $G_1, \ldots, G_{\alpha-2}$ and then also labels the dihedral group $G_{\alpha-2}$, while its final tuple begins with G_0. Relabel the subfield groups $P_1, \ldots, P_k$ and keep D, E as separate labels. With that explicit indexing correction, every permitted parameter is realized, with all intended intersections and the maximality condition proved above.

Construction 5 has the same family with e prime. In its first case, the source exponent is a product of distinct primes. The distinguished prime is $e = p_i$, and the other $m - 1$ primes are our selected a_j; thus $k = m - 1$ and the rank is $m + 1$. In its second case, one prime p_m is repeated in the ambient exponent. Take $e = p_m$ and select all m distinct primes as the a_j, including p_m itself. They remain pairwise distinct; no coprimality between e and A is needed anywhere in the proof. The rank is $m + 2$.

The literal recipes in Construction 5 require more than relabeling. Both displayed dihedral orders omit the factor $t - 1$. Its first case assigns conflicting indices. Its second case uses an undefined i in the affine product and ends its tuple at G_m despite listing G_{m+1}. The corrected dihedral and affine groups are, in both cases, D_{2M} and $Z_q \rtimes H_M$, with the product taken over all selected indices. These are the groups required by the stated rank-four intersections in the preceding discussion. We prove this corrected family, not the inconsistent literal assignment.

The method extends the previously retained rank-three split family: subfield coordinates give the intersections, and scalar irreducibility proves the affine maximality step. The uniform arithmetic lemma supplies the higher-rank argument, including the case $\ell_i \mid t - 1$. No priority claim is made for this elementary arithmetic or for the classical Suzuki subgroup structure. Arbitrary relative positions, flag transitivity, residual connectedness, a classification of all families, and compression measurements are outside this component.

Extensions of the split affine residues

Human Compression project

Abstract

We determine every rank-four residually weakly primitive extension of the specified residues R7 and R9 in the fixed higher-rank Suzuki paper. The proof uses the actual prescribed intersections and does not assume that paper's classification of all rank-three residues. The main step is a subfield containment lemma for an affine subgroup whose scalar complement generates its coefficient field.

1 Conventions and structural premises

Write $S_f = \mathrm{Sz}(2^f)$ for the standard subfield group in S_N, where $f \mid N$ and both exponents are odd. All simple subfield groups below have $f \geq 3$. Write H_m for the subgroup of order $m \mid 2^N - 1$ in the standard split torus. Its Weyl involution w inverts it. Write Z_{2^f} for the involution centre of the standard Sylow two-subgroup Q_{2^f}. In particular, Z_{2^f} has the additive coordinates of $\mathbb{F}_{2^f}$. Put $D_{2m} = H_m\langle w\rangle$.

We use the classical Suzuki matrix model, its subgroup classification, and conjugacy of subfield groups and split tori, as in the accepted preliminary papers. Specifically, a proper nonsoluble subgroup is a simple subfield group. The other maximal subgroup types are a point stabilizer, a split torus normalizer, and one of the two nonsplit torus normalizers. A nonidentity odd-order element lies in a unique torus; $N_{S_N}(H_m)$ is the full split normalizer for $m > 1$. Simple subfield groups are self-normalizing. A point stabilizer has shape $Q_q : H_{q-1}$, where $|Z_q| = |Q_q/Z_q| = q$ and $[Q_q, Q_q] = Z_q$. The torus acts on both additive quotients by scalar powers coprime to $q - 1$. A nonidentity involution fixes a unique ovoid point. These are explicit structural premises, not consequences of the classification being corrected.

The source's residual weak primitivity condition says that, for every proper subset J of the type set, some remaining intersection $G_{J\cup\{i\}}$ is maximal proper in G_J. It implies that every nested intersection is strict. Indeed, if $G_J \leq G_i$ for $i \notin J$, then $G_{I\setminus\{i\}} \leq G_J \leq G_i$, contradicting maximal proper inclusion at $I \setminus \{i\}$. We use this consequence explicitly; no geometric flag transitivity is assumed.

Lemma 1. *A fixed embedded simple subfield group S_e is contained in at most one subfield group of any prescribed degree f.*

Proof. Suppose a conjugate of S_f contains S_e. Transporting to S_f and using conjugacy of its subfield groups, adjust the transporter so that it normalizes S_e. Its ambient normalizer is S_e, which is already contained in the standard S_f. Thus the conjugate is S_f itself. □

2 A field-generating affine complement

Lemma 2. *Let $s = 2^f \geq 8$, let $f \mid N$, and let $m \mid s - 1$ such that H_m generates $\mathbb{F}_s$ as a field over $\mathbb{F}_2$. Any simple subfield group containing either D_{2m} or $Z_s : H_m$ contains the standard S_f.*

Proof. Let T have degree v. First H_m is split in T. Indeed, the order of 2 modulo m is f, an odd number. If H_m were in a nonsplit torus, then $m \mid 2^{2v}+1$, so this order would divide $4v$ but not $2v$, which is impossible for an odd order. Consequently $m \mid 2^v-1$, and field generation gives $f \mid v$.

All subgroups of order m in split tori of S_v are conjugate. If $T = gS_vg^{-1}$ contains H_m, adjust g by an element of S_v so that it normalizes H_m. The ambient normalizer is the split dihedral group. Its reflection normalizes S_v, so we may take $g = h \in H_{q-1}$, where $q = 2^N$.

If $w \in T$, then $h^{-1}wh = h^{-2}w \in S_v$, whence $h^2 \in H_{2^v-1}$. All torus orders are odd, so $h \in H_{2^v-1}$ and $T = S_v$. This treats D_{2m}.

For the affine case, $T \cap Z_q = hZ_{2^v}h^{-1}$ has additive coordinate set $\lambda\mathbb{F}_{2^v}$, where λ is the scalar by which h acts on Z_q. Containment of Z_s implies $1 \in \mathbb{F}_s \subseteq \lambda\mathbb{F}_{2^v}$, hence $\lambda \in \mathbb{F}_{2^v}$. The central scalar exponent is coprime to $q-1$. Taking its inverse power shows that $h \in H_{2^v-1}$. Again $T = S_v$, which contains S_f. □

We also use the following acquired arithmetic fact. If $t = 2^e \geq 8$, a is an odd prime, and $n \mid (t^a-1)/(t-1)$ is prime, then $H_{(t-1)n}$ generates $\mathbb{F}_{t^a}$. If $n \nmid t-1$, the order of t modulo n is a. If $n \mid t-1$, then $n = a$ and the valuation of n in t^a-1 is exactly one larger than in $t-1$. In either case no proper intermediate field contains this cyclic subgroup. The full valuation proof, including overlap with $t-1$, is in the accepted paper *Scalar fields and split torus families.*

3 All extensions of the prescribed residue

Theorem 3. *Let a rank-four residually weakly primitive coset pregeometry in $G = \mathrm{Sz}(q)$ have $G_0 = \mathrm{Sz}(s)$ maximal in G. Suppose its prescribed G_0-residue is R7 or R9. In explicit coordinates put*

$$t = 2^e \geq 8, \quad s = t^a, \quad b = t-1, \quad m_0 = bn, \qquad n \mid (s-1)/b \text{ prime},$$

where a is prime. The prescribed lattice is

$$G_{01} = \mathrm{Sz}(t), \quad G_{02} = D_{2m_0}, \quad G_{03} = Z_s : H_{m_0},$$

$$G_{012} = D_{2b}, \quad G_{013} = Z_t : H_b, \quad G_{023} = H_{m_0}, \quad G_{0123} = H_b.$$

After one simultaneous conjugation, every extension has

$$q = t^{ac}, \quad G_1 = \mathrm{Sz}(v), \quad v = t^c, \qquad a \neq c \text{ primes},$$

and, for a prime $n' \neq n$ dividing $(v-1)/b$,

$$G_2 = D_{2bnn'}, \qquad G_3 = Z_q : H_{bnn'}.$$

All remaining intersections are

$$G_{12} = D_{2bn'}, \quad G_{13} = Z_v : H_{bn'}, \quad G_{23} = H_{bnn'}, \quad G_{123} = H_{bn'}.$$

Proof. We first justify the coordinates in the statement. Conjugate G_0 and its simple subfield G_{01} to the standard copies. The full split normalizer G_{012} in G_{01} may then be made $H_b\langle w\rangle$. The odd rotation subgroup of G_{02} lies in the unique ambient split torus containing H_b, so $G_{02} = H_{m_0}\langle w\rangle$. The group G_{03} is an affine subgroup of G_0 with elementary abelian radical of order s. Its radical is the centre of a Sylow two-subgroup of G_0. Since $G_{023} = H_{m_0}$, its fixed point is one of the two torus axis points. Interchanging them by w if necessary gives

$G_{03} = Z_s : H_{m_0}$. This also gives the displayed coordinates for G_{013} and all deeper prescribed intersections.

Since G_0 is maximal, $q = s^c$ for a prime c. The proper group G_1 contains the simple $\mathrm{Sz}(t)$, hence is itself a simple subfield group. Lemma 1 makes it standard. Its degree is a proper divisor of eac, contains e, and has gcd e with ea because $G_0 \cap G_1 = \mathrm{Sz}(t)$. Therefore its degree is ec, with $a \neq c$. Thus $G_1 = \mathrm{Sz}(v)$, $v = t^c$.

The scalar group H_{m_0} generates $\mathbb{F}_s$. Lemma 2 and maximality of S_{ea} show that no proper subfield overgroup of G_{02} or G_{03} can lie outside G_0. Strictness excludes containment of G_2 or G_3 in G_0. A point stabilizer cannot contain D_{2m_0}: the odd torus fixes its two axis points and the reflection interchanges them. Nonsplit torus normalizers are also excluded by their odd orders. Hence

$$G_2 = D_{2M}, \quad M \mid q-1, \qquad m_0 = \gcd(M, s-1), \quad m_1 := \gcd(M, v-1) > b.$$

Here $G_{12} = D_{2m_1}$ and $\gcd(m_0, m_1) = b$. The elementary abelian subgroup Z_s has order at least eight, excluding both sorts of torus normalizer as overgroups of G_3. Its fixed point is unique. Thus G_3 lies in the standard point stabilizer $B_q = Q_q : H_{q-1}$.

Write $G_{23} = H_h$. This is cyclic, since the reflection coset of G_2 is outside B_q. Strictness and the prescribed intersection with G_0 give $h > m_0$ and $\gcd(h, s-1) = m_0$. The field generated by H_h contains $\mathbb{F}_s$ and is not $\mathbb{F}_s$; since $[\mathbb{F}_q : \mathbb{F}_s] = c$ is prime, it is $\mathbb{F}_q$.

Put $N = G_3 \cap Q_q$. Its intersection with Z_q contains Z_s. The scalar action of H_h is irreducible on Z_q, so $N \cap Z_q = Z_q$. Its image on Q_q/Z_q is also invariant under this irreducible scalar action. Thus $N = Z_q$ or $N = Q_q$. The second case would make $G_0 \cap G_3$ contain Q_s, contradicting the prescribed central affine group. Therefore $N = Z_q$.

The quotient G_3/Z_q maps injectively into the cyclic complement of B_q/Z_q, and is therefore abelian. It contains the nontrivial scalar subgroup H_h. In the affine quotient $Q_q/Z_q : H_{q-1}$, a translation commuting with a nonidentity scalar must be zero. Every element of G_3/Z_q therefore has zero translation coordinate. Consequently

$$G_3 = Z_q : H_d, \qquad h = \gcd(M, d), \qquad m_0 \mid h \mid d.$$

Put $d_1 = \gcd(d, v-1)$. Both $m_0 < d$ and $d_1 < d$: the former follows from $h > m_0$, and the latter from field generation by H_{m_0}, since $\mathbb{F}_s \not\subseteq \mathbb{F}_v$. The intersections G_{03} and G_{13} have proper intermediate overgroups $Z_q : H_{m_0}$ and $Z_q : H_{d_1}$ in G_3. If $h < d$, then $G_{23} = H_h < H_d < G_3$ as well. This would violate residual weak primitivity at G_3. Hence $h = d$, so $d \mid M$.

We claim $d = M$. Suppose $d < M$. Then H_d is not maximal in D_{2M}, because $H_d < H_M < D_{2M}$. Residual weak primitivity at G_2 makes either D_{2m_0} or D_{2m_1} maximal. For odd dihedral groups this means M/m_0 or M/m_1 is prime: all overgroups of a fixed reflection are obtained by enlarging its cyclic rotation subgroup. If M/m_0 is prime, the strict divisor chain $m_0 < d \mid M$ gives $d = M$, a contradiction. Otherwise M/m_1 is prime. Since $m_0/\gcd(m_0, m_1) = n$ divides it, we have $M = nm_1$. As $m_0 \mid d < M = \mathrm{lcm}(m_0, m_1)$, one has $\gcd(d, m_1) < m_1$. In the residue at $G_{12} = D_{2m_1}$, the intersection with G_3 is then a proper subgroup of H_{m_1} and cannot be maximal. The intersection with G_0, namely D_{2b}, must be maximal. Thus m_1/b is prime. But $M/m_0 = m_1/b$, again forcing $d = M$ from $m_0 < d \mid M$. This contradiction proves the claim.

Now $G_{123} = H_{m_1}$. The final residual condition makes H_b maximal in it, so $m_1 = bn'$ for a prime n'. The equality $\gcd(m_0, m_1) = b$ gives $n' \neq n$. At $G_{23} = H_M$, at least one of H_{m_0} and H_{m_1} is maximal. The prime index must respectively be n' or n, because the other subgroup is contained in H_M. Therefore $M = bnn'$. The coordinate intersections now give every asserted formula. □

4 Exact source scope and attribution

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20, fixed twenty-page edition. R7 is discussed on p. 13 and again in Section 3.3.2; R9 is on p. 14. In R7, e is prime; R9 allows composite e. The proof above works in both cases. It supplies the conditional necessity missing from a proof of the canonical Constructions 5 and 7 alone. It proves neither the source's false assertion that these and the other listed residues exhaust all cases, nor a classification of unrestricted relative positions outside the prescribed residue.

There are also literal errors in the R7 paragraph. Its final affine group is called G_4 although the types are $0, 1, 2, 3$; it is G_3. Its printed condition $n' \mid (q-1)/(s'-1)$ does not express the required additional scalar factor. The correct condition is $n' \mid (s'-1)/(t-1)$, with $s' = v$, as printed in the R9 paragraph. The theorem gives the corrected full lattice without using either of these inconsistent expressions.

The classical ambient structure and conjugacy facts are borrowed as stated. The scalar valuation argument is reused from the accepted split-family proof. The coordinate containment lemma and the necessity argument above are the present contribution. They repair a gap in our first whole-source claim ledger, which confused a refuted global classification with these separate conditional assertions. No new finite search, Lean declaration, principal refutation, or compression measurement is claimed.

Extensions of residues containing subfield groups and tori

Human Compression project

Abstract

We determine all rank-four residually weakly primitive extensions of the specified residues R10, R11 and R13 in the fixed higher-rank Suzuki paper. A torus subgroup together with its full Weyl complement determines its containing subfield groups uniquely. A second argument extracts a Boolean divisor lattice from residual weak primitivity. Neither argument assumes the source's classification of all residues.

1 Premises and residual conditions

Let $S_f = \mathrm{Sz}(2^f)$, with f odd, denote a standard subfield group of S_N, where $f \mid N$. We use the classical subgroup and conjugacy theorems for Suzuki groups, stated in the accepted preliminary papers. In particular, every proper nonsoluble subgroup is a simple subfield group. Simple subfield groups are self-normalizing, and their copies inside any given Suzuki group form one conjugacy class. The other maximal subgroups are point stabilizers and normalizers of the three cyclic torus types. A nonidentity odd-order element determines its torus uniquely. The three torus orders

$$2^f - 1, \qquad 2^f + 2^{(f+1)/2} + 1, \qquad 2^f - 2^{(f+1)/2} + 1$$

are pairwise coprime. Their normalizers have respective complements of orders $2, 4, 4$. In each case the complement contains an involution inverting the torus. Copies of the standard subfield groups intersect according to $S_a \cap S_b = S_{\gcd(a,b)}$.

The accepted torus construction supplies two compatible nonsplit families T_f^+, T_f^-, together with a common complement of order four. Their signs need not agree with the signs of the square-root formulas at every exponent. What matters here is

$$T_a^\epsilon \cap T_b^\epsilon = T_{\gcd(a,b)}^\epsilon, \qquad \gcd(|T_a^\epsilon|, |T_b^\epsilon|) = |T_{\gcd(a,b)}^\epsilon|. \tag{1}$$

For $d \mid f$, the two bottom orders divide the two different upper orders. These are reused structural premises, proved in the accepted paper *Subfield and torus geometries*.

Write $G_J = \bigcap_{j \in J} G_j$ and $G_\varnothing = G$. Residual weak primitivity means that for each proper $J \subset I$ some $G_{J \cup \{i\}}$ is maximal proper in G_J. It implies

$$G_{J \cup \{i\}} < G_J \quad (i \notin J). \tag{2}$$

Indeed, equality would imply $G_{I \setminus \{i\}} \leq G_i$; the last panel would then equal G_I, contrary to maximal proper inclusion. The arguments below only need this group condition.

2 Uniqueness of subfield overgroups

Lemma 1. *A fixed simple subfield group is contained in at most one subfield group of any prescribed degree. The same assertion holds for $H = C_m : C_c$ when $m > 1$ is odd and C_c is the full torus complement: $c = 2$ in a split torus, or $c = 4$ in a nonsplit torus.*

Proof. The first assertion follows from conjugacy of subfield copies and self-normalization. More generally, suppose $H \leq S_f$ and a conjugate gS_fg^{-1} also contains H. If the copies of H inside S_f are conjugate, adjust g by an element of S_f to normalize H. When $N_G(H) \leq S_f$, this forces $gS_fg^{-1} = S_f$.

For the stated torus subgroup, C_m is characteristic: it consists of the identity and all odd-order elements of H. Therefore its normalizer is contained in the normalizer $T : C_c$ of its unique torus. Remove the complement component of a normalizing element, leaving $h \in T$. If $z \in C_c$ is the involution, then $hzh^{-1} = h^2z$ under a choice of conjugation convention. Normalization of H implies $h^2 \in C_m$. Since $|T|$ is odd, this implies $h \in C_m$. Hence $N_G(H) = H$.

Copies of H in a Suzuki group are conjugate: the odd subgroup selects a unique torus type, the cyclic torus has a unique subgroup of order m, and its complements are conjugate in the odd-order torus extension. This proves the second assertion by the preceding transporter argument. The use of the full complement is essential; the assertion does not cover $C_m : C_2$ inside a nonsplit torus. □

Lemma 2. *Suppose two simple subfield groups of degrees a, b contain the same $H = C_m : C_c$ of Lemma 1. In the nonsplit case assume $m > 5$. Their intersection is a simple subfield group of degree $d = \gcd(a, b)$.*

Proof. The split type persists in every odd-degree subfield containing it: a nonsplit occurrence would make its odd order divide both $2^N - 1$ and $2^{2N} + 1$, whose gcd is one. In the split case $m \mid \gcd(2^a - 1, 2^b - 1) = 2^d - 1$. Thus H has a standard copy in S_d. Here $d > 1$, since $m > 1$. In the nonsplit case H is nonsplit in both groups, since its complement has order four. The identity

$$\gcd(2^{2a} + 1, 2^{2b} + 1) = 2^{2d} + 1$$

holds because a/d and b/d are odd. Thus m divides the product of the two nonsplit orders at d. It divides just one of them: each bottom order divides a different upper order at a, whereas m divides only the upper order containing its torus. Coprimality of the upper orders excludes any factor from the other bottom order. The compatible torus family therefore places a copy of H in S_d. Its action agrees with that of H: the order-four action is raising to $\pm 2^d$, and an odd multiple of d gives this action or its inverse modulo m. Reversing a complement generator identifies the two presentations. Since the only nontrivial nonsplit order at $d = 1$ is 5, the assumption $m > 5$ makes $d > 1$.

Conjugate the copy of H in $S_d \leq S_a$ onto the given one inside the first group. In these coordinates the standard S_b is an overgroup of H. Lemma 1 identifies it with the second group. Their intersection is now the standard S_d. □

Lemma 3. *Let $G_0 = S_{N/c}$ be maximal in S_N, where c is prime. Suppose a proper simple subfield group K shares an embedded simple subfield with G_0, and $G_0 \cap K = S_{N/(ca)}$, where a is prime and the common group is simple. If $K \not\leq G_0$, then $a \neq c$ and $K = S_{N/a}$ after simultaneous standardization.*

Proof. The common simple group and Lemma 1 make both groups standard. If v is the degree of K, then $v \mid N$, $v \nmid N/c$, and $\gcd(v, N/c) = N/(ca)$. Since N/c lacks just one occurrence of the prime c, the first two conditions force $v_c(v) = v_c(N)$. Consequently $a \neq c$, as otherwise the gcd would have to lack a second occurrence of c. At a the gcd equality forces $v_a(v) = v_a(N) - 1$; at every other prime different from c it forces the full valuation in N. Thus $v = N/a$. This proof allows a and c to divide the common base exponent. □

3 A divisor lattice forced by residual primitivity

Lemma 4. *Let $L = C_M : C_c$ have odd rotation order M. Let $H_i = C_{d_i} : C_c$ for $1 \le i \le r$, with a common fixed complement, and put $\gamma = \gcd(d_1, \dots, d_r)$. If these intersections satisfy residual weak primitivity, then there are distinct primes $\ell_1, \dots, \ell_r$ such that*

$$M = \gamma \prod_{i=1}^{r} \ell_i, \qquad d_i = M/\ell_i. \tag{3}$$

Primes ℓ_i may divide γ.

Proof. For $r = 1$ the conclusion is precisely maximality; assume $r \ge 2$. Every subgroup between $C_d : C_c$ and $C_M : C_c$ has the form $C_h : C_c$, $d \mid h \mid M$: it contains the complement, and removing that component from each element leaves its rotation. Thus a proper inclusion is maximal exactly when the ratio of rotation orders is prime. Successively applying residual weak primitivity gives r prime-index steps from L to the full meet. Hence $\Omega(M/\gamma) = r$, with multiplicity counted.

For each i, (2) says $\gcd(d_j : j \ne i) > \gamma$. Some prime ℓ_i therefore has

$$v_{\ell_i}(d_i) < \min_{j \ne i} v_{\ell_i}(d_j).$$

One prime cannot witness this inequality for two different indices. We have found r distinct primes dividing M/γ. Since its total number of prime factors is r, each occurs exactly once and there are no other prime factors. At the witnessing prime, d_i has the bottom valuation and every other d_j the top valuation. At all other primes every d_j has the valuation of M. These statements give (3), including when a witnessing prime divides γ. □

4 The torus residues R10 and R11

The R11 row calls n, n' two primes without expressly distinguishing them. The prescribed lattice forces them to be distinct: if $n = n'$, the subgroups G_{012}, G_{023} have the same cyclic rotation subgroup inside G_{02} and contain the same full complement G_{0123}. They coincide, so their meet has order $c\gamma n$ instead of the prescribed $c\gamma$. Thus the equal-prime case has no realization.

Lemma 5. *In an actual rank-three residually weakly primitive residue of the stated R10 or R11 form, write its ambient group as $\mathrm{Sz}(s)$, its two simple parabolics as $\mathrm{Sz}(t), \mathrm{Sz}(t')$, and their common simple intersection as $\mathrm{Sz}(u)$, where $u \ge 8$. Suppose the third parabolic is $C_{\gamma nn'} : C_c$, with its two corresponding intersections $C_{\gamma n} : C_c$ and $C_{\gamma n'} : C_c$, and common intersection $C_\gamma : C_c$. Then*

$$t = u^a, \qquad t' = u^b, \qquad s = u^{ab}$$

for distinct odd primes a, b. Thus these degree conditions follow from the residue and need not be supplied as extra hypotheses.

Proof. The common simple group standardizes the three field groups. Conjugation inside $\mathrm{Sz}(u)$ makes its full torus normalizer standard without disturbing these field groups. Their tori then belong to the same compatible family. Write h_f for its order at exponent f, and $u = 2^e$, so $h_e = \gamma$. If $f' > f \ge 3$ and $f \mid f'$, then h_f is a proper divisor of $h_{f'}$. Divisibility is (1). Strictness is immediate for the split family. For either nonsplit family it follows from $f' \ge 3f$ and

$$2^{3f} - 2^{(3f+1)/2} + 1 > 2^f + 2^{(f+1)/2} + 1 \quad (f \ge 3),$$

since the lower bound for an upper torus order exceeds the upper bound for a bottom torus order.

Let $t = 2^v$. If v/e were composite, an intermediate exponent $e < w < v$ would give a strict divisor chain $\gamma = h_e < h_w < h_v$. In particular h_v/γ could not be the prime n. Therefore $C_{\gamma n} : C_c$ would be strictly contained in the full torus normalizer of $\mathrm{Sz}(t)$, and would not be maximal. Residual weak primitivity at that simple parabolic would force $\mathrm{Sz}(u)$ to be maximal in $\mathrm{Sz}(t)$, which makes v/e prime, a contradiction. Thus $v/e = a$ is prime. The same argument gives $t' = u^b$ with b prime. If $a = b$, uniqueness of the containing field group would make the two parabolics equal, contrary to their prescribed intersection $\mathrm{Sz}(u)$. Hence $a \neq b$.

The degree of s is a multiple of eab. If it were larger, the four field exponents $e < ea < eab < \log_2(s)$ would give three strict torus-order increases. The ratio of the ambient torus order to γ would have at least three prime factors with multiplicity, so it could not equal nn'. The third parabolic would not be maximal. Residual weak primitivity in $\mathrm{Sz}(s)$ would then make one of $\mathrm{Sz}(u^a), \mathrm{Sz}(u^b)$ maximal. Its relative degree is respectively a multiple of b or a, with an additional factor, and is not prime. This is a contradiction. Thus $s = u^{ab}$, as asserted. □

Theorem 6. *Let $u = 2^e \geq 8$ and let a, b be distinct odd primes. Suppose a rank-four residually weakly primitive pregeometry in $\mathrm{Sz}(q)$ has a maximal parabolic $G_0 = \mathrm{Sz}(u^{ab})$ with the prescribed residue*

$$G_{01} = \mathrm{Sz}(u^a), \quad G_{03} = \mathrm{Sz}(u^b), \quad G_{02} = C_{\gamma nn'} : C_c,$$

$$G_{013} = \mathrm{Sz}(u), \quad G_{012} = C_{\gamma n} : C_c, \quad G_{023} = C_{\gamma n'} : C_c, \quad G_{0123} = C_\gamma : C_c, \tag{4}$$

where n, n' are distinct primes. For R10 put $c = 2$, $\gamma = u - 1$; for R11 put $c = 4$ and let γ be a nonsplit torus order at u. Then an odd prime d distinct from a, b satisfies

$$q = u^{abd}, \qquad G_1 = \mathrm{Sz}(u^{ad}), \qquad G_3 = \mathrm{Sz}(u^{bd}), \qquad G_{13} = \mathrm{Sz}(u^d).$$

There is a prime n'' distinct from n, n' such that

$$\begin{aligned} G_2 &= C_{\gamma nn'n''} : C_c, \\ G_{12} &= C_{\gamma nn''} : C_c, \qquad G_{23} = C_{\gamma n'n''} : C_c, \\ G_{123} &= C_{\gamma n''} : C_c. \end{aligned} \tag{5}$$

These formulas hold simultaneously after conjugation. In R10, $n'' \mid (u^d - 1)/(u - 1)$. In R11, $\gamma n''$ divides the compatible nonsplit torus order at u^d, and $\gamma nn'n''$ divides its ambient order. All remaining intersections are determined by (4)–(5).

Proof. The groups G_1, G_3 contain the simple groups G_{01}, G_{03}, so are simple subfield groups. All share G_{013}, allowing simultaneous standardization by Lemma 1. Maximality of G_0 gives $q = (u^{ab})^d$ for a prime d. Applying Lemma 3 to its two specified maximal subfield intersections gives $d \neq a, b$, and the asserted degrees and meet. Conjugate the common full torus normalizer G_{0123} inside S_e onto the chosen standard normalizer. This preserves all three standard simple groups and fixes their torus coordinates simultaneously.

Put $H = G_{02}$. If G_2 were a simple subfield group, Lemma 2 would make $G_0 \cap G_2$ a simple subfield group, contradicting $G_0 \cap G_2 = H$. In the nonsplit case its rotation order $\gamma nn'$ exceeds five even when $\gamma = 5$. This explains why one must use the entire H, rather than claim that its base rotation C_γ generates the base field.

Choose a minimal Suzuki overgroup of G_2, allowing the ambient group. Since G_2 is not itself a Suzuki group, it lies in a maximal subgroup of this overgroup. Minimality rules out the subfield type, so it lies in a point stabilizer or torus normalizer. A point stabilizer

cannot contain H: its quotient by its normal two-subgroup is cyclic of odd order, whereas the involution in H inverts a nontrivial odd rotation. The images of that involution and its inverse action would force this rotation to equal its inverse. It cannot disappear in the two-subgroup either. The wrong torus types are excluded by coprimality. The unique torus containing the odd part of H is therefore also the torus for G_2. Thus $G_2 = C_M : C_c$, with the same complement as H.

Its three parabolic intersections with G_0, G_1, G_3 all contain the complement from G_{0123}. Their full meet is $C_\gamma : C_c$. Lemma 4 says that M/γ is the product of three distinct primes, one omitted at each parabolic. The prescribed orders of G_{02}, G_{012}, G_{023} identify two as n, n' and force the third, n'', in precisely the positions of (5). Intersecting with $G_{13} = \mathrm{Sz}(u^d)$ proves the final divisibility condition. The inherited unique torus determines its compatible branch. Its containment in the ambient torus also gives the asserted ambient divisibility. □

5 The pure subfield residue R13

Theorem 7. *Suppose $G_0 = S_{N/d}$ is maximal in S_N and its prescribed rank-three residue has parabolics $G_{0i} = S_{N/(da_i)}$ for $i = 1, 2, 3$, where a_1, a_2, a_3 are distinct odd primes and their common intersection is simple. Every rank-four residually weakly primitive extension has $a_i \neq d$ and $G_i = S_{N/a_i}$, after one simultaneous conjugation. All intersections are the standard subfield intersections.*

Proof. Each G_i contains the simple G_{0i}, hence is a simple subfield group. The common simple subgroup standardizes every one of them by Lemma 1. Strictness excludes $G_i \leq G_0$. Lemma 3 now gives the stated degrees. Their simultaneous standard coordinates give every intersection by the gcd formula. □

6 Source scope and attribution

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20, fixed edition, pp. 14–15. The R10 paragraph interchanges the labels of its second new Suzuki parabolic and its dihedral parabolic. The consistent labels are G_1, G_3 for the Suzuki groups and G_2 for the dihedral group, as fixed by the prescribed intersections in its opening sentence.

The ambient structure and torus compatibility are borrowed as stated. The overgroup, divisor-lattice and conditional extension arguments above are supplied here to repair a gap in our first claim ledger. A refutation of the source's global classification did not settle these distinct conditional assertions. The present theorems settle the rank-four assertions for the actual prescribed residues. The accepted constructions already prove realization of their admissible canonical parameters. Neither those existence results nor this note classifies all unrestricted higher-rank relative positions. In particular, the source's separate upper rank bound for pure subfield constructions remains refuted by the previously accepted repeated-prime example. No new open-conjecture, principal-refutation, Lean or compression credit is asserted here.

Field descent and compatible tori

Human Compression project

Lang–Steinberg descent gives a common representative for pairwise incident cosets of subgroups fixed by independent field automorphisms. Cyclic coset coordinates then add a torus normalizer as another type. We prove the resulting Suzuki geometries are thick, residually connected and residually weakly primitive in every rank. The proof retains classical Suzuki structure and Lang–Steinberg, and includes the exact finite matrices used to choose compatible tori.

1 Main statements

The previously acquired rank-four configuration, described by the explicit groups below, is a flag-transitive geometry. It is firm and residually connected as well as RWPRI. The argument below establishes the common-point property for every flag, without enumerating the ambient $\mathrm{Sz}(2^{105})$.

There is also an unconditional family in every rank. Define, for positive odd d,

$$\tau_d = 2^d + 1 - \big((1+i)^d + (1-i)^d\big). \tag{1}$$

Thus $\tau_1 = 1, \tau_3 = 13, \tau_5 = 41, \tau_7 = 113, \tau_9 = 481$. Equivalently, $\tau_d = 2^d + 1 - \epsilon_d 2^{(d+1)/2}$, where $\epsilon_d = 1$ for $d \equiv 1, 7 \pmod 8$, and $\epsilon_d = -1$ for $d \equiv 3, 5 \pmod 8$.

Theorem A. Fix odd $e \geq 3$, and any distinct odd primes $a_1, \ldots, a_k$ not dividing τ_e. For each i, choose any prime divisor

$$\ell_i \mid \tau_{ea_i}/\tau_e.$$

These quotients are integers greater than one. Their prime divisors are coprime to τ_e and distinct between different indices. Put $n = e\prod_i a_i$, $m = \tau_e \prod_i \ell_i$, and $G = \mathrm{Sz}(2^n)$. There are simultaneous standard subfield subgroups

$$S_i = \mathrm{Sz}(2^{n/a_i})$$

and one subgroup $K = C_m : C_4$ for which $\Gamma(G; S_1, \ldots, S_k, K)$ is a thick, residually connected, flag-transitive RWPRI geometry of rank $k+1$. The theorem provides explicit intersection formulas in Section 6.

In particular, fixing $e = 3$ and choosing any k distinct odd primes other than 13 gives a geometry of rank $k+1$. No conjecture about infinitely many prime values of Suzuki torus orders is used.

For example, with $e = 3$ and relative field primes $3, 5, 7, 11$, one can choose $\ell_i = 37, 41, 113, 2113$. This gives a rank-five example over $\mathbb{F}_{2^{3465}}$, described by the same formulas as the smaller ones. Enumerating this field or group is unnecessary.

The construction specializes the nonsplit-torus pattern in Construction 9 of L110, Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, pp.17–18. It supplies simultaneous embeddings and flag transitivity for this family. It is not a replacement classification of all the paper's configurations or all relative positions of its subgroup types.

Two reusable principles do most of the geometric work:

- compatible cosets of fixed subgroups of coprime field automorphisms have a common point, by Lang descent;
- a subgroup with independent coset coordinates can be added as another type without losing that common-point property.

2 Cyclic descent and intersections of subfield cosets

Let $\mathbf{H}$ be a connected algebraic group over an algebraic closure of a finite field, and let F be a Steinberg endomorphism. We use Lang–Steinberg surjectivity: every $u \in \mathbf{H}$ has the form $x^{-1}F(x)$. This is retained classical theory, not reproved here; see, for example, Harper, *Totally deranged elements of almost simple groups and invariable generating sets*, Section 2.2 and equation (2.1), which use it for Steinberg endomorphisms, including the very twisted groups.

Lemma B: finite cyclic descent. Set $G = \mathbf{H}^{F^m}$, and let $\sigma = F^d|_G$, where $d \mid m$ and $h = m/d$. If $u \in G$ satisfies

$$u\,\sigma(u)\cdots\sigma^{h-1}(u) = 1,$$

then $u = x^{-1}\sigma(x)$ for some $x \in G$.

Indeed, apply Lang–Steinberg to F^d on the connected algebraic group: choose $x \in \mathbf{H}$ with $u = x^{-1}F^d(x)$. The displayed norm telescopes to $x^{-1}F^m(x)$. Its being one proves $x \in G$, which is precisely the finite-field conclusion needed. No coprimality with the characteristic is required.

Equivalently, in $G \rtimes C_h$, with the specified cyclic action, every element projecting to a generator and having order h is G-conjugate to that generator. The cyclic factor here may be treated as an abstract group of order h, so no faithfulness issue is hidden.

Theorem C: intersections of subfield cosets. Suppose $b_1, \ldots, b_k$ are pairwise coprime divisors of m. Define

$$S_i = \mathbf{H}^{F^{m/b_i}} \leq G = \mathbf{H}^{F^m}.$$

For any selection of indices, if the left cosets g_iS_i intersect pairwise, then their total intersection is nonempty.

Proof. Work in $E = G \rtimes \langle f \rangle$, with f an abstract element of order m acting as $F|_G$, and put $f_i = f^{m/b_i}$. Associate to g_iS_i the element

$$c_i = g_i f_i g_i^{-1}.$$

This association is well defined: $S_i = C_G(f_i)$. If g_iS_i and g_jS_j meet at y, both c_i, c_j are conjugated from f_i, f_j by y. Hence they commute. Their orders b_i are pairwise coprime, so the group D that they generate is cyclic of order $b = \prod_i b_i$. Its projection to $\langle f \rangle$ is an isomorphism onto $\langle f^{m/b} \rangle$.

Let c be its unique element projecting to $f^{m/b}$. Write $c = uf^{m/b}$. Since $c^b = 1$, its coefficient u satisfies the norm condition in Lemma B, with $d = m/b$. Consequently D is conjugate by an element $x \in G$ to $\langle f^{m/b} \rangle$. The conjugating element may be oriented so that

$$c_i = x f_i x^{-1} \quad \text{for every } i.$$

There is no ambiguity of generators: conjugation by G acts trivially on the quotient E/G, and the projection is injective on D. Thus $g_i^{-1}x$ centralizes f_i, so $x \in g_iS_i$ for every i. This proves the theorem.

For Suzuki groups take $\mathbf{H} = \mathrm{Sp}_4(\overline{\mathbb{F}}_2)$ and its standard exceptional Steinberg endomorphism ρ, with $\rho^2 = \phi_2$, ordinary binary Frobenius. For positive odd d, $\mathbf{H}^{\rho^d} = \mathrm{Sz}(2^d)$ in compatible standard coordinates. This is the classical Suzuki construction; see De Medts–Naert, *Suzuki–Ree groups as algebraic groups over* $\mathbb{F}_{\sqrt{p}}$, Section 2, for the exceptional endomorphism and its square. On $\mathbf{H}^{\rho^n}$, n odd, ρ and ordinary binary Frobenius generate the same cyclic field-automorphism group. Taking $F = \rho^e$, $m = \prod_i a_i$, therefore gives Theorem C for the standard S_i in Theorem A.

3 Adding a type by independent coset coordinates

The following lemma is purely group theoretic. In particular it does not identify a coset set with a quotient group.

Lemma D. Let (S_i) be a typed coset family in a group G with the common-point property of Theorem C. Suppose $K \leq G$, and put $H_i = K \cap S_i$. If the set map

$$K \longrightarrow \prod_i K/H_i, \qquad k \longmapsto (kH_i)_i \tag{2}$$

is surjective, then the enlarged typed coset family (S_i, K) also has the common-point property.

A flag not containing a K-coset is handled by the original property. Otherwise translate its K-coset to K. Each other coset meets K, so it is $k_i S_i$ with $k_i \in K$, and its intersection with K is $k_i H_i$. Surjectivity in (2) supplies a common element. Partial selections of types cause no problem: assign arbitrary cosets to the unused coordinates. Every flag therefore has a common representative and extends to a full standard-chamber translate. This proves the geometry axiom and flag transitivity.

A convenient sufficient condition is

$$K = (V_1 \times \cdots \times V_k) \rtimes B, \qquad H_i = \Big(\prod_{j \neq i} V_j\Big) \rtimes B, \tag{3}$$

where B preserves each direct factor. The restricted set map from $\prod_i V_i$ to the product of coset sets is a bijection: its i-th coordinate is just the V_i-coordinate. Neither H_i nor B needs to be normal in K.

This already settles the earlier rank-four example. There $V = C_{13} \times C_{41} \times C_{113}$, $B = C_4$, and the exact certificate gives precisely (3). The three standard subfield coset types satisfy Theorem C. Hence the four-type coset system is a flag-transitive geometry. Its RWPRI proof was already given there. Firmness and residual connectedness also follow from the argument in Section 7 below, with base $\mathrm{Sz}(2)$ and the three full prime-order tori 13, 41, 113.

4 Compatible tori in every odd degree

Here is the uniform construction behind the field-compatible cyclic groups, rather than an extrapolation from small computations.

Use the binary matrices M, r displayed in the immutable earlier note, where $M^4 + M + I = 0$, $r = U(1, 0)$, $|r| = 4$, and $r^{-1} M r = M^2$. The polynomial $X^4 + X + 1$ is irreducible. Thus $\mathbb{F}_2[M]$ is the field with sixteen elements.

For reference, the acquired seed matrices are

$$M = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix}.$$

The order-thirteen seed, in $\mathbb{F}_8 = \mathbb{F}_2[z]/(z^3+z+1)$, is

$$a = \begin{pmatrix} z^2+z+1 & 0 & 1 & z^2+1 \\ 1 & z & z^2 & 1 \\ z^2 & z^2 & z+1 & z^2+1 \\ z^2 & 1 & z^2 & z+1 \end{pmatrix}.$$

These are the literal previously checked matrices. The retained checks verify group membership, order, centralization, the exceptional endomorphism identities, and both twenty-element binary centralizers; no search is needed to verify these displayed identities.

The exact order-13 matrix $a \in \mathrm{Sz}(8)$ found in the earlier seed commutes with M. Since the multiplicative order of 8 modulo 13 is four, every nontrivial order-13 matrix in dimension four over $\mathbb{F}_8$ has irreducible minimal polynomial of degree four. In particular a has four distinct eigenlines. The same is true of M: its degree-four binary polynomial remains irreducible over $\mathbb{F}_8$, since $\gcd(4,3) = 1$. Both therefore generate the same degree-four algebra over $\mathbb{F}_8$, and their commuting regular matrices have the same general linear centralizer. Put

$$\mathbf{T} = C_{\mathbf{H}}(M) = C_{\mathbf{H}}(a).$$

As the centralizer in Sp_4 of this regular semisimple element, $\mathbf{T}$ is a connected maximal torus of dimension two. This can also be seen directly in the eigenbasis of a: its symplectic eigenvalue pairs are inverse pairs, and the centralizer consists of the corresponding two independent diagonal coordinates.

Because M is binary, ϕ_2 preserves $\mathbf{T}$. Because $\rho^3(a) = a$, one has $\rho(a) = \rho^4(a) = \phi_2^2(a)$. This is another regular element of $\mathbf{T}$, so ρ also preserves $\mathbf{T}$.

Let B be the integral endomorphism induced by ρ on the rank-two character lattice of $\mathbf{T}$. Binary Frobenius permutes the four eigenlines of M in a four-cycle, since its minimal polynomial has degree four over $\mathbb{F}_2$. The corresponding action on the four symplectic weights $\{\chi_1, \chi_2, -\chi_1, -\chi_2\}$ is a rotation w of order four, with trace zero and determinant one. Therefore

$$B^2 = 2w, \qquad (\operatorname{tr} B)^2 - 2\det B = 0, \qquad (\det B)^2 = 4.$$

Integrality forces $\det B = 2$ and $\operatorname{tr} B = \pm 2$. The remaining sign is fixed by a tiny exact binary check:

$$\mathbf{T}^{\rho} = C_{\mathrm{Sz}(2)}(M) = 1.$$

The retained seed checker verifies this against all twenty matrices of $\mathrm{Sz}(2) = \langle r, w \rangle$. Equivalently, its unique order-five subgroup is generated by $r^2 w$, which does not commute with M.

For a torus, the fixed-point count for an endomorphism whose character matrix is B is $|\det(I - B)|$ when this determinant is odd. This particular count needs only Smith normal form: diagonalizing $I - B$ over the integers turns its kernel on $(\overline{\mathbb{F}}_2^{\times})^2$ into a product of groups of roots of unity of odd orders. Here the two sign choices give one or five fixed points, so the check selects $\operatorname{tr} B = 2$.

The eigenvalues of B are consequently $1+i, 1-i$. The same Smith-normal-form argument gives, for every positive odd d,

$$|\mathbf{T}^{\rho^d}| = \det(I - B^d) = 2^d + 1 - \left((1+i)^d + (1-i)^d\right) = \tau_d.$$

These are the nonsplit Suzuki torus orders. We retain the classical fact that such finite tori are cyclic and their normalizers in the simple Suzuki groups have order $4\tau_d$ and are maximal. Write $T_d = \mathbf{T}^{\rho^d}$. The binary r normalizes $\mathbf{T}$, belongs to every $\mathrm{Sz}(2^d)$, and has order four.

Thus $\langle T_d, r\rangle = T_d : C_4$ is that full normalizer for $d > 1$. For $d = 1$, $T_1 = 1$ and $\langle r\rangle = C_4$ has index five in Sz(2); the simple-group normalizer assertion is not being applied in this exceptional case.

The benefit is simultaneous position. For all positive odd a, b,

$$T_a \cap T_b = T_{\gcd(a,b)}, \qquad \gcd(\tau_a, \tau_b) = \tau_{\gcd(a,b)}. \tag{4}$$

The first equality follows from the fixed-point equations and Euclid's algorithm. For the second, place both cyclic groups in $T_{\mathrm{lcm}(a,b)}$, then use uniqueness of subgroups of each order in a cyclic group.

4.1 Exact matrix convention for the exceptional endomorphism

There is an elementary way to check that the exceptional endomorphism used above matches the earlier matrices. Take the symplectic form with antidiagonal matrix J. On the exterior square of the natural module, quotient the kernel of symplectic contraction by its invariant line $\langle e_1 \wedge e_4 + e_2 \wedge e_3\rangle$. The exterior-product pairing descends to a nondegenerate alternating form on this quotient, with antidiagonal matrix in the basis below. In the quotient basis

$$e_1 \wedge e_2,\ e_1 \wedge e_3,\ e_2 \wedge e_4,\ e_3 \wedge e_4,$$

the induced matrix $\rho(g)$ consists of the 2×2 minors whose row and column pairs are $12, 13, 24, 34$. This is the exceptional isogeny in the given coordinates. For four independent variables, let

$$u(a, A, b, B) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ aA + b & A & 1 & 0 \\ a^2A + ab + B & b & a & 1 \end{pmatrix}.$$

Direct polynomial multiplication gives

$$\rho(u(a, A, b, B)) = u(A, a^2, B, b^2). \tag{7}$$

Consequently ρ^2 is entrywise squaring on this full unipotent group. The diagonal matrix $\mathrm{diag}(x, y, y^{-1}, x^{-1})$ maps to $\mathrm{diag}(xy, x/y, y/x, (xy)^{-1})$, giving the same square; the antidiagonal w is fixed. These groups generate the symplectic group, so the identities recover $\rho^2 = \phi_2$. The exterior-square construction makes ρ a group homomorphism; the polynomial checks are not a test of that property on a sample.

Setting $A = a^\theta, B = b^\theta$, with $\theta^2 = 2$ on the finite field, equation (7) becomes exactly $\rho(U(a, b)) = U(a, b)^\theta$, with entrywise exponent on the right. The earlier diagonal and antidiagonal generators satisfy the same fixed-point relation. This binds the usual exceptional Suzuki construction to those particular matrices. The retained checker proves (7) as an identity of sparse polynomials over $\mathbb{F}_2$, checks the symplectic identity in four independent variables, and checks the diagonal identities as Laurent-polynomial identities. It also directly verifies $\rho(a) = a^{(4)}$, $\rho^3(a) = a$, and all twenty binary centralizer tests for the actual seed.

5 The prime choices exist without a prime-value conjecture

Let $\tau'_d = 2^d + 1 + ((1 + i)^d + (1 - i)^d)$. For odd d,

$$\tau_d \tau'_d = 2^{2d} + 1, \qquad \gcd(\tau_d, \tau'_d) = 1. \tag{5}$$

Indeed the trace square is 2^{d+1}, while the difference between the two odd factors is a power of two.

Fix odd $e \geq 3$ and an odd prime $a \nmid \tau_e$. Since $T_e \leq T_{ea}$, $\tau_e \mid \tau_{ea}$. The quotient is greater than one from (1), for example by bounding τ_{ea} below by $2^{ea} + 1 - 2^{(ea+1)/2}$ and τ_e above by $2^e + 1 + 2^{(e+1)/2}$.

If an odd prime $\ell \mid \tau_e$, the elementary lifting-the-exponent identity applied to $(2^{2e})^a + 1$ gives

$$v_\ell(2^{2ea} + 1) = v_\ell(2^{2e} + 1) + v_\ell(a) = v_\ell(\tau_e).$$

The last equality uses $a \nmid \tau_e$ and (5). Divisibility $\tau_e \mid \tau_{ea}$ forces exactly that same valuation in τ_{ea}. Thus

$$\gcd(\tau_{ea}/\tau_e, \tau_e) = 1.$$

Every prime divisor of the nontrivial quotient is therefore new. For distinct a_i, a_j, equation (4) gives $\gcd(\tau_{ea_i}, \tau_{ea_j}) = \tau_e$, so selected new primes cannot repeat. This proves all the prime-choice assertions of Theorem A. There is no appeal to Zsigmondy's theorem and no exceptional case hidden in an assumed primitive-prime-divisor statement.

6 Exact subgroup intersections and RWPRI

Let $P = \prod_i a_i$, $n = eP$, and choose the cyclic subgroup $A \leq T_n$ of order $m = \tau_e \prod_i \ell_i$. It exists because each factor divides τ_n and the factors are pairwise coprime. Put $K = A : \langle r \rangle$. The action of r preserves every subgroup of the cyclic torus.

For a subset $J \subseteq \{1, \ldots, k\}$, put

$$n_J = e \prod_{i \notin J} a_i, \qquad m_J = \tau_e \prod_{i \notin J} \ell_i.$$

Then

$$\bigcap_{j \in J} S_j = \mathrm{Sz}(2^{n_J}), \qquad K \cap \bigcap_{j \in J} S_j = C_{m_J} : C_4. \tag{6}$$

The first is the standard coefficient-field intersection. For the second, T_e is always retained. The ℓ_i-factor is retained if $i \notin J$. If $i \in J$, it cannot occur: by (4) it would divide $\gcd(\tau_{ea_i}, \tau_{n_J}) = \tau_e$, contrary to its choice. Binary r belongs to every subfield group, so the same formula holds for all four cosets of the cyclic subgroup.

The RWPRI condition now follows for every proper subset of the $k + 1$ types.

- Before K is selected, select an unused S_i whenever one remains. The current and next groups are simple standard Suzuki subfield groups of relative field degree a_i, so the next is maximal. The lower field exponent remains at least $e \geq 3$.
- When all S_i have been selected, selecting K gives $T_e : C_4$, a full nonsplit torus normalizer and maximal subgroup of $\mathrm{Sz}(2^e)$.
- Once K has been selected, selecting any unused S_i removes exactly one factor ℓ_i, giving a subgroup of prime index.

These uses of subfield and torus-normalizer maximality are retained classical Suzuki facts, not consequences of the disputed source list.

For flag transitivity, write

$$V = \prod_i C_{\ell_i}, \qquad B_0 = T_e : C_4.$$

Then $K = V \rtimes B_0$, and (6) gives exactly the coordinate stabilizers in (3). Lemma D and Theorem C apply. This completes the geometry and flag-transitivity parts of Theorem A.

7 Firmness, residual connectedness, and the base-two exception

The common intersection of all parabolics is $B_0 = T_e : C_4$. A panel missing S_i has size ℓ_i. A panel missing K, with $q = 2^e \geq 8$, has size

$$|\operatorname{Sz}(q) : B_0| = q^2(q-1)\tau'_e/4 \;\geq\; 560.$$

The inequality follows from $\tau'_e \geq q + 1 - \sqrt{2q}$ and its value five at $q = 8$; all factors in the lower bound increase for $q \geq 8$. Each ℓ_i is an odd prime, hence at least three. Thus the flag-transitive geometry is thick.

For residual connectedness, flag transitivity reduces all residues to their standard subgroup systems. The incidence graph of a coset system with at least two types is connected when its marked subgroups generate the ambient residue group. This generation holds uniformly:

- If K was already selected, at least two unused field types remove different prime factors. Their two subgroups together contain every cyclic factor and B_0, hence generate the residue.
- If K is unused and at least two field types remain, their two distinct maximal simple subfields generate the current Suzuki group.
- If only one field type and K remain, the simple base subfield is maximal in the current Suzuki group. The cyclic-by-four subgroup has an extra ℓ_i-factor, so is not contained in that subfield, by (6). Together they generate.

These cases include the ambient incidence graph. They prove residual connectedness in every rank.

The earlier rank-four example has $e = 1$, so the last simple-subfield step cannot be used: $\operatorname{Sz}(2)$ is not simple and is not being declared maximal in $\operatorname{Sz}(2^{a_i})$. Instead its chosen $\tau_{a_i} = 13, 41, 113$ are full prime-order nonsplit tori for $a_i = 3, 5, 7$. At that step select the full torus normalizer, which is maximal. The remaining C_4 has index five in $\operatorname{Sz}(2)$, and the other panel indices are 13, 41, 113. The rank-two residue with one field type and K is connected because its full torus normalizer is maximal and the order-twenty base group contains an element of order five outside it. All other arguments above apply unchanged.

Thus the rank-four example is thick, residually connected, flag-transitive, and RWPRI. This is a proof supplement to the immutable earlier record, whose "unproved" statements correctly describe what had been established at that earlier stage.

8 Retained theory and scope

The substantial external inputs are Lang–Steinberg surjectivity, the standard exceptional Suzuki endomorphism/model, cyclicity of the finite nonsplit tori, and the classical maximality theorems for their normalizers and prime-degree simple subfields. The binary seed, character-lattice calculation, finite descent argument, typed Helly theorem, coordinate attachment, prime-choice argument, and all-subset RWPRI/residue calculations are given here.

The result is an existence and flag-transitivity theorem for explicitly aligned families. It does not say arbitrary subgroups with the same isomorphism types have these intersections. It does not classify all Suzuki geometries, prove a geometric intersection-property axiom, or restore the old candidate completeness claims. Classical ingredients are credited as such, and novelty of this particular assembly remains subject to a fuller literature and external mathematical review.

Subfield and torus geometries

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We realize every admissible parameter tuple in Constructions 8 and 9 of the fixed Suzuki source. A valuation calculation gives all intersections, including the cases in which a selected prime divides the base torus order. Descent and cyclic coset coordinates prove flag transitivity; the same intersections give thickness, residual connectedness and residual weak primitivity.

1 Source parameters and the base field

Write the field exponent as n, so the ambient group is $G = \mathrm{Sz}(2^n)$. The source chooses $k = \alpha - 1$ pairwise nonisomorphic maximal simple subfield groups $\mathrm{Sz}(2^{n/a_i})$, where the relative degrees a_i are distinct odd primes. Set

$$e = n/\prod_i a_i, \qquad t = 2^e.$$

The upper bound $\alpha \le \sum_j e_j = \Omega(n)$ in both constructions implies $\Omega(e) = \Omega(n) - k \ge 1$. Since n is odd, $e \ge 3$ **and** $t \ge 8$. Thus neither construction includes a nonsimple base $\mathrm{Sz}(2)$. The notation Ω here counts prime factors with multiplicity. Some a_i may also divide e; the proof allows that.

Construction 8 chooses distinct primes

$$\ell_i \mid \frac{t^{a_i} - 1}{t - 1}$$

and adds the dihedral subgroup of order

$$2(t-1)\prod_i \ell_i.$$

Construction 9 chooses a base nonsplit torus order $\gamma = t + \sqrt{2t} + 1$ or $t - \sqrt{2t} + 1$, and distinct primes ℓ_i such that $\gamma\ell_i$ divides an appropriate nonsplit torus order of $\mathrm{Sz}(t^{a_i})$. It also requires $\gamma \prod_i \ell_i$ to divide one of the two ambient nonsplit torus orders. Its added subgroup has order

$$4\gamma \prod_i \ell_i$$

and is cyclic by C_4.

The source does **not** require $\ell_i \nmid \gamma$. This note covers those cases in which a selected prime divides the base order explicitly. The arbitrary-rank existence theorem in the companion descent paper uses the extra restriction $a_i \nmid \tau_e$ only to prove an unconditional supply of parameters; that restriction is not needed for the present realization theorem for every admissible parameter tuple.

2 Both nonsplit branches have compatible binary models

The companion descent paper constructs a torus $\mathbf{T}$, stable under the exceptional endomorphism ρ, with

$$|\mathbf{T}^{\rho^d}| = \tau_d = 2^d + 1 - \big((1+i)^d + (1-i)^d\big)$$

for every odd d. To obtain the other branch, retain its binary order-four matrix r and antidiagonal w, and set

$$c = r^2 w.$$

The retained exact seed check gives

$$|c| = 5, \qquad \rho(c) = c, \qquad C_{\mathrm{Sz}(2)}(c) = \langle c \rangle.$$

It also verifies that r conjugates c to c^2 or c^3, so normalizes its centralizer. As the multiplicative order of 2 modulo 5 is four, c has four distinct eigenlines cyclically permuted by binary Frobenius. Therefore $\mathbf{T}' = C_{\mathrm{Sp}_4}(c)$ is another maximal torus with the same four-cycle Frobenius calculation as in the companion descent paper. It is ρ-stable because $\rho(c) = c$.

The character-lattice endomorphism has determinant two and trace ± 2; its five ρ-fixed points select trace **minus two**. For odd d, this gives

$$|(\mathbf{T}')^{\rho^d}| = \tau'_d = 2^d + 1 + \big((1+i)^d + (1-i)^d\big).$$

Both finite families are cyclic. Each satisfies

$$T_a \cap T_b = T_{\gcd(a,b)}, \qquad \gcd(|T_a|, |T_b|) = |T_{\gcd(a,b)}|. \tag{1}$$

The binary r supplies the order-four complement in every odd extension.

For any fixed odd base exponent e, these two branches give precisely the two possible base torus orders γ. Choose the branch whose order at e is γ, and write its order at d as h_d. Since $T_e \le T_{ea_i}$, $\gamma \mid h_{ea_i}$. The other order at ea_i is coprime to h_{ea_i}. Because $\gamma > 1$, it cannot also be divisible by γ. Consequently each source condition $\gamma\ell_i$ dividing a nonsplit torus order automatically selects this same compatible branch. The same argument identifies the prescribed ambient branch. Thus all the source's choices can be made simultaneously inside one ρ-stable torus.

For Construction 8 use the standard split tori

$$T_d = \{h(\lambda) : \lambda \in \mathbb{F}_{2^d}^{\times}\}, \qquad h_d = 2^d - 1.$$

They satisfy the same intersection and gcd identities (1). Their common complement is the binary w, of order two, acting by inversion. In the companion descent paper's standard matrix convention, the diagonal subgroups are compatible under every odd subfield embedding: when $d \mid n$ and n/d is odd, $(n+1)/2 \equiv (d+1)/2 \pmod{d}$, so the ambient exponent θ_n restricts to θ_d on $\mathbb{F}_{2^d}$.

3 One intersection calculation covers both constructions

Write $c_0 = 2$ for the split case and $c_0 = 4$ for either nonsplit case. Set $\gamma = t - 1$ in Construction 8; in Construction 9 retain the chosen nonsplit base order. Thus $\gamma = h_e$ in every case. In the chosen ambient cyclic torus T_n, let

$$A = C_m, \qquad m = \gamma \prod_i \ell_i, \qquad K = A : C_{c_0}.$$

For Construction 9, $m \mid h_n$ is a source assumption. For Construction 8 it follows from the individual divisibilities: each $\gamma\ell_i$ divides $2^n - 1$, and, because the primes are distinct, their least common multiple is m. The common binary complement normalizes A.

Take the standard subfield parabolics $S_i = \mathrm{Sz}(2^{n/a_i})$. For $J \subseteq \{1, \ldots, k\}$, put

$$n_J = e \prod_{i \notin J} a_i, \qquad m_J = \gamma \prod_{i \notin J} \ell_i.$$

Then

$$\bigcap_{j \in J} S_j = \mathrm{Sz}(2^{n_J}), \qquad K \cap \bigcap_{j \in J} S_j = C_{m_J} : C_{c_0}. \tag{2}$$

Here is the valuation argument that handles $\ell_i \mid \gamma$. For the prime ℓ_i, the exponent in m is $v_{\ell_i}(\gamma) + 1$. If $i \notin J$, $ea_i \mid n_J$, so $\gamma\ell_i \mid h_{n_J}$; the extra copy is retained. If $i \in J$, then

$$\gcd(ea_i, n_J) = e, \qquad \gcd(h_{ea_i}, h_{n_J}) = \gamma.$$

Since $\gamma\ell_i \mid h_{ea_i}$, this forces $v_{\ell_i}(h_{n_J}) = v_{\ell_i}(\gamma)$: exactly the extra copy is removed. Primes not selected among the ℓ_i occur in m only through γ, which is always retained. Thus $\gcd(m, h_{n_J}) = m_J$. The binary complement belongs to every standard subfield group, proving the full cyclic-by-complement equality in (2).

This proof allows $a_i \mid e$: factoring e from both exponents still gives $\gcd(ea_i, n_J) = e$, since the remaining relative primes are distinct from a_i.

4 Cyclic coset coordinates

Let $H_i = K \cap S_i = C_{m/\ell_i} : C_{c_0}$ and $A_i = A \cap H_i = C_{m/\ell_i}$. Every left coset of H_i in K has a representative in A, because the complement is contained in H_i, and

$$A/A_i \longrightarrow K/H_i, \qquad aA_i \longmapsto aH_i$$

is a bijection of sets.

The cyclic group A maps onto $\prod_i A/A_i$: these are cyclic quotients of the pairwise distinct prime orders ℓ_i, so this is the ordinary Chinese remainder theorem. Its kernel has order γ. This remains true when some selected primes divide γ. Hence the set map

$$K \longrightarrow \prod_i K/H_i$$

is surjective. No H_i is asserted to be normal in K. Only the subgroups A_i of the abelian A are used as group-quotient kernels.

The standard subfield types satisfy Theorem C of the companion descent paper. Its Lemma D now gives a common representative in every flag of the enlarged coset system. The system is a geometry, and G is flag-transitive.

5 RWPRI, thickness, and residual connectedness

The all-subset proof in the companion descent paper applies with (2). Before selecting K, each unused field type makes a maximal simple-subfield step; their terminal group is $\mathrm{Sz}(2^e)$, with $e \geq 3$. Selecting K there gives its full split or nonsplit torus normalizer, a maximal subgroup. After selecting K, an unused field type removes precisely one factor ℓ_i, giving prime subgroup index. These are the source's RWPRI conditions.

Panel sizes are the odd primes ℓ_i, and the index of the full base torus normalizer in the simple group $\mathrm{Sz}(2^e)$. The latter is greater than two (an index-two subgroup of a simple group would be normal). Thus the geometry is thick.

For residual connectedness, standard residues are enough because flag transitivity is proved. With K already selected, two unused field types give cyclic parts of orders m_J/ℓ_i and m_J/ℓ_j. Their least common multiple is m_J, so they generate the current cyclic-by-complement subgroup. Without K, two remaining distinct maximal simple subfields generate the current Suzuki group. If only one field type and K remain, the simple base subfield is maximal, and (2) shows that the other marked subgroup is not contained in it. They therefore generate as well. This proves connectedness of every residue of rank at least two, including the whole geometry.

We have thus rederived the complete parameterized **existence assertions** of Constructions 8 and 9, and proved flag transitivity and residual connectedness for these canonical realizations. Their inclusion in an exhaustive classification, the classification of all embeddings with those isomorphism types, and geometric intersection-property axioms are separate questions.

Pure subfield geometries and a scalar obstruction

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For standard simple subfield parabolics, the exact criterion for residual weak primitivity is that their common subfield remain simple. Descent proves that the resulting geometries are thick, residually connected and flag-transitive. In the mixed parabolic of Construction 10, maximality is instead equivalent to a scalar codimension of one; this disproves the source claim for every prescribed parameter.

1 Exact source scope and retained inputs

Write the odd field exponent as

$$n = \prod_{j=1}^{m} p_j^{f_j}, \qquad p_i \neq p_j \ (i \neq j).$$

Thus m counts distinct prime factors, whereas $\Omega(n) = \sum_j f_j$ counts factors with multiplicity. Construction 10, printed p.18, specifies rank $3 \leq \alpha \leq \min(m+1, \Omega(n))$, selects $k = \alpha - 1$ pairwise nonisomorphic maximal Suzuki subfield groups, puts $a_i = n/\log_2(s_i)$, and sets $t = 2^{n/\prod_i a_i}$. Its extra parabolic has type $(E_q \hat{\cdot} E_t) : C_{t-1}$. Construction 11 instead specifies $2 \leq \alpha \leq m-1$ and selects α pairwise nonisomorphic maximal Suzuki subfield groups. Theorem 5 immediately below claims that Constructions 1–11 give all RWPRI pre-geometries of rank at least four. The rank bound and the distinct-prime convention were checked in the page image, not only its text extraction.

The source's definition on pp.4–5 is the exact condition used here: for every proper subset J of the type set, at least one remaining intersection $G_{J\cup\{i\}}$ must be maximal in G_J. In particular, the condition applies to rank-one residues. The hat-dot on p.5 means a nonsplit extension; E_q has order q. The introduction to the constructions on p.15 instructs the reader to retain their whole intended intersection lattice, inferred from the rank-four discussion. Construction 10 comes from the R12 discussion on pp.14–15. Its intersections with the selected subfield groups have type $(E_{s_i} \hat{\cdot} E_t) : C_{t-1}$.

We retain the following classical group theory.

1. The Suzuki realization inside Sp_4, with exceptional Steinberg endomorphism ρ, $\rho^2 = \phi_2$, and compatible standard groups $S_d = \mathrm{Sz}(2^d) = \mathbf{H}^{\rho^d}$ for positive odd d. Here $S_1 = \mathrm{Sz}(2) \cong 5:4$ is allowed as an intersection, although it is not simple.
2. Prime-relative-degree inclusions $S_d < S_{ad}$, for odd prime a and $d > 1$, are maximal. Standard subfield intersections are given by the gcd of their exponents.
3. The Suzuki Sylow-2 group U_q has exponent four and its identity together with its involutions is $Z_q = Z(U_q)$, elementary abelian of order q. A nonidentity involution $z \in Z_q$ has $C_G(z) = U_q$. Consequently a normalizer of Z_q normalizes U_q, while a normalizer of U_q normalizes its center: $N_G(Z_q) = N_G(U_q) = B_q$, the ovoid point stabilizer $U_q : H_q$, where $H_q \cong C_{q-1}$.
4. Nonsplit torus normalizers have order $4(q \pm \sqrt{2q} + 1)$ and are maximal for $q \geq 8$. Odd-element centralizers and these normalizers have their usual Suzuki structure.

5. Schur–Zassenhaus complement conjugacy for the solvable group $U_q : H_t$. This puts its subgroups of order $t-1$ into the standard H_t.
6. Lang–Steinberg surjectivity on the connected algebraic group Sp_4, as used in the proved finite-descent and coset-Helly statements of Field descent and compatible tori, Section 2.

The matrix action used in Section 4 is verified directly below. The retained facts do not include any source classification of all RWPRI lattices.

2 Sharp criterion for the canonical pure-subfield family

Theorem A. Let n be positive and odd. Select $k \geq 2$ distinct prime divisors $a_1, \ldots, a_k$ of n, put

$$e = \frac{n}{\prod_{i=1}^{k} a_i}, \qquad G = S_n, \qquad G_i = S_{n/a_i}.$$

The standard coset incidence system $\Gamma(G; G_1, \ldots, G_k)$ is a thick, residually connected, flag-transitive geometry. It is RWPRI **if and only if** $e > 1$. When $e > 1$, every one-step inclusion in its intersection lattice, not only one inclusion at each subset, is maximal.

Proof. For $J \subseteq \{1, \ldots, k\}$, set

$$n_J = \frac{n}{\prod_{j \in J} a_j}.$$

Entrywise field intersection gives $G_J = S_{n_J}$, including $G_\varnothing = G$. Apply descent Theorem C with $F = \rho^e$ and $m = \prod_i a_i$. Any pairwise incident family of cosets has a common representative. It therefore extends to a translate of the standard chamber. This proves the geometry axiom and flag transitivity.

Suppose that a residue has at least two remaining types. For distinct remaining i, j, the two residue parabolics S_{n_J/a_i} and S_{n_J/a_j} are distinct maximal subgroups of S_{n_J}: their exponents exceed one, even if $e = 1$, because at least one other relative prime remains. They generate S_{n_J}. The standard coset-graph connectivity criterion therefore proves that residue connected. Flag transitivity identifies every residue with such a standard one, proving RC.

The panel of type i has $[S_{ea_i} : S_e]$ chambers. This is at least three. Indeed, S_{ea_i} is nonabelian simple and S_e is a proper nontrivial subgroup, so its index cannot be two. This includes $e = 1$, where $|S_e| = 20$. Thus the geometry is thick.

If $e > 1$, each remaining exponent n_J/a_i is at least $e > 1$. Every one-step inclusion is a prime-degree inclusion of simple subfield groups, and hence maximal. This proves RWPRI.

If $e = 1$, the distinct primes a_i, with $k \geq 2$, include at least one $a_i \geq 5$. At the subset containing every type except i, the only remaining inclusion is

$$S_1 < S_{a_i}.$$

Let C_5 be the normal subgroup of $S_1 = 5 : 4$. Inside S_{a_i}, its centralizer is a nonsplit torus. The order-four complement of S_1 normalizes this centralizer, so S_1 lies inside the full torus normalizer. Its torus order is one of $2^{a_i} \pm 2^{(a_i+1)/2} + 1$, both of which exceed five for $a_i \geq 5$. Hence this containment is strict, while the torus normalizer is proper in S_{a_i}. Consequently S_1 is not maximal, and RWPRI fails at that rank-one residue. □

The theorem concerns the standard simultaneous embeddings. It does not assert that arbitrary conjugates of the same maximal-subgroup types have the same intersections or are flag transitive. If another family has a common prescribed simple subgroup S_e with $e > 1$, however, the retained split-dihedral normalizer/transporter argument in Earlier subfield and

scalar proofs, Section B, forces simultaneous standard position after conjugating S_e. Thus within that prescribed common-subfield lattice there is only one ambient conjugacy class for a fixed labelled selection of exponents.

3 Construction 11 and its omitted boundary

Construction 11 has $k \le m-1$, so an unselected prime remains in e. In particular $e > 1$, and Theorem A supplies canonical realizations for **every literal admissible parameter**. It also answers the source's separate flag-transitivity question for this family. It does not establish flag transitivity of all arbitrary relative positions sharing only the maximal-parabolic isomorphism types.

The exact corrected parameter condition for the canonical pure-subfield family is

$$2 \le k \le m, \qquad \frac{n}{\prod_{i=1}^{k} a_i} > 1. \tag{1}$$

For squarefree n, this recovers $k \le m-1$. If any prime factor of n is repeated, it also permits $k = m$. The source's bound omits this latter boundary.

For example take

$$n = 3^2 \cdot 5 \cdot 7 \cdot 11 = 3465, \qquad (a_1, a_2, a_3, a_4) = (3, 5, 7, 11).$$

Then

$$(G_1, G_2, G_3, G_4) = (S_{1155}, S_{693}, S_{495}, S_{315}), \qquad G_{1234} = S_3 = \mathrm{Sz}(8).$$

Theorem A proves this is a rank-four thick, RC, FT, RWPRI geometry, without enumerating its enormous ambient group. Here $m = 4$, whereas Construction 11 allows only ranks two and three. Constructions 1–10 each have at least one soluble maximal parabolic, while all four parabolics here are nonabelian simple. Consequently this stated-action subgroup family is absent from all eleven literal constructions.

This contradicts the **literal exhaustive list** in Theorem 5, as a theorem about the subgroup systems specified in the paper. It is a boundary omission, distinct from proving that no natural extension of the constructions could produce the example: allowing appropriate truncations of larger constructions or replacing Construction 11's bound by (1) would include it. The statement does not classify all unlabelled geometries or all possible actions of G.

4 A scalar-center normal form

Here n and e are odd, $e \mid n$, $e > 1$, $q = 2^n$, and $t = 2^e$. Use standard Suzuki unipotents $U(a, b)$, with $Z_q = \{U(0, b) : b \in \mathbb{F}_q\}$, and diagonal matrices

$$h(\lambda) = \mathrm{diag}(\lambda^{\theta+1}, \lambda, \lambda^{-1}, \lambda^{-\theta-1}), \qquad \theta = 2^{(n+1)/2}.$$

Multiplying these diagonal and unipotent matrices gives

$$h(\lambda)^{-1} U(a, b) h(\lambda) = U(\lambda^{\theta} a, \lambda^{\theta+2} b). \tag{2}$$

Both exponents are coprime to $q-1$. For $\theta+2$, if an odd divisor c divided both $q-1$ and $2^{(n-1)/2}+1$, squaring the latter congruence would give $2^{n-1} \equiv 1 \pmod{c}$, whereas $2^n \equiv 1 \pmod{c}$; thus $2 \equiv 1 \pmod{c}$, forcing $c = 1$. Since $\theta + 2 = 2(2^{(n-1)/2} + 1)$, the claim follows. In particular the standard H_t acts on both the first unipotent parameter and the center by all nonzero scalars in $\mathbb{F}_t$. Any additive invariant subgroup is therefore an $\mathbb{F}_t$-subspace.

Lemma B: normal form. Every subgroup $Q \leq S_n$ of type $(E_q \widehat{\cdot} E_t) : C_{t-1}$ is ambient-conjugate to

$$Q_0 = Z_q U_t H_t. \tag{3}$$

Proof. Its normal Sylow-2 subgroup P has order qt and contains the elementary abelian kernel of order q. Place P inside an ambient U_q. All its involutions lie in Z_q, so that elementary abelian kernel must equal Z_q. Moreover Z_q is exactly the identity together with the involutions of P; it is characteristic in P and normal in Q. Therefore $Q \leq N_G(Z_q) = B_q$.

Project to $B_q/U_q = H_q$. The odd complement of Q maps isomorphically to the unique subgroup of order $t-1$, namely H_t. Complement conjugacy in $U_q : H_t$ allows an ambient U_q-conjugation putting it equal to H_t. Now P/Z_q is an additive subgroup W of the first unipotent parameter, of order t, normalized by H_t. By (2), $W = v\mathbb{F}_t$ for some nonzero v. A diagonal conjugation, which centralizes H_t, scales this line to $\mathbb{F}_t$. Because P contains the full center, it is the full preimage of that line in U_q; thus $P = Z_q U_t$. This proves (3). □

Lemma C: exact maximality test. Suppose $H < Q_0$ has odd part $t-1$ and a normal Sylow-2 subgroup which is not elementary abelian. Then H surjects onto Q_0/Z_q. Put $W = H \cap Z_q$, regarded as an $\mathbb{F}_t$-subspace of $\mathbb{F}_q$. The overgroups of H in Q_0 correspond exactly to the $\mathbb{F}_t$-subspaces containing W, by

$$V \longmapsto Z_V H, \qquad L \longmapsto L \cap Z_q. \tag{4}$$

In particular, H is maximal in Q_0 exactly when $\dim_{\mathbb{F}_t}(\mathbb{F}_q/W) = 1$.

Proof. A non-elementary Sylow-2 subgroup inside U_q contains an element of order four. Its image modulo Z_q is nonzero, so the image of H in Q_0/Z_q has a nonzero normal additive subgroup. The odd projection has full order $t-1$ and acts irreducibly on the one-dimensional $\mathbb{F}_t$-space U_t/Z_t. It follows that the image contains all this additive group, and therefore equals Q_0/Z_q.

Thus $Q_0 = Z_q H$. The action of H on Z_q is exactly the scalar action of H_t, so W and every center intersection $L \cap Z_q$ are $\mathbb{F}_t$-subspaces. Conversely such a subspace V is normalized by H, making $Z_V H$ a subgroup. Writing every element as zh proves $L = (L \cap Z_q)H$. This gives (4), with the stated maximality criterion. □

This is an overgroup classification inside the specified mixed parabolic. It reduces maximality to one scalar-module dimension; no ambient Suzuki group enumeration or search over relative positions is required.

5 Uniform obstruction for Construction 10

Theorem D. For every parameter allowed by Construction 10, the intended bare subgroup-intersection lattice has a canonical embedding. No embedding having its prescribed mixed parabolic and pairwise mixed intersections is RWPRI.

Proof. Put $k = \alpha - 1 \geq 2$, select distinct odd relative primes a_i, and set

$$P = \prod_i a_i, \quad n = eP, \quad t = 2^e, \quad q = 2^n, \quad s_i = 2^{n/a_i}.$$

The source's bound $\alpha \leq \Omega(n)$ gives $\Omega(e) = \Omega(n) - k \geq 1$, hence $e > 1$. Choose canonical subfield groups $G_i = S_{n/a_i}$ and $Q = Z_q U_t H_t$. For every selection J of subfield types,

$$\bigcap_{j \in J} G_j = S_{n_J}, \qquad Q \cap \bigcap_{j \in J} G_j = Z_{2^{n_J}} U_t H_t. \tag{5}$$

These are exactly the intended field intersections and mixed-extension orders, including common Borel $U_t H_t$. The extension $Z_{2^{n_J}} U_t$ by its elementary abelian center kernel is nonsplit: every element outside that center has order four, so a nontrivial elementary abelian complement could not exist. Thus (5) also has the literal hat-dot type. This proves bare existence.

For any other proposed embedding, Lemma B puts its mixed parabolic into Q_0. Its pairwise intersections H_i have the prescribed type $(E_{s_i} \hat{\cdot} E_t) : C_{t-1}$. Their Sylow-2 groups cannot be elementary abelian: an extension of elementary abelian groups whose total group is elementary abelian splits as a vector-space extension. Lemma C therefore applies, and order division gives

$$|H_i \cap Z_q| = s_i, \qquad \dim_{\mathbb{F}_t}(\mathbb{F}_q/(H_i \cap Z_q)) = P - P/a_i.$$

Since $k \geq 2$ and every $a_i \geq 3$, this dimension exceeds one. Every H_i is therefore nonmaximal in Q_0. Explicitly, extend $H_i \cap Z_q$ by one $\mathbb{F}_t$-dimension; (4) gives a proper intermediate subgroup. Taking the source's RWPRI subset J consisting only of the mixed type leaves no maximal remaining intersection. This contradicts RWPRI. $\square$

The negative conclusion uses only the mixed parabolic and the specified pairwise intersection types. It does not require simultaneous alignment of a common simple subfield. It strengthens the earlier rank-three R12 argument in Earlier subfield and scalar proofs, Section C. It does not prove that arbitrary choices of the listed maximal parabolic types, with different intersections, cannot satisfy RWPRI. Nor does failure of RWPRI imply failure of the geometry or FT axioms.

Diagrams of the surviving prescribed families

Human Compression project

Abstract

The diagram remarks in the fixed higher-rank Suzuki source concern flag-transitive realizations of its prescribed intersection lattices. A subgroup-product test verifies those remarks for the surviving families. The test does not assume the source's exhaustive list and makes no additional assertion of flag transitivity.

1 A product test

Lemma 1. *Let a flag-transitive coset geometry have standard rank-two residue $\Gamma(H; A, B)$. This residue is a generalized digon if and only if $H = AB$, equivalently $|H|\,|A \cap B| = |A|\,|B|$.*

Proof. All objects of the two types are incident precisely when every left A-coset meets every left B-coset. By translation this is equivalent to A meeting hB for every $h \in H$, hence to $h \in AB$ for every h. The product has cardinality $|A||B|/|A \cap B|$. The geometry diagram omits exactly the pairs with a digon residue. □

Write $e \geq 3$ odd, $t = 2^e$, and let $a_1, \ldots, a_k$ be distinct odd primes. Put $n = e \prod_i a_i$ and $S_d = \mathrm{Sz}(2^d)$. In all families below the field parabolics are $P_i = S_{n/a_i}$ in the acquired simultaneous position. The intersection formulas proved in the companion manuscripts apply to every subset of the types.

2 The families with a torus parabolic

Proposition 2. *For the canonical realizations of Constructions 8 and 9, the diagram is a star with centre the torus-normalizer type K and leaves the k field types.*

Proof. Use the notation γ, ℓ_i and $c \in \{2, 4\}$ of the torus family theorem. Thus $K = C_{\gamma \prod_i \ell_i} : C_c$, and $K \cap S_J = C_{\gamma \prod_{i \notin J} \ell_i} : C_c$. Flag transitivity has already been proved by descent and cyclic coset coordinates, so the standard subgroup residues are the actual residues.

For two field types i, j, all other types, including K, have been selected. The current group has order $c\gamma\ell_i\ell_j$; its two parabolics have orders $c\gamma\ell_i, c\gamma\ell_j$, and their intersection has order $c\gamma$. Lemma 1 makes this residue a digon.

For the pair i, K, the current group is $\mathrm{Sz}(t^{a_i})$; the parabolics are $\mathrm{Sz}(t)$ and $C_{\gamma\ell_i} : C_c$, with intersection of order $c\gamma$. Their product has size $|\,\mathrm{Sz}(t)|\ell_i$. Its two-part is t^2, whereas the two-part of $|\,\mathrm{Sz}(t^{a_i})|$ is t^{2a_i}. They cannot be equal. Every field type is therefore adjacent to K, and no two field types are adjacent to one another. □

3 The families with two soluble parabolics

The corrected Constructions 5 and 7 use

$$M = (t-1)\prod_i \ell_i, \quad P_D = C_M : C_2, \quad P_E = Z_{2^n} : C_M,$$

where the split-torus scalar subgroup has order M. These subgroups and every intersection have been constructed in the companion scalar field theorem. That theorem proves the source's subgroup maximality condition, but does not prove flag transitivity.

Proposition 3. *If a realization with these prescribed intersections is a flag-transitive geometry, its diagram is a star with centre D; its leaves are E and all the field types.*

Proof. Put $\gamma = t - 1$. For two field types i, j, after selecting both D, E and all other fields, the current group is cyclic of order $\gamma\ell_i\ell_j$. Its two parabolics have orders $\gamma\ell_i, \gamma\ell_j$, with meet of order γ. Their product is the entire group.

For types i, E, the current group has order $2\gamma\ell_i$; the two parabolics have orders 2γ and $\gamma\ell_i$, and their meet has order γ. Again the residue is a digon. For types i, D, the current group has order $t^{a_i}\gamma\ell_i$; its parabolics have orders $t\gamma$ and $\gamma\ell_i$, with meet γ. Their product has order $t\gamma\ell_i$, strictly smaller. Finally, for D, E the current group is $\mathrm{Sz}(t)$, and the product of its parabolics has order $2t\gamma$, smaller than $t^2(t^2+1)\gamma$. These are exactly the asserted edges. □

4 Pure subfield geometries

Proposition 4. *The canonical pure-subfield geometry has complete diagram on its k field types. In particular its diagram is not a star when $k > 3$.*

Proof. For a pair of remaining relative primes a, b, the current group is $\mathrm{Sz}(t^{ab})$, its parabolics are $\mathrm{Sz}(t^a), \mathrm{Sz}(t^b)$, and their intersection is $\mathrm{Sz}(t)$. Here the bottom $t = 2^e$ may equal two; the group-order formula remains valid then. Write $f(x) = x^2(x^2+1)(x-1)$. For $x \geq 2$,

$$\tfrac{1}{2}x^5 < f(x) < x^5.$$

Consequently

$$\frac{f(t^{ab})f(t)}{f(t^a)f(t^b)} > \frac{t^{5(ab+1-a-b)}}{4} > 1,$$

since $a, b \geq 3$ and $t \geq 2$. The product criterion fails for every pair, as claimed. Flag transitivity and the residue identification were proved in the pure-subfield theorem. □

Source scope and attribution. D. Leemans, *The residually weakly primitive pregeometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), pp. 18–19, observes that the listed higher-rank families other than Construction 11 have star diagrams, conditional on being flag-transitive geometries. The preceding proofs establish that statement for the surviving prescribed families. The claimed pentagon-and-circles diagram of Construction 1 is conditional on an intersection lattice now proved impossible, and therefore supplies no existing geometry. Constructions 2–4 and 6 have the same obstruction to their prescribed lattices; Construction 10 fails the source maximality axiom. None of these conditional diagram calculations restores the false exhaustive list. The additional statement that almost all Suzuki geometries have this form has no specified counting measure and is not a classification conclusion established here. The source's future request to study flag transitivity is answered for the canonical families 8, 9 and 11; flag transitivity of arbitrary embeddings and of the corrected families 5 and 7 is not asserted by this manuscript.

Scalar subspaces and flag transitivity

Human Compression project

Abstract

The previously acquired scalar-subspace family satisfies the literal algebraic local two-transitivity condition in arbitrary rank. Combining its intersection formula with the proved descent theorem shows that these pregeometries are thick, residually connected, flag-transitive geometries. The additional argument is a surjectivity calculation on independent scalar coordinates.

Let $e \geq 3$ be odd, put $t = 2^e$ and $K = \mathbb{F}_t$, and choose $k \geq 2$ distinct odd primes $a_1, \ldots, a_k$. Set

$$A = \prod_i a_i, \quad F = \mathbb{F}_{t^A}, \quad F_i = \mathbb{F}_{t^{A/a_i}}, \quad G = \mathrm{Sz}(t^A), \quad S_i = \mathrm{Sz}(F_i).$$

Choose $x_i \in \mathbb{F}_{t^{a_i}} \setminus K$ and put

$$T = \mathrm{span}_K(1, x_1, \ldots, x_k), \qquad Q = Z_T B_t,$$

where $B_t = U_t H_t$ is the standard point stabilizer of $\mathrm{Sz}(t)$ and Z_T is the central unipotent subgroup with parameter in T. The earlier scalar-subspace proof establishes that $1, x_1, \ldots, x_k$ are independent over K and, for $J \subseteq \{1, \ldots, k\}$,

$$T \cap F_J = T_J = \mathrm{span}_K(1, x_i : i \notin J), \qquad Q \cap \bigcap_{j \in J} S_j = Z_{T_J} B_t, \tag{1}$$

where $F_J = \mathbb{F}_{t^{A/\prod_{j \in J} a_j}}$. For completeness, apply $\sigma_i : y \mapsto y^{t^{A/a_i}}$ to a linear combination lying in F_i. This automorphism fixes every x_j with $j \neq i$, fixes K, and moves x_i. Subtracting the two expressions forces its x_i coefficient to vanish. The same argument with a zero linear combination proves independence, and intersection over $i \in J$ gives the displayed formula. The coordinate multiplication law makes $Z_T B_t$ a subgroup: U_t centralizes Z_T, and H_t acts by all nonzero K-scalars.

Theorem 1. *The coset system $\Gamma(G; S_1, \ldots, S_k, Q)$ is a thick, residually connected, flag-transitive geometry of rank $k + 1$. It satisfies the source's residual weak primitivity condition and local two-transitivity, now on its actual rank-one residues.*

Proof. Put $T_i = \mathrm{span}_K(1, x_j : j \neq i)$ and $H_i = Q \cap S_i = Z_{T_i} B_t$. Every left H_i-coset in Q has a representative in Z_T, and

$$T/T_i \longrightarrow Q/H_i, \qquad b + T_i \longmapsto z(b) H_i \tag{2}$$

is a bijection of sets. Indeed $Q = Z_T H_i$ and $Z_T \cap H_i = Z_{T_i}$, by (1). No quotient-group structure on Q/H_i is needed. Given arbitrary coordinates $c_i \in K$, the single vector $\sum_i c_i x_i \in T$ represents them in every T/T_i. Thus

$$Q \longrightarrow \prod_i Q/H_i$$

is surjective.

The standard subfield cosets have the common-representative property by the acquired descent theorem, with Steinberg endomorphism ρ^e and independent relative degrees a_i. Its attachment lemma and (2) therefore give that property for all the types including Q. Every flag extends to a translate of the standard chamber; hence this is a flag-transitive geometry. In particular all its actual residues are isomorphic to the standard subgroup-intersection residues.

The earlier proof of residual weak primitivity can now be read in these actual residues. Before selecting Q, an unused field type makes a maximal prime-degree inclusion of simple Suzuki groups, since the bottom field exponent is at least $e > 1$. After every field type has been selected, the remaining inclusion is $B_t < \mathrm{Sz}(t)$, which is maximal. Once Q is selected, an unused field type replaces T_J by a hyperplane containing K. The overgroups of $Z_{T_J \cap F_i} B_t$ inside $Z_{T_J} B_t$ correspond to K-subspaces between those two parameter spaces: the proof is the elementary equality $L = (L \cap Z_{T_J}) B_t$, together with the scalar action of H_t. The codimension is one, so every such inclusion is maximal.

The full intersection is B_t. The panel missing Q has $[\mathrm{Sz}(t) : B_t] = t^2 + 1$ chambers and the usual doubly transitive Suzuki action. The panel missing S_i has $[Z_{K+Kx_i} B_t : B_t] = t$ chambers. Its quotient action on $(K + Kx_i)/K$ contains all translations and all nonzero K-scalars; it is the doubly transitive affine action on K. Since $t \geq 8$, every panel has at least three chambers. This proves thickness and actual local two-transitivity.

Finally consider a standard residue of rank at least two. If Q has been selected, at least two field types remain. Their two parameter hyperplanes sum to T_J, so their subgroups generate $Z_{T_J} B_t$. If Q is unselected and at least two field types remain, two distinct maximal simple subfield groups generate the current Suzuki group. If precisely one field type and Q remain, the base $\mathrm{Sz}(t)$ is maximal in $\mathrm{Sz}(t^{a_i})$, and $Z_{K+Kx_i} B_t$ is not contained in it because $x_i \notin K$. Again they generate the residue group. The coset incidence graph is connected exactly when its parabolics generate; flag transitivity transports this conclusion to every residue. Thus the geometry is residually connected. □

Attribution and source scope. The scalar family and its intersection proof were already independently accepted in the project; the Frobenius-isolation step there was suggested by the root reviewer. The descent and coset-attachment lemmas are earlier project proofs retaining classical Lang–Steinberg. The new step here is the explicit coordinate bijection (2) and its geometric consequences. It uses no new finite acquisition and does not change any frozen proof.

The fixed higher-rank source, D. Leemans, Note di Matematica 20(1) (2000/2001), p. 2, quotes the earlier assertion that no rank at least four geometry satisfies both residual weak primitivity and local two-transitivity. Taking $k \geq 3$ disproves this statement even with firmness, residual connectedness and flag transitivity imposed. For $k = 2$, the rank is three while the ambient field degree is composite, and every panel is thick. These are further consequences of the same already acquired family, not new original or principal-refutation counts. No full Lean formalization or complete classification is claimed.

Preliminary statements and their precise scope

Human Compression project

Abstract

We record the complete preliminary dependency boundary for the fixed 1999 rank-three Suzuki article. Several literal arithmetic assertions need correction: integer divisibility is insufficient for subfield divisibility, the residue class controlling the factor divisible by five is misstated, and a prime extension must remain in the same torus. We give small exact witnesses and the corrected general statements. The standard group structure and the explicitly cited linear-diagram theorem remain attributed inputs.

1 Definitions and quoted structural facts

Throughout, the source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63, in the fixed twenty-one-page edition. Write

$$q = 2^{2e+1}, \qquad \theta = 2^{e+1}, \qquad \alpha_q = q + \theta + 1, \quad \beta_q = q - \theta + 1.$$

A simple Suzuki subfield group has size parameter at least eight. The source uses $a \mid_M b$ to mean that b/a is prime, and $a \mid_P b$ to mean that a is a proper divisor of b (printed p. 47). These symbols must not be replaced by ordinary divisibility in a source join.

Source Lemma 2.1 states that a rank-three coset pregeometry is a geometry. Indeed an incident pair of cosets has a common representative g, and adding gG_k of the missing type gives a chamber. A singleton similarly extends using any representative; the empty flag extends to the base chamber. Thus every flag extends to a chamber. No flag-transitivity assertion follows from this argument.

The classical Suzuki maximal-subgroup list in source Table 1 is an explicitly attributed premise from Lüneburg and Suzuki. Source Lemmas 3.1 and 3.2 are explicitly quoted from the earlier rank-two paper. Their required content is independently derived in the accepted L115 preliminary note: the maximal subgroups of $B_q, D_q, N_+(q), N_-(q)$, and pairwise coprimality of $q-1, \alpha_q, \beta_q$. The resulting bounds on pairwise intersection orders, respectively $2, 2, 4$, follow from divisibility of subgroup orders. No additional finite census is used here.

Source Lemma 3.8 is expressly stated to follow from the main theorem of M. Aschbacher, *Flag structures on Tits geometries*, Geometriae Dedicata 14 (1983), 21–31. We preserve this as a cited premise in its source form: a rank-three geometry with a linear diagram is flag transitive. This note does not reconstruct the cited theorem. In particular, no arbitrary intersection pattern is declared to have such a diagram without the relevant residue and diagram hypotheses.

2 Field degrees and the arithmetic statements

The classical list gives one conjugacy class of maximal simple subfield groups for each prime $p \mid n$, where $q = 2^n$, provided $n/p \geq 3$. If n is prime there are none. If n is composite all these quotients are at least three. Consequently there is exactly one class if and only if $n = p^k$, $k \geq 2$. This proves source Lemma 3.3.

The repunit argument in the accompanying normalizer-restriction note proves the implication in Lemma 3.4: if $\mathrm{Sz}(s)$ is maximal in $\mathrm{Sz}(q)$ and $(q-1)/(s-1)$ is prime, then $q = s^p$, $s = 2^{p^a}$ for a prime p and a positive integer a. Therefore, if there are at least two maximal subfield classes, $s-1$ is *not a maximal proper divisor* of $q-1$. This is the consequence needed in the source.

Lemma 3.5: the printed word "prime". Printed p. 48 instead says that $s-1$ is not a prime divisor of $q-1$. That literal assertion is false. Take $q = 2^{15} = 32768$ and $s = 8$. The subfield group $\mathrm{Sz}(8)$ is maximal, of field index five, and there is also the different maximal class $\mathrm{Sz}(32)$, of field index three. Yet $s-1 = 7$ is prime and

$$32767 = 7 \cdot 31 \cdot 151.$$

Its quotient $32767/7 = 4681$ is composite, in agreement with the corrected maximal-divisor assertion. This is a wording correction to an ancillary lemma, not an additional principal-refutation count.

Lemma 3.6: divisibility and the residue variable. Part (1) is $5 \mid q^2+1$, not $s^2+1 \mid q^2+1$. It follows at once from $q^2 = 2^{4e+2} \equiv -1 \pmod 5$. For part (2), the native statement assumes merely $s \geq 8$ divides q. This is insufficient. At $s = 8, q = 32$ it holds as integer divisibility, but

$$(\alpha_8, \beta_8) = (13, 5), \qquad (\alpha_{32}, \beta_{32}) = (41, 25),$$

and 13 divides neither 41 nor 25. The corrected hypothesis is $s = 2^d$, $d \mid 2e+1$, equivalently $q = s^r$ for an odd integer r. Under that hypothesis the compatible-torus divisibilities, with the possible interchange of the two signs, are proved in the accepted L115 preliminary note. Only that version is imported in our proofs.

Part (3) prints the residue class of $2e+1$ modulo four. The correct variable is e:

$$5 \mid \alpha_q \iff e \equiv 0, 3 \pmod 4, \qquad 5 \mid \beta_q \iff e \equiv 1, 2 \pmod 4.$$

To prove this for all e, observe that both 2^{2e+1} and 2^{e+1} are periodic modulo five with period four in e. For $e = 0, 1, 2, 3$, the respective residues of (α_q, β_q) are $(0,1), (3,0), (1,0), (0,3)$. The printed formula already fails at $q = 8$, where $2e+1 = 3$ but $\alpha_q = 13$ is not divisible by five.

Lemma 1. *For odd positive integers m, n,* $\gcd(2^{2m}+1, 2^{2n}+1) = 2^{2\gcd(m,n)}+1$.

Proof. Put $x = 4$ and $d = \gcd(m, n)$. The Euclidean algorithm for powers gives $\gcd(x^{2m}-1, x^{2n}-1) = x^{2d}-1$. The common divisor $D = \gcd(x^m+1, x^n+1)$ therefore divides $(x^d-1)(x^d+1)$. It is coprime to x^d-1: the latter divides x^m-1, and an odd common divisor of x^m-1, x^m+1 is one. Hence $D \mid x^d+1$. Conversely $m/d, n/d$ are odd, so x^d+1 divides both x^m+1 and x^n+1. This proves the equality and source Lemma 3.7. □

3 Unique extensions and proper overgroups

For source Lemma 3.9(i), let $H = C_d \rtimes C_2$, $d > 1$ dividing q^2+1. The cyclic odd subgroup determines one ambient nonsplit torus T and its normalizer $N = T \rtimes C_4$. Every group of the form $C_d \rtimes C_4$ containing H normalizes its characteristic odd subgroup and hence lies in N. An involution of N is contained in exactly one cyclic subgroup of order four: that cyclic subgroup is its centralizer in N, since the involution inverts T. Adjoining this C_4 to the unique order-d subgroup of T gives the unique extension. This proves (i), including actual existence.

For (ii), write $|T| = a_\epsilon(q)$. If $H = C_d \rtimes C_4$ and $p \mid |T|/d$ is prime, the unique subgroup of T of order dp, together with the given C_4, supplies a unique $C_{dp} \rtimes C_4$ containing H. Every possible extension must have this torus and this cyclic odd subgroup, proving uniqueness.

Lemma 3.9(ii): the required compatible factor. The native condition is instead $p \mid (q^2+1)/d$, which can choose a prime from the opposite, coprime nonsplit factor. At $q = 8, d = 5, p = 13$, the subgroup $C_5 \rtimes C_4$ exists, but $C_{65} \rtimes C_4$ does not: it would contain an element of order65, whereas the odd elements of Sz(8) lie in cyclic tori of orders $5, 7, 13$. The source condition therefore needs the compatible-factor restriction just proved. The retained torus partition is the only group-theoretic premise in this small obstruction; no enumeration is required.

The full split-torus overgroup argument in the normalizer-restriction note gives the precise version of source Lemma 3.10: the *strict proper* overgroups of C_{q-1} are its dihedral normalizer, the two point stabilizers, and their two $E_q \rtimes C_{q-1}$ subgroups. The torus itself and the whole ambient group must be added if neither strictness nor properness is imposed. We state this qualification explicitly rather than assigning an unintended meaning to "containing".

Finally let $S = \mathrm{Sz}(s)$ be maximal in $G = \mathrm{Sz}(q)$ and let $H = D_{2(s-1)}$ be its full split normalizer. Every proper overgroup of H is either a dihedral subgroup D_{2u} of its ambient split normalizer, with $s-1 \mid u \mid q-1$, or a simple subfield group $\mathrm{Sz}(t)$. Point stabilizers are excluded by the inversion argument, and nonsplit normalizers by coprimality. In the subfield case the rotation subgroup is split, so $s-1 \mid t-1$. Writing $s = 2^d, t = 2^f$ gives $d \mid f$; this follows from the Euclidean identity $\gcd(2^d-1, 2^f-1) = 2^{\gcd(d,f)}-1$. Maximality of the field degree d in the ambient degree forces $f = d$ for a proper subfield. The accepted uniqueness of the subfield containing this embedded full split normalizer then gives $\mathrm{Sz}(t) = S$. This proves source Lemma 3.11 for proper overgroups. The ambient group itself remains excluded by that explicit qualification.

4 Attribution and remaining work

The new source joins above distinguish literal errors, intended corrected statements, and earlier quoted inputs. They do not replace the primary classification counterexamples already accepted for L114, and add no principal-refutation or ordinary-recovery count. The arithmetic checks attached to this note only verify the displayed small integers; the general proofs and imported subgroup premises require distinct mathematical review. The source's general rank-three thinness assertion under its literal algebraic local-action convention and the final source-case joins remain separate obligations. No Lean verification or compression measurement is claimed.

A rank obstruction in prime-degree Suzuki groups

Human Compression project

Abstract

We prove the nonexistence assertion in Theorem 5.4 of the fixed 1999 rank-three Suzuki paper. The argument uses its subgroup definition of residual weak primitivity directly, so assumes neither flag transitivity nor a geometric interpretation of every residue. Two successive maximal intersections suffice. The required proper-overgroup information is taken from the classical Suzuki subgroup classification, with the elementary point-stabilizer calculation imported from the separately reviewed note.

1 The subgroup condition

Let I be finite and let $(G_i)_{i\in I}$ be subgroups of a finite group G. Write $G_J = \bigcap_{j\in J} G_j$ and $G_\varnothing = G$. The condition used in the source [1], printed page 46, is

$$\text{for every } J \subsetneq I \text{ there is } i \in I \setminus J \text{ such that } G_{J\cup\{i\}} \text{ is maximal proper in } G_J. \tag{1}$$

Here $J = \varnothing$ is included. The native statement of Theorem 5.4, printed page 55, excludes rank *at least* three, not merely rank strictly greater than three.

Lemma 1. *Under* (1)*, for every $J \subsetneq I$ and every $i \notin J$, $G_{J\cup\{i\}} < G_J$. In particular all G_i are proper, no G_i is contained in a different G_j, and adjoining any further type to an intersection strictly decreases it.*

Proof. If $G_J \le G_i$, then $G_{I\setminus\{i\}} \le G_J \le G_i$. Consequently $G_{I\setminus\{i\}} = G_I$, contradicting (1) at $I \setminus \{i\}$, where the only available type is i. This proves the asserted strictness and its consequences. □

2 Classical inputs and the point stabilizer

Put $q = 2^n$, with $n \ge 3$ odd, and $G = \mathrm{Sz}(q)$. We retain the classical Suzuki subgroup and ovoid structure used in the source and specified in the accepted notes [2, 3]. The maximal subgroups have types

$$B = P \rtimes C_{q-1}, \quad D = C_{q-1} \rtimes C_2, \quad N_\epsilon = C_{a_\epsilon} \rtimes C_4, \quad \mathrm{Sz}(2^d), \qquad a_\epsilon = q + \epsilon 2^{(n+1)/2} + 1,$$

where the last type has $d \ge 3$ and n/d prime. Every proper subgroup is contained in one of these groups. The three odd torus orders are pairwise coprime. Every nonidentity odd element is in a unique cyclic torus; its centralizer is that torus. A split torus has normalizer D, whose involution inverts it. In N_ϵ the order-four complement acts without nonidentity fixed points on its odd kernel, and its square acts by inversion. The group P has order q^2 and centre Z elementary abelian of order q. A nonidentity involution fixes exactly one ovoid point; B is a point stabilizer. These are attributed structural premises, not new classification claims.

The reviewed elementary proof in [2] shows that every maximal subgroup of B contains Z. We record its consequence for (1), separately from any flag-transitivity hypothesis.

Lemma 2. *If $|I| \geq 2$ and (1) holds, no G_i is a point stabilizer B.*

Proof. Suppose $G_0 = B$. Choose $i \neq 0$ such that $J = B \cap G_i$ is maximal in B. Then $Z \leq J$. No dihedral or nonsplit torus normalizer contains Z: their Sylow two-subgroups have order two or are cyclic of order four. Nor does a proper simple subfield group $\mathrm{Sz}(s)$ contain Z, since $q \geq s^3 > s^2$, the order of its Sylow two-subgroups. A point stabilizer containing Z must equal B, because any fixed nonidentity involution in Z has a unique fixed ovoid point. Thus B is the only maximal subgroup of G containing J. The proper subgroup G_i lies in a maximal subgroup, so $G_i \leq B$, contrary to Lemma 1. □

3 Two maximal intersections

For the remainder assume n is prime. There are then no proper simple subfield groups. This hypothesis is essential to the following proper-overgroup arguments.

Lemma 3. *If x has nontrivial odd order dividing a_ϵ, every proper subgroup containing x lies in the unique N_ϵ associated with its torus. If a dihedral subgroup E has a nontrivial odd rotation subgroup in a split torus, every proper subgroup containing E lies in the normalizer D of that split torus.*

Proof. For the first assertion, take a maximal subgroup containing the given proper subgroup. Coprimality excludes point stabilizers, split normalizers and the opposite nonsplit type. In a nonsplit normalizer an odd-order element belongs to its odd kernel. Uniqueness of the torus containing x therefore identifies that maximal subgroup.

For the second assertion, coprimality excludes both nonsplit types. A point stabilizer cannot contain E. Indeed, its quotient by its normal two-subgroup is abelian of odd order. An odd rotation has nontrivial image in this quotient, whereas the involution inverting it has trivial image. The inversion relation would force that odd image to equal its inverse, a contradiction. The remaining maximal subgroup is a split normalizer, and the rotation subgroup determines its unique torus, hence determines this normalizer. □

Lemma 4. *If n is prime and $|I| \geq 3$, no G_i is a full nonsplit torus normalizer $N = C_a \rtimes C_4$.*

Proof. Suppose $G_0 = N$. For $i \neq 0$, the subgroup $K_i = G_0 \cap G_i$ cannot contain a nonidentity odd-order element: Lemma 3 would give $G_i \leq N$, violating Lemma 1. Thus K_i is a two-group, cyclic of order at most four. Choose i with K_i maximal in N. It cannot have order one or two, since it lies in a Sylow two-subgroup of order four. Hence $K_i \cong C_4$.

Apply (1) at $\{0, i\}$ to choose $k \notin \{0, i\}$ such that G_{0ik} is maximal in K_i; it has order two. An involution $z \in N$ lies in a unique cyclic subgroup of order four. To see this, conjugate its Sylow subgroup to the specified complement V. Then z is its involution and $\mathrm{C}_N(z) = V$, because z inverts the odd kernel. Any cyclic group of order four containing z centralizes z and therefore equals V. It follows that the cyclic two-subgroup K_k containing G_{0ik} is contained in K_i. Hence $G_{0ik} = G_{0k}$, again contradicting Lemma 1. □

Lemma 5. *Suppose n is prime and $q - 1$ is composite. If $|I| \geq 3$, no G_i is the full split normalizer $D = C_{q-1} \rtimes C_2$.*

Proof. Suppose $G_0 = D$, and write $A = C_{q-1}$. Each $K_i = G_0 \cap G_i$, $i \neq 0$, is either contained in A or is a subgroup of order at most two. Indeed, any other subgroup of the odd dihedral group D is dihedral with a nontrivial odd rotation subgroup; Lemma 3 would force $G_i \leq D$. Choose i such that K_i is maximal in D. The maximal subgroups of D are A and dihedral groups of order $2(q-1)/p$, where p is a prime divisor of $q - 1$. The latter have nontrivial odd rotation subgroups because $q - 1$ is composite, so $K_i = A$. Now choose $k \notin \{0, i\}$ with G_{0ik}

maximal in A. Its order is $(q-1)/p > 1$ for some prime p, so is odd and nontrivial. The preceding restriction on K_k forces $K_k \leq A = K_i$. Thus $G_{0ik} = G_{0k}$, contrary to Lemma 1. □

Theorem 6. *Let $n \geq 3$ be an odd prime and suppose $2^n - 1$ is composite. There is no subgroup family for $G = \mathrm{Sz}(2^n)$ satisfying (1) with $|I| \geq 3$. In particular there is no residually weakly primitive pre-geometry of rank at least three in the source's sense.*

Proof. The case $J = \varnothing$ in (1) makes at least one G_i maximal in G. The classical list leaves B, D, N_+, N_-, since n is prime. Lemmas 2, 4 and 5 exclude all four possibilities. □

This proves exactly the source's Theorem 5.4, including its full rank range. The point-stabilizer exclusion also proves the subgroup form of its quoted Lemma 4.1. The nonsplit and split arguments above are only claimed in prime field degree; they do not by themselves dispose of all composite-degree assertions in Lemmas 4.2–4.6. No sufficient realization has been used as a necessity statement. No finite group search, new conjecture resolution, Lean verification or compression measurement is claimed for this component.

References

[1] D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63. Fixed twenty-one-page edition, especially printed pp. 46 and 55.

[2] Human Compression project, *Maximal subgroups and elementary reductions for Suzuki geometries*, accepted preliminary component for L115, with its explicit classical Suzuki premises.

[3] Human Compression project, *Normalizer fibres in rank-two Suzuki geometries*, accepted L115 component. Its structural inputs are attributed there to Suzuki, Tits and Lüneburg.

Maximal intersections in Suzuki torus normalizers

Human Compression project

Abstract

The subgroup definition of residual weak primitivity can be applied before assuming flag transitivity. For cyclic groups and odd dihedral groups it forces every parabolic to be maximal. In Suzuki groups, two successive maximal intersections then reduce the possible torus normalizers to compatible maximal subfields. Elementary size estimates exclude the remaining pairs. A separate normalizer calculation handles two subfields of the same size. These arguments prove the maximal-parabolic restrictions in Lemmas 4.2–4.6 and the full split-normalizer case of Lemma 5.1 in the fixed rank-three source.

1 The subgroup condition in cyclic and dihedral groups

For a family $(H_i)_{i\in I}$ in a finite group H, write $H_J = \bigcap_{j\in J} H_j$ and $H_\varnothing = H$. We use the condition

$$\text{for every } J \subsetneq I \text{ some } i \notin J \text{ makes } H_{J\cup\{i\}} \text{ maximal proper in } H_J. \tag{1}$$

The previously accepted strict-intersection lemma says that adding *any* further type strictly decreases an intersection. Its proof is short: $H_J \le H_i$ would imply $H_{I\setminus\{i\}} = H_I$, contrary to (1) at $I \setminus \{i\}$. In particular no distinct parabolics contain one another. The family induced inside any H_J again satisfies (1).

Lemma 1. *If H is cyclic and the family satisfies* (1)*, every H_i is maximal in H. Their indices are distinct primes.*

Proof. Induct on $|I|$. For one type this is the definition. Choose H_0 maximal, of prime index p. For $i \neq 0$, noncontainment in H_0 means that $|H_i|$ has the full p-part of $|H|$. Consequently $|H_i \cap H_0| = |H_i|/p$. The induced family in H_0 satisfies (1), so induction says that $H_i \cap H_0$ is maximal there. Its index is

$$[H_0 : H_i \cap H_0] = \frac{|H|/p}{|H_i|/p} = [H : H_i],$$

which is prime. Equal indices would give equal subgroups in a cyclic group, contrary to strict intersections. □

Lemma 2. *Let $H = D_{2m}$ with m odd. Every parabolic in a family satisfying* (1) *is maximal in H. At most one is cyclic. If the rank is at least three, at least two are dihedral.*

Proof. Again induct on the rank. Write $H = \langle r\rangle \rtimes \langle t\rangle$, with $|r| = m$, $t^2 = 1$, and $r^t = r^{-1}$. If a chosen maximal parabolic is $A = \langle r\rangle$, every other parabolic is dihedral by noncontainment. Its intersection with A is its rotation subgroup, which is maximal in A by Lemma 1 applied to the induced family. The parabolic therefore has prime index in H.

Otherwise choose $H_0 = \langle r^p, t\rangle$ for a prime $p \mid m$, after changing the reflection t. Every $H_0 \cap H_i$ is maximal in H_0 by induction (or Lemma 1 when $H_0 = C_2$). If $H_i = C_d$ is cyclic, noncontainment in H_0 gives the full p-part in d, and its intersection has order d/p. The only maximal cyclic subgroup of H_0 is its rotation subgroup, so $d/p = m/p$ and $H_i = A$.

If $H_i = \langle r^{m/d}, r^k t\rangle$ is dihedral, first suppose d has the full p-part of m. Then $p \nmid m/d$. The two reflection congruences have a simultaneous solution, so $H_0 \cap H_i$ is dihedral of order $2d/p$. Its index in H_0 is m/d; since it is maximal, that index is prime. Thus H_i is maximal in H. If d has a smaller p-part, then $d \mid m/p$. Noncontainment forces its reflection coset to be disjoint from H_0, so $H_0 \cap H_i = C_d$. Maximality in H_0 forces $d = m/p$, and again H_i is maximal, now of index p. This covers every subgroup. A maximal cyclic subgroup must be A, so there is at most one; the last assertion follows immediately. □

This proves directly the dihedral consequence cited from reference [20] in the source's Lemmas 4.3 and 4.4. It does not claim a new classification of all dihedral geometries or import their unexplained incidence properties.

2 Structural inputs and a subfield uniqueness lemma

Put $q = 2^n$, with $n \geq 3$ odd. We retain the classical Suzuki subgroup, ovoid and matrix description specified in the accepted L115 notes. Every proper subgroup either lies in a point stabilizer, a split torus normalizer, a nonsplit torus normalizer, or is itself a simple subfield group. Write

$$B_q = U_q \rtimes \mathbb{F}_q^*, \quad D_q = C_{q-1} \rtimes C_2, \quad N_\epsilon(q) = C_{a_\epsilon(q)} \rtimes C_4, \quad a_\epsilon(q) = q + \epsilon\sqrt{2q} + 1.$$

The three odd torus orders are pairwise coprime, and a nonidentity odd element lies in a unique torus. A proper simple subfield size satisfies $q \geq s^3$. The accepted elementary estimates imply, also directly,

$$q/2 \leq a_\epsilon(q) < 2q \quad (q \geq 8). \tag{2}$$

Indeed $q + 1 - \sqrt{2q} \geq q/2$ and $q + 1 + \sqrt{2q} < 2q$ in this range. The uniqueness of an embedded $D_{2(s-1)}$ or $C_{a_\delta(s)} \rtimes C_4$ inside a fixed-size subfield group was proved in the accepted normalizer-fibre note.

Lemma 3. *For $s = 2^d \geq 8$, $d \mid n$, every subgroup conjugate to the standard $Z_s \rtimes \mathbb{F}_s^*$ is contained in exactly one subgroup isomorphic to $\mathrm{Sz}(s)$. Here Z_s is the centre of a Sylow two-subgroup of the standard subfield group.*

Proof. We prove that the standard $H = Z_s \rtimes \mathbb{F}_s^*$ is self-normalizing in $G = \mathrm{Sz}(q)$. Its characteristic Sylow two-subgroup Z_s determines its unique fixed ovoid point. A normalizer therefore lies in B_q. Use the accepted coordinates $u(a, b)$ for its Sylow subgroup and $h(v)$ for the diagonal torus. Conjugation by $h(v)$ multiplies the central coordinate by $v^{\theta+2}$, where $\theta = 2^{(n+1)/2}$. Preserving the central line $\mathbb{F}_s$ forces $v^{\theta+2} \in \mathbb{F}_s^*$. Since $\theta+2$ is coprime to $q-1$, this is equivalent to $v \in \mathbb{F}_s^*$. Remove this diagonal factor.

For $u(a, b)$, preservation of H forces every commutator with $h(k)$, $k \in \mathbb{F}_s^*$, to have first coordinate zero. That coordinate is a nonzero scalar multiple of $(k^\theta - 1)a$. Taking $k \neq 1$ therefore gives $a = 0$. The remaining commutator coordinate is a nonzero scalar multiple of $(k^{\theta+2} - 1)b$ and must lie in $\mathbb{F}_s$. Again take $k \neq 1$; its coefficient is a nonzero element of $\mathbb{F}_s$, so $b \in \mathbb{F}_s$. Thus $\mathrm{N}_G(H) = H$.

The standard subfield group $S = \mathrm{Sz}(s)$ is self-normalizing, as proved in the accepted note. Its copies of the indicated affine subgroup form one conjugacy class: the Sylow centres are conjugate, and the odd complements in their point stabilizers are conjugate by the elementary affine-complement argument. The same incidence count used in that note therefore gives

$$\#\{S^g : H \leq S^g\} = \frac{|\mathrm{N}_G(H)|}{|\mathrm{N}_S(H)|} = 1.$$

Conjugation gives the assertion for every conjugate of H. □

We also need an elementary overgroup observation valid for every s. In $S = \mathrm{Sz}(s)$, the proper overgroups of its full split torus $A_s = C_{s-1}$ are exactly

$$A_s, \quad D_s, \quad Z_P A_s, \quad Z_Q A_s, \quad B_P, \quad B_Q, \tag{3}$$

where P, Q are the two fixed ovoid points. The classical list excludes nonsplit normalizers by coprimality and proper simple subfields by their smaller torus orders. Inside either point stabilizer an overgroup is VA_s. Full scalar multiplication acts irreducibly on both $Z(U)$ and $U/Z(U)$ over $\mathbb{F}_2$. Thus $V = 1, Z(U)$ or U: if its image on $U/Z(U)$ is nonzero, then $VZ(U) = U$, and the accepted identity $Z(U) = \Phi(U)$ gives $V = U$. This proves (3).

3 Excluding a full nonsplit normalizer

Theorem 4. *A family satisfying* (1) *in* $\mathrm{Sz}(q)$ *cannot have a full nonsplit torus normalizer as a parabolic when its rank is at least three.*

Proof. The prime-degree case is already accepted, so assume n composite. Both nonsplit torus orders are then composite by the accepted subfield arithmetic. Suppose $G_0 = N = C_m \rtimes C_4$, with $m = a_\epsilon(q)$. Choose $J = G_{01}$ maximal in N. Its possibilities are D_{2m} and $C_{m/p} \rtimes C_4$, $p \mid m$ prime. No proper subgroup outside N contains D_{2m}: the classical list and the torus partition exclude the other solvable types, and a proper simple subfield has every torus order less than $2s < q/2 \le m$. The first possibility is therefore excluded.

Set $d = m/p > 1$ and $J = C_d \rtimes C_4$. The proper overgroup G_1 is not contained in N by strict intersections. The classical list forces $G_1 = \mathrm{Sz}(s)$. Inside it the odd subgroup C_d lies in a nonsplit torus of order $a_\delta(s)$, which divides m and is smaller than m. Since it is a multiple of m/p, it equals d. Thus J is its full normalizer in G_1. Moreover G_1 is a maximal subfield group: an intermediate field would supply a strictly intermediate compatible torus order between d and m, impossible when $m/d = p$ is prime. Write $q = s^a$ with a an odd prime.

Choose another type 2 so that $K = G_{012}$ is maximal in J. If $K = D_{2d}$, then G_{02} contains it. A subgroup of N containing K has odd kernel of order d or m. In the first case it is contained in the unique $C_d \rtimes C_4$ containing K, namely J. In the second case it contains D_{2m}, which cannot be in the proper subgroup $G_2 \not\le N$. Both contradict strict intersections.

The other possibility is $K = C_{m/(pr)} \rtimes C_4$ for a prime $r \mid d$. Since G_{02} contains this fixed C_4, its odd kernel has order divisible by $m/(pr)$ and dividing m. Noncontainment in J requires the full p-part. If $r = p$, this forces the whole N, which is impossible. If $r \neq p$, it forces $G_{02} = C_{m/r} \rtimes C_4$. The same argument as for G_1 gives $G_2 = \mathrm{Sz}(t)$, with $m/r = a_\eta(t)$ and $q = t^b$, b prime.

If $s = t$, the two torus orders are distinct and hence are the opposite ones. Their product is $s^2 + 1$, whereas

$$m = \frac{(m/p)(m/r)}{m/(pr)} \le s^2 + 1.$$

This contradicts $m \ge q/2 \ge s^3/2 > s^2 + 1$. If $s \neq t$, the distinct maximal field indices a, b give $q = 2^{hab}$, $s = 2^{hb}$ and $t = 2^{ha}$ for a positive integer h. Distinct odd primes satisfy $ab - a - b \ge 7$, so $q/(st) \ge 128$. But p divides $m/r = a_\eta(t)$, while (2) gives

$$p = \frac{m}{a_\delta(s)} > \frac{q}{4s} \ge 32t > a_\eta(t).$$

This final contradiction proves the theorem. □

4 The full split normalizer

Theorem 5. *Suppose $q-1$ is composite and a family of rank at least three satisfying* (1) *has $G_0 = D_q$. At most one G_{0i} is dihedral.*

Proof. Apply Lemma 2 to the family induced in G_0. Every G_{0i} is maximal in D_q. Suppose two are dihedral: $G_{01} = D_{2m/p}$ and $G_{02} = D_{2m/r}$, where $m = q-1$ and p, r are primes. Their rotation subgroups are nontrivial. A proper overgroup of such a split dihedral subgroup cannot lie in a point stabilizer: its involution has trivial image in the abelian odd quotient, so cannot invert the nontrivial odd image of a rotation. Coprimality excludes nonsplit normalizers, and the torus partition identifies the only possible split normalizer as D_q. Consequently $G_1 = \mathrm{Sz}(s)$ and $G_2 = \mathrm{Sz}(t)$. Their split torus orders are multiples of m/p and m/r, respectively, and proper divisors of m. Hence

$$s - 1 = m/p, \qquad t - 1 = m/r.$$

As in the preceding proof, the prime indices imply that both are maximal subfield groups.

If $p \neq r$, then $s \neq t$. Write their distinct field indices as a, b. The same calculation gives $q/(st) \geq 128$. The prime p divides $m/r = t - 1$, but

$$p = \frac{q-1}{s-1} > \frac{q}{2s} \geq 64t > t - 1,$$

a contradiction.

If $p = r$, the two distinct dihedral subgroups have the same rotation subgroup $A_s = C_{s-1}$ and disjoint reflection cosets. Thus $G_{012} = A_s$, and $s = t$. Strictness requires $L = G_1 \cap G_2 > A_s$. Apply (3) inside G_1. If $L = D_s$, it equals the normalizer of A_s in G_1, namely G_{01}. Then $G_{012} = G_{01}$, a contradiction. Each other possibility contains one of the groups $Z_P A_s$ or $Z_Q A_s$. By Lemma 3, this subgroup is contained in a unique copy of $\mathrm{Sz}(s)$, so $G_1 = G_2$, again impossible. This treats the same-size case separately; distinct field indices were not assumed merely from distinct parabolics. □

Corollary 6. *A family satisfying* (1) *with a full split normalizer as a parabolic has rank at most three. In rank at least four, every parabolic maximal in* $\mathrm{Sz}(q)$ *must be a simple subfield group.*

Proof. If $q - 1$ is prime, the earlier accepted rank bound applies. If it is composite and the rank is at least four, the residue in D_q has rank at least three. Lemma 2 gives at least two dihedral parabolics in it, contrary to Theorem 5. For the last assertion, the accepted point-stabilizer exclusion and Theorem 4 exclude the other solvable maximal subgroups. The classical maximal-subgroup list leaves only simple subfields. □

Corollary 7. *Suppose n is composite and a rank-three family satisfying* (1) *has $G_0 = D_q$. After interchanging types 1 and 2, its pattern is*

G_0	G_1	G_2
$D_{2(q-1)}$	$E_q \rtimes C_{q-1}$	$\mathrm{Sz}(s)$

$$G_{01} = C_{q-1}, \quad G_{02} = D_{2(s-1)}, \quad G_{12} = E_s \rtimes C_{s-1}, \quad G_{012} = C_{s-1},$$

where $(q-1)/(s-1)$ is prime. The degree n is a power of one odd prime. This is a necessary pattern, not an existence assertion.

Proof. The two proper intersections in D_q are maximal by Lemma 2. At most one is dihedral by Theorem 5, and there cannot be two cyclic ones. Thus one is C_{q-1} and the other is $D_{2(q-1)/p}$. The proof of Theorem 5 gives $G_2 = \mathrm{Sz}(s)$ with $s - 1 = (q-1)/p$. The full-torus overgroups (3), now in $\mathrm{Sz}(q)$, and strictness leave $G_1 = Z_P C_{q-1}$ or $Z_Q C_{q-1}$; the full point stabilizers have already been excluded. Their intersection with G_2 properly contains C_{s-1}. Applying (3) inside G_2, it cannot be dihedral, since a split dihedral group cannot lie in a point stabilizer. It cannot be the full B_s, whose Sylow two-subgroup has exponent four, whereas every two-subgroup of G_1 is elementary abelian. The remaining option is $E_s \rtimes C_{s-1}$.

For completeness, the prime quotient forces the degree assertion. Write $q = s^a$. If a were composite, the geometric sum $(s^a - 1)/(s - 1)$ would factor into two integers greater than one; thus a is prime. Write $s = 2^d$. If a prime $b \neq a$ divided d, put $x = 2^{d/b}$. The smaller integer $M = 1 + x + \cdots + x^{a-1} > 1$ divides $1 + x^b + \cdots + x^{b(a-1)}$: modulo M, $x^a = 1$, and multiplication by b permutes the residues modulo a. This would again make $(q-1)/(s-1)$ composite. Therefore d is a power of a, and so is $n = ad$. □

5 Attribution and scope

The source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63. Theorem 4 proves its Lemma 4.2 for every permitted field degree. Lemma 2, Theorem 5 and Corollary 6 prove Lemmas 4.3–4.6; Corollary 7 proves the precise necessary assertion in Lemma 5.1. The cyclic and dihedral induction replaces the explicitly cited consequence of the earlier dihedral classification. The same-size subfield argument is supplied explicitly rather than inferring distinct field degrees from distinct subgroups. These are elementary arguments; no priority claim is made for the methods.

The classical subgroup classification, the faithful Suzuki coordinates, ovoid action and subfield embeddings remain borrowed structural inputs. The accepted L115 notes supply the Frattini identity, scalar actions, torus arithmetic and normalizer-incidence counting. The accepted L114 prime-degree proof supplies strict intersections and the point-stabilizer and prime-degree exclusions. No finite computation or Lean certificate is asserted in this component. It does not repair the refuted complete candidate list in Theorem 5.5, prove the remaining thinness assertion, or confer whole-original or new-conjecture credit.

The necessary patterns when the split torus has prime order

Human Compression project

Abstract

We prove the necessary-list assertion of Theorem 5.3 in the fixed 1999 rank-three Suzuki paper. Proper covers of a subgroup of prime order reduce the possibilities to three patterns. A sparse matrix entry excludes one apparent additional configuration without assuming an unstated intersection property of ovoid circles. The prime-degree rank bound and the rank-two locally doubly transitive cases follow as well. We distinguish the source's actions on residues defined by subgroup intersections from actions on full incidence residues. Under the latter hypothesis we also prove geometric thinness; it does not follow merely by identifying the former actions.

1 Definitions and acquired restrictions

Let $q = 2^n \geq 8$, suppose $p = q - 1$ is prime, and put $G = \mathrm{Sz}(q)$. Then n is prime. We use the source's subgroup condition: for every proper $J \subsetneq I$, some $G_{J\cup\{i\}}$ is maximal proper in $G_J = \bigcap_{j\in J} G_j$, with $G_\varnothing = G$. The preceding note *A rank obstruction in prime-degree Suzuki groups* proves directly from this condition that every additional type strictly decreases an intersection. It excludes a point stabilizer as a parabolic in every rank at least two, and a full nonsplit torus normalizer in every rank at least three. We import those arguments; they do not assume flag transitivity. Their distinct mathematical review has been adopted in the accompanying prime-degree root review; the exact accepted inputs are bound below.

We retain the same classical Suzuki subgroup classification, torus partition, ovoid action and faithful matrix model. A split torus $A = C_p$ fixes exactly two ovoid points, denoted P, Q; its normalizer is $D = A \rtimes \langle w\rangle$, where w interchanges P, Q and inverts A. Write Z_P, Z_Q for the centres of their Sylow two-subgroups. Every nonidentity involution has one fixed ovoid point, and its centralizer is the corresponding Sylow two-subgroup. There are no proper simple subfield groups because n is prime.

Lemma 1. *The following proper overgroups in G have the indicated forms.*

1. *If C_2 is maximal in $L < G$, then L has order four or $L = D_{2r}$ for an odd prime r dividing $|G|$.*

2. *The proper overgroups of a full split torus $A = C_p$ are $A, D, Z_P A, Z_Q A, B_P, B_Q$, where B_P, B_Q are the two point stabilizers containing A.*

3. *If C_r is maximal in $L < G$ for an odd prime r, and L contains an involution, then $L = D_{2r}$ or $r = p$ and $L = Z_R A'$ for a split torus A' of order p fixing R.*

Proof. For (1), first suppose L lies in a point stabilizer $B = U \rtimes T$. Its normal two-subgroup $V = L \cap U$ contains the given C_2. Maximality gives $V = L$ or $V = C_2$. In the latter case L normalizes C_2 and hence centralizes its involution, so again L is a two-group. A maximal subgroup of a finite two-group has index two, giving $|L| = 4$. If L lies in a cyclic torus normalizer, intersect it with the odd cyclic kernel. A nontrivial intersection K is normal and $KC_2 > C_2$, so maximality forces $L = KC_2$. The involution inverts K. Its order must be

prime, since otherwise a proper nontrivial rotation subgroup gives an intermediate dihedral subgroup. If $K = 1$, L is a two-group and the same index argument applies. These cases exhaust the proper subgroups by the classical classification.

For (2), coprimality excludes the nonsplit normalizers. A split normalizer containing A is D, by uniqueness of the torus. A point stabilizer containing A fixes P or Q. In either one, write an overgroup L of A as VA, with $V = L \cap U$. The action of A on both $Z(U)$ and $U/Z(U)$ is the full scalar action of $\mathbb{F}_q^*$, so is irreducible over $\mathbb{F}_2$. If $V \leq Z(U)$, then V is zero or the whole centre. Otherwise $VZ(U) = U$, and the acquired identity $Z(U) = \Phi(U)$ forces $V = U$. This gives precisely the displayed list.

For (3), a point-stabilizer case has $r \mid p$, so $r = p$ and (2) shows that the only cover there is $Z_R A'$. In a cyclic torus normalizer, let K be the intersection of L with the odd kernel. It contains C_r and is proper in L, because L has an involution. Maximality gives $K = C_r$. The quotient embeds in a cyclic group of order two or four. An order-four quotient would give an intermediate subgroup, so the quotient has order two. Its involution acts by inversion on the odd torus. Thus $L = D_{2r}$. □

2 An excluded triangle

Lemma 2. *There is no family satisfying the source's subgroup condition with*

$$G_0 = D, \qquad G_{01} = A, \qquad G_{02} \cong C_2, \qquad G_{12} \cong C_p, \qquad G_{012} = 1.$$

Proof. By Lemma 1(2) and the point-stabilizer exclusion, G_1 is $Z_P A$ or $Z_Q A$. Interchanging P, Q if necessary makes $G_1 = Z_P A$. Conjugation by A, which preserves G_1, sends the reflection in G_{02} to w: squaring is bijective on the odd group A. Every order-p subgroup of $Z_P A$ is a complement and is conjugate to A by Z_P. This is the elementary affine fixed-point argument already used in the accepted preliminary note. Thus $G_{12} = A^z$ for some $1 \neq z \in Z_P$; $z = 1$ would contradict $G_{012} = 1$.

Apply Lemma 1(2) to A^z. The group G_2 is a proper overgroup containing w. Point stabilizers are excluded and G_2 cannot equal A^z, so the possibilities are D^z or one of the two groups $Z_R A^z$ at its fixed points. Those fixed points are P, Q^z. The group at P is G_1, which is excluded by strict intersections.

The possibility $w \in D^z$ is also impossible. Such a w would interchange P, Q^z, whereas w interchanges P, Q. Hence $Q^z = Q$. The nonidentity involution z fixes only P, a contradiction. For the remaining possibility, $w \in B_{Q^z}$ would give $zwz \in B_Q$, since $z^2 = 1$. Use the retained faithful matrix model with B_P lower triangular, B_Q upper triangular, w antidiagonal, and

$$z = z(b) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ b & 0 & 1 & 0 \\ b^\theta & b & 0 & 1 \end{pmatrix}, \qquad b \in \mathbb{F}_q^*, \quad \theta = 2^{(n+1)/2}.$$

Direct multiplication gives $(z(b)wz(b))_{21} = b \neq 0$. It cannot be upper triangular. This excludes the last possibility. □

The matrix calculation is a quantified identity in the field; it is not a finite test. It replaces the source's ovoid-orbit intersection step in this particular configuration. The matrix model itself is classical and is the same one used in the retained Suzuki realization manuscripts.

3 All necessary rank-three patterns

Theorem 3. *Every rank-three subgroup family satisfying the source's condition has, after permuting its types, one of the following intersection patterns:*

G_0	G_1	G_2	G_{01}	G_{02}	G_{12}
D_{2p}	$E_q \rtimes C_p$	$E_q \rtimes C_p$	C_2	C_2	C_p
D_{2p}	$E_q \rtimes C_p$	D_{2r}	C_p	C_2	C_2
D_{2p}	D_{2r}	D_{2s}	C_2	C_2	C_2

The triple intersection is trivial. In the second row r is an odd prime divisor of $(q-1)(q^2+1)$. In the third row s is such an odd prime, and r is either such an odd prime or 2; the notation D_4 means C_2^2. Equal odd primes are allowed. These are precisely the necessary patterns of source Theorem 5.3.

Proof. Some parabolic is maximal in G. The imported restrictions exclude point stabilizers and full nonsplit normalizers, leaving $G_0 = D$. Inside D_{2p} the proper nontrivial subgroups are $A = C_p$ and its reflection groups C_2. Strict intersections and the subgroup condition therefore leave exactly two possibilities for its rank-two residue: $(G_{01}, G_{02}, G_{012}) = (A, C_2, 1)$ or $(C_2, C_2, 1)$. The triple intersection is maximal in G_{12} as well, so G_{12} is cyclic of prime order.

In the first case, Lemma 1(2), strict intersections and the point-stabilizer exclusion give $G_1 = Z_P A$, after exchanging the two fixed points. The prime order of G_{12} is either 2 or p. Lemma 2 excludes p. Thus G_2 contains two distinct involutions, with one of their order-two groups maximal in G_2 by the subgroup condition at $\{2\}$. Lemma 1(1) gives an odd prime dihedral group or a four-group; a cyclic group of order four has only one involution. The four-group possibility is impossible: the involution in $G_{12} \leq Z_P$ fixes P, whereas the reflection in $G_{02} \leq D$ exchanges P, Q. Commuting involutions would lie in the centralizer Sylow two-subgroup of the first and would both fix P. We obtain the second row.

In the second residue case, write $G_{12} = C_r$. If $r = 2$, each of G_1, G_2 contains two distinct involutions and has an order-two subgroup maximal in it. Lemma 1(1) makes each a prime dihedral group or a four-group. They cannot both be four-groups: if u, v are the two distinct reflections in G_0 and z is the involution of G_{12}, both u, v would centralize z. Then $\langle u, v\rangle = D_{2p}$ would lie in the two-group $\mathrm{C}_G(z)$, impossible. This gives exactly the third row.

Suppose instead that r is odd. At each G_i, $i = 1, 2$, either its intersection C_2 with G_0 or its intersection C_r with the other parabolic is maximal. Lemma 1(1),(3) gives $G_i = D_{2r}$, or $r = p$ and G_i is $E_q \rtimes C_p$. If $r \neq p$, both are D_{2r} and lie in the unique nonsplit normalizer determined by G_{12}. That normalizer would contain the two reflections generating D_{2p}, contradicting coprimality of its odd order with p. Thus $r = p$. If both are D_{2p}, their common rotation subgroup G_{12} determines the same full split normalizer, so they coincide, contrary to strict intersections. If exactly one is dihedral, permuting types gives the excluded triangle of Lemma 2. Both must therefore be $E_q \rtimes C_p$, giving the first row. □

This is a necessity proof. The previously reviewed phase-five realizations show sufficiency at the level of the specified subgroup intersections, but are not used anywhere in this proof. Their additional geometric scope must be stated separately when imported.

Corollary 4. *Under the same prime-$q-1$ hypothesis no such family has rank at least four.*

Proof. A maximal parabolic must again be D_{2p}. Two successive applications of the subgroup condition give a maximal proper subgroup of D_{2p}, then a maximal proper subgroup of that group. The first has prime order p or 2, and the second is trivial. If rank is at least four, the condition at this three-type intersection would demand a proper maximal subgroup of the trivial group. This is impossible. □

4 Local actions and incidence panels

On printed page 45, the source defines Γ_F for a flag $F \subseteq C = \{G_0, G_1, G_2\}$ by the intersection subgroups. On printed page 46 its local double transitivity condition refers to flags $F \subset C$ and to this Γ_F, and it expressly qualifies agreement with the earlier definition in its reference [10] by flag transitivity. We preserve that literal convention and keep the full incidence panel separate. For a two-type base flag of types i, j, the algebraic rank-one residue is G_{ij}/G_{012}. In the three patterns above its denominator is trivial, so its action is regular. A regular action of a nontrivial group is doubly transitive precisely when the group has order two: its point stabilizer is trivial, hence is transitive on the remaining points precisely when there is only one such point. Thus the third row, and only that row, has all the source's algebraic rank-one actions doubly transitive.

The corresponding geometric implication under the stronger interpretation can be proved without any assumption of flag transitivity.

Lemma 5. *Suppose a rank-three family in Theorem 3 has the property that the stabilizer of each two-type base flag is doubly transitive on its full incidence panel. Then the geometry is thin and flag transitive.*

Proof. For the base flag $\{G_i, G_j\}$, its stabilizer is G_{ij}. The orbit of the remaining base coset G_k has stabilizer $G_{012} = 1$ and therefore has size $|G_{ij}|$. Transitivity on the full panel forces this orbit to be the entire panel. Consequently the action on it is regular, and double transitivity forces $|G_{ij}| = 2$. Every two-type flag is a translate of the base flag of its type: the two incident cosets share a representative. All panels therefore have exactly two elements, proving thinness. Moreover every chamber can first be translated so that two of its elements are base cosets; transitivity of their stabilizer on the panel then moves its third element to the remaining base coset. This proves flag transitivity. □

For rank two, the algebraic residue agrees with the actual incidence residue. The accepted complete normalizer families give a short independent check of the remaining prime branches of Theorem 6.1. The split-normalizer family with intersection C_p has local actions of degrees two and q; the second is the natural affine action of $E_q \rtimes \mathbb{F}_q^*$ and is doubly transitive. This is one equivalence class. Every other split-normalizer family has intersection C_2, and its action on $p \geq 7$ cosets is not doubly transitive: a point stabilizer of order two cannot be transitive on $p - 1 > 2$ points.

In a nonsplit family the maximal parabolic is $C_t \rtimes C_4$, where $t = a_\epsilon(q)$ is prime, and the intersection is C_4. Its action on t cosets is doubly transitive exactly when $t = 5$. Indeed the point stabilizer has order four, acts without fixed nonzero vectors in the cyclic kernel, and is transitive there exactly when $t - 1 = 4$. The equality $a_\epsilon(q) = 5$ occurs only for $q = 8$ and the minus sign: it holds there, whereas $q \geq 32$ gives $a_\epsilon(q) \geq q/2 + 1 \geq 17$. At $q = 8$ the other parabolic must be another $C_5 \rtimes C_4$ or $C_4 \times C_2$. Both local actions are doubly transitive (degree five or degree two). Their accepted class counts are each $(q-2)/(2n) = 1$. The opposite-torus family has an action of degree thirteen and fails the same test. Hence there is exactly one rank-two class when $q > 8$, and exactly three when $q = 8$, in the source's specified-group equivalence convention.

The source's thinness conclusion. Under the literal algebraic definition, the source further calls all these rank-three locally doubly transitive cases thin. The two cosets in an algebraic residue need not a priori exhaust the third-type elements incident with a pair of cosets in the full incidence structure. Lemma 5 proves thinness if local double transitivity is required on the full panel; it does not identify the two hypotheses. The retained diagnostic in the next section checks one Sz(8) marking and finds exactly two elements in each panel. It

does not distinguish the two hypotheses in that example and is not a counterexample. The literal source's general thinness conclusion remains a separate obligation.

5 One retained marking over the field of eight elements

The sole finite acquisition in this component revisits one previously constructed marking. It searches no additional marking and enumerates no ambient Suzuki group. Put

$$\mathbb{F}_8 = \mathbb{F}_2[t]/(t^3 + t + 1), \quad a = w, \quad b = z(t)wz(t), \quad c = z(1),$$
$$G_0 = \langle a, b\rangle, \qquad G_1 = \langle a, c\rangle, \qquad G_2 = \langle b, c\rangle,$$

where w is the antidiagonal matrix and $z(b)$ is the matrix in Lemma 2, now with $\theta = 4$. The returned certificate gives the complete element lists of the three subgroups, of orders $14, 10, 10$. An independent reader implements multiplication in this displayed field, verifies every subgroup product and inverse, and checks that the indicated generators produce the lists. Their pair intersections have order two and their triple intersection is trivial. The first subgroup is the full split torus normalizer in Sz(8) by the classical subgroup description; each of its two distinct involutions inverts its cyclic subgroup of order seven. The three parabolics therefore satisfy the source's subgroup condition: G_0 is maximal in G, each rank-one intersection is maximal in either incident prime dihedral parabolic, and the trivial group is maximal in every order-two intersection. This statement uses the retained classical Suzuki model and maximal-subgroup classification, not a new ambient census.

Here is the complete test of actual panels. Fix the base flag $\{G_j, G_k\}$ and let i be the missing type. A left coset of G_i incident with G_j has a representative in G_j, because incidence means nonempty intersection. Form the distinct sets gG_i for all $g \in G_j$, and retain exactly those meeting G_k. This gives the entire panel, with no assumption of flag transitivity. Such a coset belongs to the base chamber's $G_j \cap G_k$ orbit precisely when it meets $G_j \cap G_k$. The actual returned values are

missing type	candidate cosets	actual panel	cosets in the standard orbit
0	5	2	2
1	7	2	2
2	7	2	2.

The certificate contains every coset as a literal matrix set and all its relevant intersections. The independent reader reconstructs the entire displayed test and compares these sets, not just their counts. Six damaged certificates are rejected; optimization modes are prohibited because the historical acquisition code used assertions.

Every two-type flag is a translate of a base flag. It follows that the incidence geometry is thin and that the stabilizer acts transitively on each full panel. The argument of Lemma 5 then gives flag transitivity; its order-two panel actions are doubly transitive. Rank three is a geometry because every edge has a common representative and can be completed using that representative for the third type. The residues of rank two are connected: their two order-two stabilizers generate the displayed dihedral parabolic. The whole geometry is connected because G_0 is maximal and $G_1 \not\leq G_0$, so $\langle G_0, G_1\rangle = G$. This one marking consequently satisfies both readings of local double transitivity. It yields no source correction and proves no assertion about all markings.

6 Attribution and scope

The target is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* Sz(q), Note di Matematica 19(1) (1999), 43–63: Theorem 5.3, its prime-field rank bound, and the rank-two and algebraic local-action parts of the prime branches of Theorem 6.1. The structure of

Suzuki groups, their faithful matrices and ovoid action remain classical inputs. The accepted L115 preliminary and normalizer-family notes supply the affine complement argument, Frattini identity and complete rank-two families with their equivalence counts. Our derivation combines these through strict intersections and prime subgroup covers; the matrix entry in Lemma 2 replaces one implicit geometric step. No novelty claim is made for the elementary lemmas. The one bounded computation supplies actual panels for an already retained marking. No new conjecture credit, whole-source credit, Lean verification or compression measurement is asserted.

Realizations and obstructions for the displayed patterns

Human Compression project

Abstract

We assemble the previously reviewed proofs concerning the individual patterns in the rank-three Suzuki article. Each of its three prime-case patterns has realizations with the prescribed intersections and RWPRI property. Among its ten composite-case patterns, four have impossible intersections, five have RWPRI realizations, and one has the displayed intersections but fails RWPRI. These conclusions concern the listed patterns; the source's claim that its list is exhaustive is false and is not a premise of the arguments below.

1 Coordinates and the two rigidity arguments

Throughout $q = 2^n$ with n odd. We use the classical Suzuki matrix model, its unique Bruhat coordinates, the usual subfield conjugacy classes, involution centralizers, cyclic torus centralizers and their normalizers. Prime relative degree makes an inclusion of simple Suzuki subfield groups maximal. These are the attributed structural premises of the original reviewed arguments. The source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63.

Write $S_d = \mathrm{Sz}(2^d)$ for the standard subgroup obtained by restricting matrix parameters to $\mathbb{F}_{2^d}$. This is initially a field-exponent index. Unique coordinates give

$$S_a \cap S_b = S_{\gcd(a,b)}.$$

In particular $S_1 = \mathrm{Sz}(2) \cong C_5 \rtimes C_4$ has order 20; it is not a simple Suzuki group. Let H_m be the unique subgroup of order m in the standard split torus, let w invert that torus, and put $D_d = H_{2^d-1} \rtimes \langle w \rangle$ for $d > 1$ dividing n. The split-normalizer uniqueness argument, also used in the accepted normalizer and subfield notes, gives

$$N_G(D_d) = D_d,$$

and exactly one standard-size subfield group of any prescribed degree containing d contains this embedded D_d.

For clarity, the argument for this uniqueness is as follows. A normalizer of D_d normalizes its characteristic odd subgroup and hence lies in the full split normalizer. A torus element h normalizes D_d exactly when $h^2 \in H_{2^d-1}$. The quotient torus has odd order, so this is equivalent to $h \in H_{2^d-1}$. Within a subfield group, every copy of D_d is conjugate to the standard one: its odd rotation group lies in a split torus, and squaring in that odd torus aligns the reflection. Pulling an inclusion back into the standard subfield, then making this internal conjugation, leaves a conjugator in $N_G(D_d)$. It therefore does not change the subfield group. This proves the asserted uniqueness.

Lemma 1. *Suppose conjugates S, T of S_a, S_b have full split dihedral subgroups lying in a common split dihedral subgroup of G, and these two dihedrals share an involution. Then S, T are simultaneously conjugate to S_a, S_b. In particular $S \cap T \cong \mathrm{Sz}(2^{\gcd(a,b)})$.*

Proof. Conjugate the common ambient split normalizer to the standard one, and then conjugate its shared reflection to w using the torus. The two rotation subgroups are the unique cyclic subgroups of their orders. The preceding uniqueness argument now forces both subfield groups to be standard. □

The involution version is equally useful. Fix $z = u(0,1)$ in the standard centre Z_q of a Sylow two-subgroup U_q. Every conjugate of S_d containing z is S_d^u for some $u \in C_G(z) = U_q$, because involutions form one class inside S_d. Its involutions commuting with z, together with the identity, are precisely Z_{2^d}: conjugation by U_q fixes the entire Z_q pointwise. Thus if conjugates of S_a, S_b share z, their common involutions commuting with z, together with the identity, form $Z_{2^{\gcd(a,b)}}$. If a, b are coprime, their intersection cannot contain a Klein four group.

2 The four impossible composite patterns

In Theorem 5.5, patterns 1, 2, 3 require distinct odd prime degrees a, b, subgroups $S \cong \mathrm{Sz}(2^a)$, $T \cong \mathrm{Sz}(2^b)$, and a split dihedral group containing their full split normalizers with a common reflection. They prescribe $S \cap T$ to have type D_{10}, C_2^2, or C_4. Lemma 1 instead forces $S \cap T = \mathrm{Sz}(2)$, of order 20. Hence all three displayed intersections are impossible. This argument does not rely on either Mersenne number being prime.

Pattern 5 requires the same coprime field degrees and an intersection $C_4 \times C_2$. That group contains a Klein four group, contradicting the preceding involution argument. This also independently excludes pattern 2. These are obstructions for every permitted relative position, not merely failures of a preferred coordinate construction.

3 The three patterns when $q - 1$ is prime

Here $q - 1$ is prime and $q \geq 8$. Put $z(x) = u(0, x)$, and $\theta = 2^{(n+1)/2}$. We use the classical facts that every nonidentity odd-order element belongs to a cyclic torus of order $q - 1, \alpha_q = q + \sqrt{2q} + 1$, or $\beta_q = q - \sqrt{2q} + 1$, with normalizer quotients of orders two, four, four. The nonidentity elements in a Sylow two-subgroup have trace zero.

Lemma 2. *The $q - 1$ matrices $wz(x)$, $x \neq 0$, represent all nonidentity odd-order conjugacy classes, one each.*

Proof. Their traces are the distinct nonzero values x^θ. An element of even order with nontrivial odd part would centralize that odd part, whose centralizer is an odd cyclic torus; this is impossible. Thus an even-order element is a two-element, and its trace is zero. Nonzero trace therefore forces odd order. The displayed matrices belong to distinct odd-order classes. Torus centralizers and normalizers give the total number of these classes as

$$\frac{q-2}{2} + \frac{\alpha_q - 1}{4} + \frac{\beta_q - 1}{4} = q - 1.$$

Indeed the normalizer quotients act freely on nonidentity torus elements, and conjugacy inside a torus is controlled by its normalizer. This proves exhaustion. In particular every prime dividing $(q-1)(q^2+1)$ occurs as $|wz(x)|$, and $(q-2)/2$ choices give order $q - 1$. □

For the first source pattern, choose involutions a, b with $|ab| = q - 1$. Their unique ovoid fixed points P, Q are distinct; otherwise they would lie in the same elementary abelian involution centre and commute. Let T be the pointwise stabilizer of P, Q and let Z_P, Z_Q be the centres at those points. Set

$$G_0 = \langle a, b \rangle, \qquad G_1 = Z_P T, \qquad G_2 = Z_Q T.$$

Then $G_1 \cap G_2 = T$. No nonidentity rotation of G_0 fixes P: its two fixed points are exchanged by a, whereas a fixes P. Two reflections fixing P would commute, impossible in an odd dihedral group. Hence $G_0 \cap G_1 = \langle a \rangle$; similarly $G_0 \cap G_2 = \langle b \rangle$, and the triple intersection is trivial.

For the second pattern, choose x with $|wz(x)| = p$, where p is the prescribed odd prime, and take

$$G_0 = H_{q-1} \rtimes \langle w \rangle, \qquad G_1 = Z_q H_{q-1}, \qquad G_2 = \langle w, z(x) \rangle.$$

The intersections are $H_{q-1}, \langle w \rangle, \langle z(x) \rangle$, and the triple is trivial. To see that the last two intersections are no larger, a subgroup of D_{2p} containing a reflection is either that subgroup of order two or the whole dihedral group. The latter would put $z(x)$ in the split normalizer or w in the point stabilizer, respectively, both impossible.

For the third pattern, the parabolics are generated by the pairs of three involutions. The two prescribed half-orders cannot both be two, by the necessary patterns proved in the preceding note. If one is two, let the other p' be odd, choose t, x with $|wz(t)| = q - 1$, $|wz(x)| = p'$ and $x \neq t$, and use $w, z(t), z(x)$. If $p' = q - 1$ there are at least three available parameters by Lemma 2. The orders are $2(q-1), 4, 2p'$. The two central involutions commute; an odd dihedral group cannot contain two distinct commuting involutions. Consequently all pair intersections are the indicated groups of order two, and the triple intersection is trivial.

For distinct odd p, p', choose x, y with $|wz(x)| = p$, $|wz(y)| = p'$. Choose λ satisfying $\lambda^{\theta+1} x^\theta = y^\theta$ and put

$$a = w, \qquad b = h(\lambda) w, \qquad c = z(x).$$

The exponent $\theta + 1$ is coprime to $q - 1$, so λ exists uniquely, and $\lambda \neq 1$. Thus $ab = h(\lambda)^{-1}$ has order $q - 1$. The traces give $|ac| = p$, $|bc| = p'$ by Lemma 2. An intersection properly containing a shared reflection would identify the corresponding prime dihedral groups, which is impossible from their different orders or would put c in the split normalizer. Thus the required intersections again hold.

For equal odd $p = p'$, choose distinct t, x with $|wz(t)| = q - 1$, $|wz(x)| = p$ and take

$$a = w, \qquad b = z(t) w z(t), \qquad c = z(x).$$

Now $ab = (wz(t))^2$ has order $q - 1$, and bc is conjugate to ac because $z(t)$ commutes with c. Also $\langle a, b \rangle = \langle w, z(t) \rangle$, while c does not belong to that dihedral group, since it is distinct from and commutes with $z(t)$. The same maximal-reflection argument gives all intersections, including $p = 5$.

In each pattern the full split normalizer is maximal in G. The reflection intersections have prime index in each odd dihedral parabolic. The torus complement is maximal in $Z_q H_{q-1}$ by its irreducible scalar action. Finally all nontrivial pair intersections over the trivial triple have prime order. These facts verify every subset in the source RWPRI axiom. We have proved existence of the displayed subgroup patterns, without inferring thinness or flag transitivity for all their possible embeddings.

4 The realizable composite patterns

For this section only, S_s denotes the standard group $\mathrm{Sz}(s)$ with a field-size index. Patterns 4 and 6 are the following one family. Let $q = s^r$, $s \geq 8$, r an odd prime, choose a prime $\ell \mid (q-1)/(s-1)$ and put $m = (s-1)\ell$. Take

$$G_0 = S_s, \qquad G_1 = H_m \rtimes \langle w \rangle, \qquad G_2 = Z_q H_m.$$

The intersections are respectively $D_{2(s-1)}, Z_s H_{s-1}, H_m$, with full intersection H_{s-1}. Unique coordinates prove these equalities. The subfield and its full split normalizer give maximal

inclusions, and the cyclic extension indices are ℓ. The complement H_m is maximal in $Z_q H_m$: its scalar field contains $\mathbb{F}_s$, cannot equal $\mathbb{F}_s$ because $m > s-1$, and has degree dividing the prime r. Thus it is $\mathbb{F}_q$, and the scalar action is irreducible. These facts verify all RWPRI subsets. The torus action on Z_q uses a power coprime to $q-1$, which does not change the generated scalar field.

For patterns 7, 8, 9 put $q = t^{ab}$, $s = t^a$, $s' = t^b$, where $t \geq 8$ and a, b are distinct odd primes. For pattern 7 choose distinct primes $p \mid (s-1)/(t-1)$, $p' \mid (s'-1)/(t-1)$ and put $m = (t-1)pp'$. Use $S_s, S_{s'}$ and $D = H_m \rtimes \langle w \rangle$. The cyclic gcd identities are

$$\gcd(m, s-1) = (t-1)p, \qquad \gcd(m, s'-1) = (t-1)p'.$$

They hold also when a selected prime divides $t-1$: its additional valuation occurs in exactly its selected relative prime degree. Equivalently, these are the two-index instances of the scalar-field arithmetic lemma in the separately reviewed split-family note. The pair intersections are $S_t, D_{2(t-1)p}, D_{2(t-1)p'}$, and the full intersection is $D_{2(t-1)}$. It is maximal in S_t, and has prime index in each larger dihedral; all subfield inclusions have prime relative degree. Hence RWPRI holds.

For pattern 8 choose a full nonsplit torus A_t of S_t, of order $m = t \pm \sqrt{2t} + 1$, and an element r of order four in its normalizer. Its ambient centralizer is one cyclic nonsplit torus A_q. Put $A_s = C_{S_s}(A_t)$ and $A_{s'} = C_{S_{s'}}(A_t)$. Their intersection is A_t, so their orders have gcd m. Choose primes p, p' with $mp \mid |A_s|$, $mp' \mid |A_{s'}|$. They are distinct, and the corresponding subgroup $C_{mpp'}$ exists in A_q. Then

$$K = \langle C_{mpp'}, r \rangle, \qquad K \cap S_s = C_{mp} \rtimes \langle r \rangle, \qquad K \cap S_{s'} = C_{mp'} \rtimes \langle r \rangle,$$

with full intersection $A_t \rtimes \langle r \rangle$. The same prime-index and maximal-normalizer checks prove RWPRI. The compatible nonsplit torus sign is determined by A_t, rather than chosen independently in the two subfields.

For pattern 10 return to exponent indices. Choose distinct prime divisors a, b, c of n, with $n/(abc) > 1$ as in the printed parameter condition. The standard subfields $S_{n/a}, S_{n/b}, S_{n/c}$ have exactly the displayed pair and triple intersections by field gcds. All one-step inclusions have prime relative degree between simple subfield groups, so every RWPRI condition holds.

5 The mixed pattern: intersections without maximality

Pattern 9 prescribes $S_s, S_{s'}$ with common S_t and a third group of type $E_q \hat{:} E_t \rtimes C_{t-1}$. Its intersections with the two simple groups have types $E_s \hat{:} E_t \rtimes C_{t-1}$ and $E_{s'} \hat{:} E_t \rtimes C_{t-1}$; the full intersection is the point stabilizer $B_t = U_t H_{t-1}$. Here the first elementary abelian factor has the literal full order q, as printed; it is not an unspecified smaller subspace.

Conjugate the common S_t to standard position. It contains its full split normalizer, so the uniqueness argument forces the two containing simple groups to be the standard $S_s, S_{s'}$. An internal conjugation in S_t puts their common Borel at B_t. The third parabolic has a normal Sylow two-subgroup P of order qt containing an elementary abelian group of order q. In an ambient Sylow two-subgroup every involution is central, so this elementary abelian group is its full centre. It consists of the identity and all involutions of P, hence is characteristic in P and normal in the third parabolic. Its unique fixed ovoid point is therefore preserved. Containment of B_t forces this point to be the standard point at infinity. Thus the normal elementary abelian group is Z_q, and containment and orders force the third parabolic and its intersections to be

$$Z_q U_t H_{t-1}, \qquad Z_s U_t H_{t-1}, \qquad Z_{s'} U_t H_{t-1}.$$

Conversely these standard groups have all the printed intersections, so the bare subgroup pattern exists.

For any $\mathbb{F}_t$-subspace V with $\mathbb{F}_s < V < \mathbb{F}_q$, the group $Z_V U_t H_{t-1}$ lies strictly between the first intersection and the third parabolic. Such a subspace exists: adjoining one vector to $\mathbb{F}_s$ gives dimension $a + 1 < ab$. The same argument gives an intermediate group above the second intersection, using $b + 1 < ab$. Hence neither remaining intersection is maximal in this parabolic. The RWPRI condition for its singleton label fails. The forced coordinates show that this is an obstruction for every realization of the printed type, not just the one displayed here.

6 What these arguments settle

The original final paragraph asks about actual existence of the displayed intersections. The retained results are

source patterns	intersections exist	RWPRI realization
Theorem 5.3, all three	yes	yes
Theorem 5.5, $1, 2, 3, 5$	no	no
Theorem 5.5, $4, 6, 7, 8, 10$	yes	yes
Theorem 5.5, 9	yes	no

These are the previously reviewed phase-five arguments, reassembled with the now accepted uniqueness and scalar-field interfaces made explicit. Their old sufficient-pattern proofs did not establish the source's completeness claim or Theorem 5.4; the present source treatment uses separate reviewed proofs and counterexamples for those assertions. No classification of all relative positions or all flag-transitive geometries is inferred from this table. The saved finite examples support the constructions but are not substituted for the uniform proofs, and their acquisitions are not repeated here.

Compatible tori and a missing Suzuki geometry

Human Compression project

Abstract

A common binary matrix algebra places two nonsplit tori in compatible subfield groups. The resulting rank-three geometry has maximal parabolics Sz(8), Sz(32) and $C_{533} : C_4$, and is absent from the stated exhaustive classification. We rework the previously reviewed explicit example, replacing the finite intersection census by a coprime-order argument. A separate corollary of the scalar-subspace construction contradicts the same source's rank-two uniqueness assertion. Neither argument gives a corrected complete classification.

1 Statement and retained inputs

For subgroups G_i of G, the elements of the coset pregeometry are the labelled left cosets gG_i; incidence is nonempty intersection. Write $G_J = \bigcap_{j\in J} G_j$ and $G_\varnothing = G$. The definition on printed page 46 of [1] calls this system residually weakly primitive if for every proper subset J of the type set some unused i makes $G_{J\cup\{i\}}$ a maximal proper subgroup of G_J. We use exactly this definition.

Theorem 1. *In $G = \mathrm{Sz}(32768)$ there are standard subfield subgroups $S = \mathrm{Sz}(8)$, $T = \mathrm{Sz}(32)$ and a subgroup $K = C_{533} : C_4$ with*

$$S \cap T = C_5 : C_4, \quad S \cap K = C_{13} : C_4, \quad T \cap K = C_{41} : C_4, \quad S \cap T \cap K = C_4. \tag{1}$$

Their rank-three coset geometry is residually weakly primitive, thick, flag transitive and residually connected. Its intersection lattice is absent from the applicable cases of Theorem 5.5 of [1].

The standard faithful Suzuki matrices are denoted by $U(a, b)$, $h(t)$ and w, where w is the antidiagonal identity matrix. Their formulas, Bruhat forms and exact coefficient-field intersections are given in [2, Lemma 1]. In particular, for standard embeddings, $\mathrm{Sz}(2^a) \cap \mathrm{Sz}(2^b) = \mathrm{Sz}(2^{\gcd(a,b)})$, including the degree-one group $\mathrm{Sz}(2) = C_5 : C_4$. We retain the classical Suzuki order formula

$$|\mathrm{Sz}(q)| = q^2(q^2 + 1)(q - 1), \tag{2}$$

maximality of simple subfield groups of prime relative field degree, and maximality of the full normalizers of nonsplit cyclic tori of orders $q \pm \sqrt{2q} + 1$. Such a normalizer has order four times the torus order. These structural facts are among the inputs used by [1, Section 3]; completeness of its geometry lists is not used here. Our common proof [2, Lemma 1], rather than a new subfield-intersection calculation, is an explicit prerequisite.

2 One algebra, two compatible tori

All displayed integer entries in this section encode binary polynomial coefficients in

$$F = \mathbb{F}_2[z]/(z^{15} + z + 1).$$

Thus, for example, 5682 denotes $\sum_{i=0}^{14} \epsilon_i z^i$ with binary digits ϵ_i of 5682; it is not an integer reduced modulo 32768. The polynomial is irreducible. The supplied arithmetic control also checks this directly: successive multiplication by z visits all 32767 nonzero polynomial classes before returning to one.

Put

$$M = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix} = U(1,0).$$

Direct multiplication gives

$$M^4 + M + I = 0, \quad |M| = 15, \quad |r| = 4, \qquad r^{-1}Mr = M^2. \tag{3}$$

The polynomial $X^4 + X + 1$ has no binary root and is not divisible by $X^2 + X + 1$, the only irreducible quadratic over $\mathbb{F}_2$. Consequently $\mathbb{F}_2[M] \cong \mathbb{F}_{16}$. Its scalar extension to F is a field because $\gcd(4, 15) = 1$.

Take $\lambda_8 = 5682$ and $\lambda_{32} = 624$ in the stated encoding, and set

$$a = (M + \lambda_8 I)^{315}, \qquad b = (M + \lambda_{32} I)^{25575}. \tag{4}$$

Here $315 = (8^4 - 1)/13$ and $25575 = (32^4 - 1)/41$. The following table gives their full open Bruhat parameters:

g	u	v	t	x	y	$\lvert g\rvert$
a	893	0	5682	893	893	13
b	5992	4671	1131	5992	1831	41

$$g = U(u, v)h(t)wU(x, y). \tag{5}$$

Equations (3)–(5), the scalar identities $\lambda_8^8 = \lambda_8$, $\lambda_{32}^{32} = \lambda_{32}$, and the indicated orders are finite polynomial identities in F. They are checked by the short witness program supplied with this note, using repeated squaring and the displayed modulus. Each entry of the first parameter row is fixed by eighth powering; each entry of the second is fixed by thirty-second powering. Thus $a \in S$ and $b \in T$. The table is also a direct membership certificate, independently of any enumeration of a Suzuki group.

Since a and b are polynomials in M, they commute. Their distinct prime orders give $|ab| = 13 \cdot 41 = 533$ and $\langle ab\rangle = \langle a, b\rangle$. With $c = ab$, the remaining matrix identity is

$$r^{-1}cr = c^{278}, \qquad 278^2 \equiv -1 \pmod{533}. \tag{6}$$

It follows that $K = \langle c, r\rangle = C_{533} : C_4$: the two cyclic factors have coprime orders, and (6) is an automorphism of order four. Every element has a unique form $a^i b^j r^k$, with $0 \le i < 13$, $0 \le j < 41$ and $0 \le k < 4$. In particular $|K| = 2132$. This K is not the full normalizer of C_{533} in the ambient group; no such maximality is asserted.

Lemma 2. *The intersections are exactly those in (1).*

Proof. The standard-subfield result gives $S \cap T = \mathrm{Sz}(2)$ of order twenty. Both a and r belong to S. If $a^i b^j r^k \in S$, multiplication on the left by a^{-i} and on the right by r^{-k} gives $b^j \in S$. But 41 does not divide $|S| = 29120$. Since b has order 41, we must have $j = 0$. Hence $K \cap S = \langle a, r\rangle$, of order 52. Similarly 13 does not divide $|T| = 32537600$, so $K \cap T = \langle b, r\rangle$, of order 164. Their common intersection contains $\langle r\rangle$ and has order dividing $\gcd(52, 164) = 4$, proving the final equality. The groups of orders 52 and 164 are the full nonsplit torus normalizers in S and T: (6) normalizes their cyclic factors, and the retained normalizer order is respectively $4 \cdot 13$ and $4 \cdot 41$. □

3 The geometry and the source contradiction

We verify all seven instances of residual weak primitivity. The subgroup S is maximal in G because the relative field degree is $15/3 = 5$. Inside S and T, respectively, choose the maximal torus normalizers $S \cap K$ and $T \cap K$. Inside K, choose $K \cap S$, of prime index 41. At each pairwise intersection the final subgroup $\langle r \rangle$ has prime index, respectively 5, 13 and 41. These are all proper type subsets. All eight intersections are distinct, by their orders. This verifies the exact subgroup condition of the source.

For clarity, the geometry properties can be proved without its general rank-three assertion. We use the following elementary observation. If A, B, C are subgroups and

$$A = (A \cap B)(A \cap C), \tag{7}$$

every triple flag of their left-coset system has a common point. Translate the A-coset to A, and write the other two cosets as bB, cC with $b, c \in A$. Factor $b^{-1}c = xy$ with $x \in A \cap B$, $y \in A \cap C$. Then $bx = cy^{-1}$ lies in all three cosets. Every flag of fewer types also extends to a chamber: an incident pair has a common point, and that point supplies the missing coset. Thus (7) gives a geometry and transitivity on every type of flag.

In our example the product set $(K \cap S)(K \cap T)$ has cardinality

$$\frac{52 \cdot 164}{4} = 2132 = |K|.$$

It is contained in K, so equals K. No assumption that an arbitrary subgroup product is a subgroup is needed. Apply (7) with $A = K$. The three panel sizes are 5, 13 and 41, so the geometry is thick.

For residual connectivity, the two residue subgroups in S generate S: one is its maximal torus normalizer of order 52, while the other, $S \cap T$, contains an element of order five and is not contained in that normalizer. The same argument, using order 164, works in T. The product formula gives generation in K. Finally $\langle S, T \rangle = G$, because S is maximal and 31 divides $|T|$ but not $|S|$. The usual coset-graph argument now proves connectedness of the whole incidence graph and of every rank-two residue: a component containing the base flag consists exactly of the cosets reached by words in its incident stabilizer subgroups. Those subgroups generate the relevant group, as just shown. Flag transitivity transports this statement to all residues.

It remains to compare the actual source conclusion. Here the odd field degree is $15 = 3 \cdot 5$, whose prime exponents sum to two. Moreover the subfield $S = \mathrm{Sz}(8)$ has $8 - 1$ prime. The applicable portion of [1, Theorem 5.5, printed pp.58–59] is therefore its first five shapes. Only shape 5 has two Suzuki maximal parabolics and one odd cyclic group extended by C_4. It prescribes the intersection of the two Suzuki parabolics as $C_4 \times C_2$, of order eight; our intersection has order twenty. A permutation of types cannot change this discrepancy, since K is soluble and S, T are not. Shape 8 belongs to the separate branch where the sum of prime exponents exceeds two, and cannot be used in this branch. This proves Theorem 1.

The source's closing qualification on page 61 says that its lists are necessary subgroup lattices, without establishing realizability of every listed shape. Our example contradicts necessity, so that qualification does not remove the counterexample. We do not claim the geometric intersection property. Nor is this rank-three example locally doubly transitive: in the local action on a panel of size thirteen, the stabilizer of one point has order four and cannot be transitive on the other twelve points.

4 A separate rank-two consequence

The preceding example should be distinguished from the locally doubly transitive geometries in [3].

Corollary 3. *The uniqueness assertion for composite $q-1$ in [1, Theorem 6.1, printed p.61] is false at $q = 32768$.*

Proof. Here $q-1 = 32767 = 7 \cdot 31 \cdot 151$ is not prime. The source asserts that the unique residually weakly primitive and locally doubly transitive pregeometry has rank two and parabolics $D_{2(q-1)}$ and $q : (q-1)$, with common subgroup of order $q-1$. Its two valencies are 2 and q.

Apply [3, Theorem 3] to $F = \mathbb{F}_{32768}$ and $K = \mathbb{F}_8$. For each two-dimensional K-subspace T containing K, it gives a rank-two geometry with parabolics $\mathrm{Sz}(8)$ and Q_T, where $|Q_T| = 3584$ and their intersection has order 448. It is residually weakly primitive and locally doubly transitive, with valencies 65 and 8. These differ from 2 and 32768, even after interchanging types. Thus even the underlying incidence geometry is absent from the uniqueness assertion. The forty-one classes counted in [3, Corollary 5] use that paper's stated equivalence under automorphisms of the specified group; a count of forty-one bare incidence isomorphism classes is not needed or asserted here. □

5 Attribution and verification

The binary algebra, fixed scalars, matrices and rank-three construction were obtained in the project's earlier manuscript on compatible Suzuki tori, and were separately reviewed there. We retain that proof, its certificate, original program and review as provenance. The present note reuses the exact witness; its intersection proof uses group orders in place of the earlier enumeration of 2132 elements. The short new replay reads the fixed witness, verifies (3)–(6) and the arithmetic used in Lemma 2, and never searches for M or enumerates an ambient group. Classical Suzuki structure is retained explicitly.

The exact subfield-intersection lemma and the scalar-subspace theorem are acquired from [2, 3]; their complete proofs and reviews accompany this note. The rank-two corollary is a direct use of the latter theorem, not a separate discovery. The earlier higher-rank extension is not a premise. No other source cases or future realization questions are declared settled, and no complete corrected classification, publication priority, new conjecture solution or full Lean proof is claimed.

References

[1] D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63.

[2] Human Compression project, *A missing boundary in the Suzuki subfield construction*, reviewed project manuscript, 2026, Lemma 1.

[3] Human Compression project, *Scalar subspaces in rank-two Suzuki geometries*, reviewed project manuscript, 2026, Theorem 3 and Corollary 5.

An algebraic local action does not force thinness

Human Compression project

Abstract

We give three explicit involutions in Sz(8) whose parabolics have orders $14, 10, 26$. The resulting rank-three coset geometry satisfies RWPRI and the algebraic definition of local 2-transitivity used in the fixed source. Its actual panels have three elements. A short matrix identity exhibits a third chamber, and a separate literal certificate checks every panel. This refutes the $q = 8$ thinness clause of the source's Theorem 6.1; it makes no assertion about the separate $q > 8$ clause.

1 A small matrix certificate

Work in $F = \mathbb{F}_2[X]/(X^3 + X + 1)$. Integers from zero to seven encode coefficients in the basis $1, X, X^2$, so $2 = X$, $6 = X^2 + X$, and addition is exclusive-or. Put

$$a = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 6 & 2 & 0 & 1 \\ 2 & 0 & 1 & 0 \\ 7 & 5 & 0 & 2 \\ 7 & 7 & 2 & 6 \end{pmatrix}, \quad c = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

These are elements of the classical Suzuki matrix group. To make their membership explicit, use $w = a$, $z(t) = u(0, t)$, and

$$u(s,t) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ s & 1 & 0 & 0 \\ s^5 + t & s^4 & 1 & 0 \\ s^6 + st + t^4 & t & s & 1 \end{pmatrix}.$$

Then $b = z(2)wz(2)$ and $c = u(1,1)wz(1)wu(1,1)^{-1}$. The classical group model and the maximality of D_{14} in Sz(8) are explicit structural premises. All the calculations below use ordinary multiplication of these matrices.

The three matrices are distinct nonidentity involutions. Direct multiplication gives

$$|ab| = 7, \qquad |ac| = 5, \qquad |bc| = 13, \qquad (ac)^2(ab)^3 a = (bc)^6. \tag{1}$$

For the order statements it is enough to check the indicated prime power is the identity and the product itself is not. An independent reader performs these checks by polynomial multiplication modulo $X^3 + X + 1$. Thus (1) is a small certificate that can be checked without constructing the ambient group.

2 The geometry and the extra chamber

Let $G = \mathrm{Sz}(8)$ and put

$$G_0 = \langle a, b \rangle, \qquad G_1 = \langle a, c \rangle, \qquad G_2 = \langle b, c \rangle.$$

We use left cosets, in agreement with the source's convention. Two involutions whose product has odd prime order p generate D_{2p}. Consequently the orders of our parabolics are $14, 10, 26$. The gcd of any two orders is two; since the corresponding parabolics share their indicated involution, we obtain

$$G_0 \cap G_1 = \langle a \rangle, \quad G_0 \cap G_2 = \langle b \rangle, \quad G_1 \cap G_2 = \langle c \rangle, \quad G_0 \cap G_1 \cap G_2 = 1. \tag{2}$$

The last equality follows from $a \neq b$.

Theorem 1. *The coset geometry $\Gamma(G; G_0, G_1, G_2)$ is RWPRI and locally 2-transitive under the literal algebraic-residue convention of L114. It is not thin and is not flag transitive.*

Proof. Two incident cosets have a common representative, which extends the pair to a chamber by its coset of the missing type. The same observation extends smaller flags. Hence this rank-three coset pregeometry is a geometry.

For the empty flag, $G_0 = D_{14}$ is maximal in G. Within each parabolic, each intersection subgroup in (2) has prime index, respectively seven, five or thirteen, and hence is maximal. The trivial triple intersection is maximal in each order-two pair intersection. These exhaust the subsets required by the source's RWPRI definition. Its algebraic rank-one residues are precisely the regular actions $C_2/1$ on two points, so are 2-transitive.

Consider the base edge (G_1, G_2). The cosets G_0 and cG_0 give its two standard extensions. There is also the coset xG_0, where $x = (ac)^2$. It meets G_1 at x, and (1) shows that it meets G_2 at $(bc)^6$. It is distinct from G_0: an element of order five cannot belong to $G_0 \cap G_1 = \langle a \rangle$.

It is distinct from cG_0 as well. If they were equal, then $cx \in G_0 \cap G_1 = \langle a \rangle$. The equation $cx = 1$ is impossible since x has order five and c has order two. The equation $cx = a$ would give $(ac)^2 = ca = (ac)^{-1}$, contradicting $|ac| = 5$. There are therefore at least three chambers over this edge, disproving thinness. The extra chamber has no common representative: such a representative would lie in $G_1 \cap G_2 = \{1, c\}$, which would identify its third coset with one of the two standard cosets. A translate of the base chamber always has a common representative. Thus the extra chamber also disproves flag transitivity. □

3 Complete panels and exact source scope

The retained finite data contain the full lists of the three small parabolics. The reader verifies every group product, inverse and generated element, independently of the acquisition code. For missing type i, every coset incident with G_j has a representative in G_j. Listing the distinct gG_i, $g \in G_j$, and retaining those meeting G_k therefore checks the entire actual panel. In this example all three panels have exactly three elements, with representatives

missing type	standard representatives	extra representative
0	$1, c$	$(ac)^2$
1	$1, b$	$(ab)^3$
2	$1, a$	$(ba)^3$

Each retained coset comes with its full intersections with both base parabolics and with their intersection. Only the two standard cosets meet the latter. The short proof above needs only one extra coset; the complete panel calculation supplies the exact sizes.

The source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63. Printed p. 45 defines the residue at a subflag of the distinguished chamber from intersections of parabolics. Printed p. 46 uses that residue in its local 2-transitivity condition. It separately defines thinness by the number of actual chambers over an edge and explicitly qualifies agreement with earlier definitions by flag transitivity.

The $q = 8$ clause of Theorem 6.1 on p. 61 claims that its rank-three geometries are thin. The theorem above refutes that literal clause. It is not a counterexample to the different assertion obtained by requiring flag transitivity or 2-transitivity on the full actual panels.

The discovery run stopped after the seventy-first prescribed candidate; its unmodified prefix and terminal receipt are retained. The independent reader checks this one witness, not the discovery prefix or a classification of all embeddings. The nine earlier positive markings remain valid. The rank bound and the three rank-two classes for $q = 8$ are independently accepted elsewhere. The separate thinness clause for prime $q - 1 > 7$ remains an obligation, as do the final source joins.

Attribution and credit. The group model, incidence definitions and maximal-subgroup premise are borrowed as stated. The explicit witness, identity (1), source comparison and independent finite reader are the contribution of this component. No priority or later-edition claim is made. This resolves one additional claim in an original whose principal refutation is already registered; it adds no original, principal-refutation, conjecture, Lean or compression count. General source application still requires distinct mathematical review.

A second failure of the thinness assertion

Human Compression project

Abstract

The thinness clause for prime $q-1>7$ is separate from the clause for $q=8$. We settle it by an explicit example in Sz(32), using the same coset argument as for the previously verified Sz(8) example. Its parabolics have orders $62, 62, 82$, and each actual panel has three elements although the corresponding algebraic residue has two.

Work in $F=\mathbb{F}_2[X]/(X^5+X^2+1)$. Integers from zero to31 encode coefficients in the basis $1, X, X^2, X^3, X^4$; field addition is exclusive-or. The independent arithmetic reader checks all31 nonzero polynomial classes have inverses. In the classical Suzuki model put

$$a=\begin{pmatrix}0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{pmatrix},\quad b=\begin{pmatrix}12&3&0&1\\3&0&1&0\\20&4&0&3\\30&20&3&12\end{pmatrix},\quad c=\begin{pmatrix}26&5&0&1\\5&0&1&0\\29&16&0&5\\19&29&5&26\end{pmatrix}.$$

Their membership in the ambient group is explicit. With $w=a$ and $z(t)=u(0,t)$, where

$$u(s,t)=\begin{pmatrix}1&0&0&0\\s&1&0&0\\s^9+t&s^8&1&0\\s^{10}+st+t^8&t&s&1\end{pmatrix},$$

we have $b=z(3)wz(3)$ and $c=u(1,5)wz(1)wu(1,5)^{-1}$. The ambient matrix construction and maximality of the full split normalizer D_{62} in Sz(32) remain explicit classical premises.

Literal polynomial multiplication verifies that a, b, c are distinct nonidentity involutions and that

$$|ab|=31,\qquad |ac|=31,\qquad |bc|=41,\qquad (ac)^{15}(ab)^{15}a=(bc)^{20}. \tag{1}$$

Define $G_0=\langle a,b\rangle$, $G_1=\langle a,c\rangle$ and $G_2=\langle b,c\rangle$. Their respective orders are $62, 62, 82$. Coprimality gives $G_0\cap G_2=\langle b\rangle$ and $G_1\cap G_2=\langle c\rangle$. The subgroup $G_0\cap G_1$ contains $\langle a\rangle$, which is maximal in the prime dihedral group G_0. If the intersection were larger, then $G_0=G_1$, placing b, c in a group of order62 and contradicting $|bc|=41$. Thus $G_0\cap G_1=\langle a\rangle$, and the triple intersection is trivial.

Theorem 1. *The rank-three coset geometry $\Gamma(\mathrm{Sz}(32); G_0, G_1, G_2)$ is RWPRI and satisfies the literal algebraic local 2-transitivity condition of L114. It is not thin and is not flag transitive.*

Proof. An incident pair of left cosets has a common representative and hence extends to a chamber. Thus the pregeometry is a geometry. For the empty subset in the RWPRI definition, G_0 is maximal in $G=\mathrm{Sz}(32)$. Each pair intersection has prime index31 or41 in the corresponding parabolic, and the trivial triple intersection is maximal in each pair intersection. These verify every required subset. Each source algebraic rank-one residue is the regular two-point action $C_2/1$, so is 2-transitive.

The base edge (G_1, G_2) has the standard extensions G_0, cG_0. Put $x=(ac)^{15}$. The coset xG_0 meets G_1 at x and, by (1), meets G_2 at $(bc)^{20}$. It differs from G_0, since x has order31

and $G_0 \cap G_1 = \langle a \rangle$. If $xG_0 = cG_0$, then $cx \in \langle a \rangle$. The alternative $cx = 1$ contradicts the orders of c, x. The alternative $cx = a$ gives $(ac)^{15} = ca = (ac)^{30}$, contradicting $|ac| = 31$. We have found a third chamber, so the geometry is not thin. This chamber has no common representative: such a representative would lie in $G_1 \cap G_2 = \{1, c\}$ and identify its third coset with one of the standard cosets. It therefore cannot be a translate of the base chamber, proving failure of flag transitivity. □

The independent reader checks every product and inverse in the three returned subgroup lists and verifies generation, intersections and all cosets in each panel. It obtains exactly three cosets in each case, with the following representatives:

missing type	standard representatives	extra representative
0	$1, c$	$(ac)^{15}$
1	$1, b$	$(ab)^{15}$
2	$1, a$	$(ba)^{15}$

The reader explicitly matches each displayed representative with its complete returned coset. Thus the table is certified along with (1).

Source scope and attribution. The source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* Sz(q), Note di Matematica 19(1) (1999), 43–63. Its definitions on printed pp. 45–46 use algebraic residues formed from parabolic intersections, and count actual chambers for thinness. The example has $q - 1 = 31 > 7$ prime and therefore refutes the thinness clause in the second branch of Theorem 6.1 on printed p. 61. The previously accepted rank bound and rank-two count in that branch are unaffected. A version requiring flag transitivity or local 2-transitivity on full actual panels has different hypotheses.

The field and split pair were retained from earlier work. The new contribution is this third involution, its short coset identity and the exact source application, reusing the preceding Sz(8) argument. The bounded discovery stopped after156 prescribed candidates. Its unaltered output is retained; the independent reader verifies this witness alone and makes no classification claim. No new original, principal-refutation count, open-conjecture, Lean or compression credit is assigned. Distinct mathematical review is required before adoption.

Scalar subspaces and unbounded algebraic local rank

Human Compression project

Abstract

The algebraic residue convention in the fixed rank-three Suzuki article admits locally doubly transitive, residually weakly primitive pregeometries of arbitrarily large rank. A coordinate subspace supplies the soluble parabolic; deleting its coordinates matches intersections of independently chosen maximal subfields. This refutes the article's separate universal rank assertion. No claim about full incidence residues in higher rank is needed.

1 The assertion and acquired structure

The introduction of D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63, asserts on printed pp. 44–45 that no pregeometry of rank greater than three satisfies both residual weak primitivity and local double transitivity, and that rank three occurs only when $q-1$ is prime. These also follow from the complete list stated in Theorem 6.1. We use the definitions on pp. 45–46: the residue at a flag of the distinguished chamber is formed from the parabolic intersections. The local condition is imposed on these algebraic rank-one residues, not on all elements in an actual incidence panel. This definition applies to every finite type set; it is not restricted to rank three. Flag transitivity is not assumed.

We retain the classical faithful Suzuki coordinates and subgroup theorem. The standard group over a subfield consists of matrices with parameters in that subfield. Its point stabilizer is $B_t = U_t H_t$, of order $t^2(t-1)$. The standard involution centre is $Z_F = \{u(0,b) : b \in F\}$, with addition in its parameter. It centralizes U_t, and H_t acts by the full scalar group $\mathbb{F}_t^*$. Indeed the standard action is $b \mapsto k^{\theta+2}b$ and $\gcd(\theta+2, t-1) = 1$. These identities and the overgroup correspondence below are acquired from the accepted scalar-subspace and subfield-family notes. A simple subfield group is maximal at prime relative field degree. The action of $\mathrm{Sz}(t)$ on $\mathrm{Sz}(t)/B_t$ is its classical doubly transitive action on t^2+1 points.

2 Independent field coordinates

Let $t = 2^e \geq 8$, with e odd. Choose distinct odd primes $a_1, \ldots, a_k$, with $k \geq 2$, and put

$$A = \prod_{i=1}^{k} a_i, \quad F = \mathbb{F}_{t^A}, \quad K = \mathbb{F}_t, \quad F_i = \mathbb{F}_{t^{A/a_i}}.$$

Choose $x_i \in \mathbb{F}_{t^{a_i}} \setminus K$, and set

$$T = K + Kx_1 + \cdots + Kx_k, \qquad T_J = K + \sum_{i \notin J} Kx_i \quad (J \subseteq \{1, \ldots, k\}).$$

Lemma 1. *The vectors $1, x_1, \ldots, x_k$ are linearly independent over K. Moreover*

$$F_J := \bigcap_{j \in J} F_j = \mathbb{F}_{t^{A/\prod_{j\in J} a_j}}, \qquad T \cap F_J = T_J,$$

where the empty intersection means F.

Proof. The field intersection formula follows from the gcd of the degrees. The automorphism $\sigma_i : y \mapsto y^{t^{A/a_i}}$ fixes F_i and all x_j for $j \neq i$. Its restriction to $\mathbb{F}_{t^{a_i}}$ has fixed field K, since a_i is coprime to A/a_i. Hence $\sigma_i(x_i) \neq x_i$. Applying $\sigma_i - 1$ to a relation $c_0 + \sum c_j x_j = 0$, with $c_j \in K$, gives $c_i(\sigma_i(x_i) - x_i) = 0$. Thus every $c_i = 0$, and then $c_0 = 0$. A vector in $T \cap F_i$ must likewise have zero x_i coefficient. Imposing this for all $i \in J$ proves the second equality. □

For a K-subspace $W \supseteq K$ of F, write

$$Q_W = Z_W B_t, \qquad Z_W = \{u(0, b) : b \in W\}.$$

This is a group: U_t centralizes Z_W, and H_t preserves W. Its order is $t|W|(t-1)$, since $Z_W \cap B_t = Z_K$ has order t. Every overgroup of Q_V inside Q_W is Q_D for a K-subspace $V \leq D \leq W$. Indeed every element has form zb, with $z \in Z_W$, $b \in B_t$. A subgroup containing B_t is therefore of this form with an additive subgroup of W containing K. Normalization by H_t makes that subgroup a K-subspace. Consequently

$$Q_V \text{ is maximal in } Q_W \iff \dim_K(W/V) = 1. \tag{1}$$

For any intermediate field $E \supseteq K$, unique coordinates give

$$\mathrm{Sz}(E) \cap Q_W = Q_{W \cap E}. \tag{2}$$

In fact multiplying by an element of $B_t \leq \mathrm{Sz}(E)$ reduces membership to the central parameter in $W \cap E$.

3 The construction and every residue check

Theorem 2. *In $G = \mathrm{Sz}(t^A)$ take the $k+1$ parabolics*

$$G_i = \mathrm{Sz}(F_i) \quad (1 \leq i \leq k), \qquad G_* = Q_T.$$

Their coset pregeometry satisfies the source's residual weak primitivity and algebraic local double transitivity conditions. Its rank is $k+1$. For $k = 2$ it is an incidence geometry of rank three, and $q-1$ is composite. Thus the source's general rank bound and its restriction of rank three to prime $q-1$ are false.

Proof. For $J \subseteq \{1, \ldots, k\}$, unique field coordinates, Lemma 1, and (2) give all intersections:

$$G_J = \mathrm{Sz}(F_J), \qquad G_{J \cup \{*\}} = Q_{T_J}. \tag{3}$$

The intersection of all parabolics is $B_t = Q_K$. Every parabolic is proper: each G_i is a prime-degree subfield, and Q_T is contained in the ambient point stabilizer. The intersections in (3) decrease strictly when any type is added.

We verify residual weak primitivity for every proper type subset. If the subset omits $*$ and some i, adjoining i is a prime-degree simple subfield inclusion, hence maximal. If it contains all $1, \ldots, k$ and omits only $*$, its group is $\mathrm{Sz}(t)$ and adjoining $*$ gives its maximal point stabilizer B_t. If the subset contains $*$ and omits an i, adjoining i removes exactly the independent coordinate Kx_i from T_J. Equation (1) gives maximality. This exhausts every proper subset. The last index is t, not a prime; maximality follows from scalar irreducibility as in (1).

The rank-one algebraic residue obtained by omitting $*$ is the coset action of $\mathrm{Sz}(t)$ on $\mathrm{Sz}(t)/B_t$, doubly transitive by the classical action. Omitting i gives Q_{K+Kx_i} on its cosets of B_t. These cosets identify with $(K + Kx_i)/K \cong K$. The centre provides all translations, while

H_t provides every nonzero scalar multiplication. Its induced action is therefore the affine group $K \rtimes K^*$, doubly transitive on t points. These are all the source's required local actions.

For $k = 2$, every rank-three coset pregeometry is a geometry: an incident pair has a common representative, which supplies the third coset and extends it to a chamber. Singletons extend in the same way. No flag transitivity follows from this observation. Finally $q - 1 = (t-1)(1+t+\cdots+t^{A-1})$ has both factors greater than one. The rank-three example thus has composite $q - 1$. Since k is arbitrary, (3) gives pregeometries of arbitrarily large rank. □

Concrete parameters are $t = 8$, $(a_1, a_2) = (3, 5)$, giving rank three at $q = 2^{45}$; and $t = 8$, $(a_1, a_2, a_3) = (3, 5, 7)$, giving rank four at $q = 2^{315}$. These are symbolic examples: any x_i in its displayed subfield outside K suffices. No ambient Suzuki group is enumerated.

Attribution and scope. Suzuki structure and its natural doubly transitive action are classical premises. The correspondence $W \mapsto Q_W$ and its maximality criterion reuse the accepted scalar-subspace theory. The contribution here is the independent field-coordinate construction and its application to the source's separate universal rank claim. No full incidence thinness, higher-rank geometry property, or flag transitivity is asserted. This result does not dispose of a different statement using full actual panels or additional geometric hypotheses. It is an additional correction within the already refuted original, not new manuscript or principal refutation credit. No computation, Lean declaration or compression factor is introduced.

Scalar subspaces in rank-two Suzuki geometries

Human Compression project

Abstract

A maximal subfield need not be maximal as an invariant additive subspace. This distinction changes a family in the rank-two Suzuki classification. We give the exact overgroups in the relevant mixed parabolic, construct the resulting geometries, and determine their equivalence by field automorphisms. For Sz(32768) this single family has forty-one isomorphism classes in the source's convention, whereas its table gives fourteen classes in total. The argument reuses the scalar-center correspondence already established in this project's work on higher-rank Suzuki configurations. It does not classify the remaining families.

1 An elementary overgroup correspondence

We first state the reusable group-theoretic fact. All groups in this note are finite. An action is called primitive when it is transitive and has no nontrivial invariant partition.

Lemma 1. *Let V be an abelian normal subgroup of Q, let $H \le Q$, and suppose $Q = VH$. Put $W = V \cap H$. The overgroups of H in Q correspond bijectively to the H-invariant subgroups of V containing W, by*

$$L \longmapsto L \cap V, \qquad T \longmapsto TH.$$

The coset action on Q/H is primitive precisely when V/W has no proper nonzero H-invariant subgroup. It is doubly transitive precisely when H is transitive on the nonzero elements of V/W.

Proof. If $H \le L \le Q$, write an element of L as vh. Since $h \in L$, we have $v \in L \cap V$, and therefore $L = (L \cap V)H$. Conversely, an H-invariant subgroup $T \le V$ makes TH a subgroup; because $W \le T$, its intersection with V is T. These operations are inverse. The usual correspondence between blocks containing the base coset and overgroups of its stabilizer proves the assertion about primitivity. Finally, V/W acts regularly on the cosets, identified with its additive set, and the base stabilizer acts by conjugation. Its transitivity away from zero is exactly double transitivity. □

If V is a vector space over a field K and the image of H on V is the full group of nonzero scalar multiplications, the invariant additive subgroups are exactly the K-subspaces. Thus an overgroup TH covers H precisely when $\dim_K(T/W) = 1$. On those cosets its permutation group is the affine group of the line over K. This is the scalar-center argument from the earlier project manuscript; the statement above isolates the group-theoretic reason for it.

2 Application to the Suzuki root group

Let $n > d \geq 3$ be odd integers, with $d \mid n$, and put $F = \mathbb{F}_{2^n}$, $K = \mathbb{F}_{2^d}$, $q = 2^n$, $s = 2^d$ and $\theta = 2^{(n+1)/2}$. We use the standard faithful four-dimensional Suzuki matrices, with

$$U(a,b) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ a^{\theta+1} + b & a^\theta & 1 & 0 \\ a^{\theta+2} + ab + b^\theta & b & a & 1 \end{pmatrix}.$$

Multiplication gives the root-group law

$$U(a,b)U(c,e) = U(a+c, b+e+ac^\theta), \qquad Z_F = \{U(0,b) : b \in F\}.$$

The diagonal torus consists of

$$h(t) = \mathrm{diag}(t^{\theta+1}, t, t^{-1}, t^{-\theta-1}), \qquad h(t)^{-1}U(a,b)h(t) = U(t^\theta a, t^{\theta+2}b). \tag{1}$$

Let U_K be given by $a, b \in K$, let $H_K = \{h(t) : t \in K^*\}$, and put $B = U_K H_K$. These are the standard root group, torus and point stabilizer in $S = \mathrm{Sz}(s) \leq G = \mathrm{Sz}(q)$. The ambient exponent θ restricts to the Suzuki exponent on K: the two exponent indices differ by a multiple of d.

We retain the classical Suzuki facts that this matrix model is faithful, that B is the stabilizer in the doubly transitive action of S on $s^2 + 1$ ovoid points, that S is maximal in G when n/d is prime, and that $\mathrm{Aut}(G) = G \rtimes \langle \phi \rangle$ with $\phi : x \mapsto x^2$ of order n. These structural results are due to Suzuki and Tits, and are also the structural inputs to the source classification: see its Section 2.2 and the structural list in Section 4, which cite Suzuki's *On a class of doubly transitive groups* (Annals of Mathematics 75 (1962), 105–145), Tits and Lüneburg. We do not reprove these classical facts here.

Both exponents in (1) are relatively prime to $q - 1$. This is immediate for θ. For the other, an odd common divisor of $2^n - 1$ and $2^{(n-1)/2} + 1$ would give simultaneously $2^n = 1$ and $2^{n-1} = 1$ modulo that divisor, forcing it to be one. The action of H_K on Z_F is consequently the full group of K-scalar multiplications.

Theorem 2. *Put $Q = Z_F B$. The overgroups of B in Q are exactly*

$$Q_T = Z_T B \quad (K \leq T \leq F), \tag{2}$$

where T ranges over K-subspaces and $Z_T = \{U(0,t) : t \in T\}$. Their orders are $|Q_T| = |T|s(s-1)$. In particular, the groups covering B are exactly those with $\dim_K T = 2$, and their order is $s^3(s-1)$. For every $c \in F \setminus K$,

$$\langle B, U(0,c) \rangle = Z_{K+Kc} B. \tag{3}$$

Proof. Here Z_F is abelian and normal in Q, and $Z_F \cap B = Z_K$. The root group U_K centralizes Z_F; the diagonal action is the scalar action just proved. Lemma 1 gives (2), the order formula, and the maximality criterion. Conjugating $U(0,c)$ by H_K generates exactly the additive space Kc, and B already contains Z_K. This proves (3). □

The source's Lemma 6.9, printed page 112, asserts instead that (3) has order $qs(s-1)$ and contains B maximally because K is a maximal subfield. If $n/d = r$ is an odd prime, then $r \geq 3$. There are K-subspaces strictly between K and F, regardless of the absence of intermediate fields. Thus B is never maximal in this Q. This invalidates the corresponding mixed-parabolic row of Tables 8 and 11. The distinction is visible already at $s = 8$, $q = 512$:

$$|B| = 448 < 3584 < |Q| = 28672. \tag{4}$$

3 The actual geometries and their equivalence

Suppose from now on that $r = n/d$ is prime. For every two-dimensional K-subspace T containing K, form the rank-two coset geometry

$$\Gamma_T = \Gamma(G; S, Q_T). \tag{5}$$

As in the source, elements are left cosets and incidence means nonempty intersection. Isomorphism means an automorphism of the specified group G carrying one subgroup configuration to the other, with type interchange allowed. We make no assertion here about isomorphisms that do not preserve the specified group action.

Theorem 3. *Every geometry (5) is firm, residually connected, flag transitive, residually weakly primitive, and locally doubly transitive. Its two local degrees are $s^2 + 1$ and s. Two members are conjugate under G precisely when their spaces T are equal. Their isomorphism classes in the source's convention are exactly the orbits of ϕ on*

$$\mathbb{P}_K(F/K). \tag{6}$$

There are $(s^{r-1} - 1)/(s - 1)$ conjugacy classes before this field fusion.

Proof. Coordinates in the standard subfield matrices give $S \cap Q_T = B$. More explicitly, every element of Q_T has a unique expression $U(a, b)h(t)$ with $a \in K$, $b \in T$, $t \in K^*$; such an element is in S exactly when $b \in K$. The subgroup S is maximal in G. Since $T \neq K$, the group Q_T is not contained in S, and therefore $\langle S, Q_T \rangle = G$. The incidence graph is consequently connected. A rank-two coset system is a geometry and is flag transitive: every element has an incident partner, and each incident pair is a translate of the base pair. Its two valencies are $[S : B] = s^2 + 1$ and $[Q_T : B] = s$, so it is firm. In rank two, connectedness supplies all the residual connectivity conditions of rank at least two.

The stabilizer B is maximal in S by the classical primitive ovoid action, and is maximal in Q_T by Theorem 2. Along with maximality of S in G, these are exactly the source's rank-two residual weak primitivity conditions. The action of S on its local cosets is doubly transitive. By Lemma 1, the other local action is the affine action of $K \rtimes K^*$ on K, also doubly transitive. This proves the first assertions.

The group S is nonsoluble whereas Q_T is soluble, so an isomorphism cannot exchange their types. The joint normalizer of S and B in G is B: maximality and simplicity give $N_G(S) = S$, and B is self-normalizing in S because it fixes a unique ovoid point. Every element of B preserves each T, by the scalar action. Hence no two different spaces in (2) give conjugate configurations.

Both S and B are preserved by ϕ. Their joint normalizer in $\mathrm{Aut}(G)$ is thus $B \rtimes \langle \phi \rangle$. After the trivial B action is removed, equivalence is exactly field fusion of the spaces T. Such spaces are the inverse images of lines in F/K, giving (6) and its stated cardinality. □

4 Two exact field counts

The following counts use the full automorphism group of G, not just inner conjugacy. They require no Suzuki group census.

Lemma 4. *Let σ be an odd-order field automorphism of F preserving K. Every σ-fixed element of F/K has a representative in F^σ. Consequently*

$$(F/K)^\sigma = F^\sigma/K^\sigma.$$

Proof. If σ has odd order a and the class of v is fixed, the sum $v + \sigma(v) + \cdots + \sigma^{a-1}(v)$ is a fixed representative of that same class: its image in F/K is $av = v$ in characteristic two. The kernel of the map from fixed representatives is K^σ. □

First take $K = \mathbb{F}_8$, $F = \mathbb{F}_{512}$. There are nine lines in F/K. The map ϕ^3 is K-linear of order three. An invariant line would have an eigenvalue in K^* of order dividing three, hence eigenvalue one. But its fixed space is zero by Lemma 4, since $F^{\phi^3} = K$. Therefore ϕ^3 fixes no line. The orbits of the order-nine map ϕ can only have lengths one, three or nine; the first two would be fixed by ϕ^3. All nine lines form one orbit.

Now take $K = \mathbb{F}_8$, $F = \mathbb{F}_{32768}$. There are

$$(8^4 - 1)/7 = 585 \tag{7}$$

lines in F/K. The K-linear map ϕ^3 has order five and no invariant line: the same argument uses $\gcd(5, 7) = 1$ and $F^{\phi^3} = K$. Hence any power ϕ^j with $5 \nmid j$ fixes no line, because its generated group contains ϕ^3.

Put $\sigma = \phi^5$, of order three. Its fixed field is $\mathbb{F}_{32}$, and its fixed field on K is $\mathbb{F}_2$. Thus Lemma 4 gives a fixed subspace of F/K with sixteen elements. On K, σ acts as the fourth power. If a K-line is invariant, choose a nonzero vector v on it and write $\sigma(v) = \lambda v$. There is a unique $k \in K^*$ with $k^3 = \lambda^{-1}$, because $\gcd(3, 7) = 1$. Then kv is fixed. Conversely a fixed nonzero vector spans an invariant line. Each such line contains exactly one nonzero fixed vector, since the fixed scalar field is $\mathbb{F}_2$. There are therefore fifteen fixed lines for each of ϕ^5 and ϕ^{10}. Burnside's orbit count now gives

$$\frac{585 + 15 + 15}{15} = 41. \tag{8}$$

There are three orbits of length five and thirty-eight of length fifteen.

Corollary 5. *For $G = \mathrm{Sz}(32768)$, the geometries (5) with $S = \mathrm{Sz}(8)$ give forty-one isomorphism classes in the source's convention. All are locally doubly transitive. Theorem 7.1 and its Table 11 omit this family, and the total fourteen in Table 12 at $e = 7$ is false.*

Proof. The count is (8). The group Q_T has order 3584 and a nonabelian normal Sylow two-subgroup. None of the other rows with the nonsoluble parabolic Sz(8) has this subgroup type: their other parabolic is Suzuki, dihedral, a cyclic group extended by a group of order four, or the mixed group of order $32768 \cdot 8 \cdot 7$. Thus (5) is absent from the asserted exhaustive list. Forty-one classes in one family also contradict the printed total fourteen. The same order argument already gives an omitted family at $q = 512$. □

Attribution and scope. The target is D. Leemans, *The Rank 2 Geometries of the Simple Suzuki Groups* Sz(q), Beiträge zur Algebra und Geometrie 39(1) (1998), 97–120: Lemma 6.9, Theorem 7.1 and Tables 8, 11, 12. The source's classification is considered under its specified-action equivalence, defined on printed page 100. The matrix model, ovoid action, subgroup maximality and ambient automorphism description are retained classical inputs. The scalar-center argument was already proved and separately reviewed in the project's earlier Suzuki subfield manuscript, Section 4; the new contribution here is its application to this rank-two row, the genuine replacement family, its local double transitivity and its equivalence calculation. No historical priority is asserted. The remaining rows and the complete corrected global count are not established by this note. This corrects a published classification; it is not an answer to a named open conjecture.

The supplementary program independently constructs the fields of orders 512 and 32768, verifies their defining polynomials by Rabin's criterion, lists all two-dimensional $\mathbb{F}_8$-subspaces containing $\mathbb{F}_8$, and checks every field orbit. It agrees with (7)–(8). These finite controls are not premises of the written proof.

Maximal subgroups and elementary reductions for Suzuki geometries

Human Compression project

Abstract

We give complete proofs of the preliminary arithmetic and subgroup arguments used in the rank-two Suzuki classification. The maximal subgroups of a point stabilizer follow from its Frattini quotient and an affine fixed point argument; the two cyclic normalizer cases follow from the same argument. A general obstruction then excludes a point stabilizer from every firm residually weakly primitive geometry of rank greater than one. These are surviving results of the source manuscript, whose full classification requires a separate correction. Neither the remaining families nor their global counts are established here.

1 Arithmetic of the three odd factors

Write $q = 2^n$, where $n \geq 3$ is odd, and set

$$t_q = 2^{(n+1)/2}, \qquad a_\epsilon(q) = q + \epsilon t_q + 1 \quad (\epsilon \in \{1, -1\}).$$

Thus $t_q^2 = 2q$ and

$$a_+(q)a_-(q) = q^2 + 1. \tag{1}$$

Lemma 1. *The integers $q-1, a_+(q), a_-(q)$ are pairwise coprime. If $2^n - 1$ is prime, then n is prime. Suppose $s = 2^d \geq 8$, with d odd, and $q = s^r$, with r odd. Then*

$$s^2 + 1 \mid q^2 + 1, \qquad s^2 + 1 \nmid a_+(q), \qquad s^2 + 1 \nmid a_-(q).$$

If also $r > 1$, all three integers $q-1, a_+(q), a_-(q)$ are composite.

Proof. Every integer in the first assertion is odd. A common divisor of $a_+(q)$ and $a_-(q)$ divides $2t_q$, a power of two, so is one. If an odd integer h divides $q-1$ and $a_\epsilon(q)$, reduction modulo h gives $q = 1$, $\epsilon t_q = -2$, and $t_q^2 = 2$. Squaring the second equality gives $4 = 2$, so $h \mid 2$ and $h = 1$. If $n = uv$ with $u, v > 1$, the identity

$$2^{uv} - 1 = (2^u - 1)(1 + 2^u + \cdots + 2^{u(v-1)})$$

is a nontrivial factorization. This proves the assertion about a prime $2^n - 1$, and proves directly that $q - 1$ is composite when $q = s^r$, $s > 2$, $r > 1$.

Put $m = s^2 + 1$ and $t = t_s$. Since $s^2 = -1 \pmod m$ and r is odd, $q^2 + 1 = s^{2r} + 1$ is zero modulo m. Also $t_q = ts^{(r-1)/2}$. Reducing the odd integer r modulo eight gives the following representatives for $a_\epsilon(q)$ modulo m:

$r \pmod 8$	1	3	5	7
$a_\epsilon(q)$	$1 + s + \epsilon t$	$1 - s + \epsilon st$	$1 + s - \epsilon t$	$1 - s - \epsilon st.$

All four expressions, for either sign, are nonzero. Indeed, $s > t > 1$, while $st > s - 1$. Their absolute values are less than m: $s + t + 1 < s^2 + 1$, and $st + s - 1 < s^2 + 1$ because $t + 1 \leq s$ for $s \geq 8$. None can therefore represent zero modulo m.

Finally assume $r \geq 3$. Both $a_\epsilon(q)$ exceed s^2+1. To see this without an asymptotic estimate, $t_q \leq q/2$ for $q \geq 8$, so

$$a_\epsilon(q) \geq q - t_q + 1 \geq s^3/2 + 1 > s^2 + 1.$$

If one of these two integers were prime, it would be coprime to m, since it exceeds m. By (1) and $m \mid q^2+1$, the other factor would then be divisible by m, contrary to the preceding paragraph. Both are composite. □

For completeness, the remaining elementary congruence used in the source is Fermat's $p \mid 2^p - 2$ for prime p. For odd p, multiplication by 2 permutes the nonzero residues modulo p. Multiplying those residues and cancelling $(p-1)!$ gives $2^{p-1} = 1 \pmod{p}$. The case $p = 2$ is immediate.

Lemma 1 proves the mathematical content of source Lemmas 3.1–3.6, and the last paragraph gives Lemma 3.7. In Lemma 3.2 the proper-subfield condition must be explicit: $r > 1$ and $s > 2$. The printed statement does not state $r > 1$. Taken literally it fails at $q = s = 8$, $r = 1$. This is a correction of its domain, not an additional refutation of the source's principal classification. The source attributes Lemmas 3.1 and 3.7 to Cataldi and Fermat, and the argument for Lemma 3.5 to J. A. Thas. We retain those attributions; the residue table above is a common proof of the same arithmetic assertions, not a claim of priority.

2 Two elementary subgroup arguments

We write $\Phi(P)$ for the intersection of the maximal subgroups of a finite group P. Only the following elementary property of finite p-groups is needed. If M is maximal in a finite p-group P, then M is normal and $[P : M] = p$. Here is a proof. A proper subgroup K acts on the left cosets P/K. Its fixed cosets are exactly $N_P(K)/K$. Every other orbit has size divisible by p, and $|P : K|$ is divisible by p. There is at least one fixed coset, so there are at least p, and $N_P(K) > K$. For maximal M this forces $N_P(M) = P$. The quotient P/M has no proper nontrivial subgroup, so has order p. Consequently every square in a finite two-group belongs to its Frattini subgroup. We also use the immediate generating property: if $J \leq P$ and $J\Phi(P) = P$, then $J = P$, since otherwise a maximal subgroup containing J would contain $J\Phi(P)$.

Lemma 2. *Let $P \lhd B$ be a finite p-group, and let $Z \leq \Phi(P)$ be normal in B. Every maximal subgroup of B contains Z.*

Proof. If a maximal subgroup M omits Z, then $MZ = B$ because Z is normal. As $Z \leq P$, intersecting with P gives $P = (M \cap P)Z$. Thus $P = (M \cap P)\Phi(P)$, and the generating property gives $P \leq M$. This contradicts $Z \not\leq M$. □

Lemma 3. *Let V be a finite vector space in characteristic p, and let H be a finite group acting linearly on V, with order prime to p. The action need not be faithful. Every complement to V in $V \rtimes H$ is conjugate by a translation to H. If H acts irreducibly on V over its prime field, the maximal subgroups of $V \rtimes H$ are the inverse images of the maximal subgroups of H, together with the conjugates of H.*

Proof. A complement K has order $|H|$ and acts affinely on V. The point

$$v = |K|^{-1} \sum_{k \in K} k(0)$$

is fixed by every element of K: affine maps preserve affine averages, and left multiplication permutes K. Translating v to zero conjugates K into H; equality follows from their orders.

A maximal subgroup M containing V is the inverse image of a maximal subgroup of H. Otherwise $MV = V \rtimes H$. The additive subgroup $M \cap V$ is normalized by M and by V, hence by the whole group. It is a vector subspace over the prime field, invariant under H. Irreducibility here is understood over that prime field; it gives $M \cap V = 0$, so M is a complement. Conversely, every proper overgroup L of H satisfies $L = (L \cap V)H$. Its intersection with V is invariant, so is either zero or V. Thus H is maximal. □

The prime-field qualification in this lemma matters. We use it for $V = (\mathbb{F}_q, +)$ and the full scalar group $\mathbb{F}_q^*$, for which it holds: an additive invariant subgroup containing a nonzero v contains every scalar multiple of v, hence all of $\mathbb{F}_q$.

Corollary 4. *Let m be odd and $d \in \{2, 4\}$. In any semidirect product $C_m \rtimes C_d$, the maximal subgroups are, up to conjugacy,*

$$C_m \rtimes C_{d/2}, \qquad C_{m/p} \rtimes C_d \quad (p \text{ a prime divisor of } m).$$

For $d = 2$, the first group is C_m. For $m = 1$ only the first group occurs. The statement allows an action that is not faithful.

Proof. Write $A = C_m$. A maximal subgroup containing A is the inverse image of the unique maximal subgroup of C_d. Otherwise MA is the whole group. Put $K = M \cap A$. Every subgroup of A is characteristic in A, so K is normal in the whole group. In the quotient by K, the image of M is a complement to the cyclic normal subgroup A/K, and is maximal. If A/K had a proper nontrivial subgroup T, that subgroup would be invariant and $T(M/K)$ would be a proper overgroup of M/K, a contradiction. Thus $|A : K| = p$ is prime. Lemma 3, applied to $A/K \cong (\mathbb{F}_p, +)$, shows that this complement is conjugate to the image of the specified C_d. Taking inverse images proves the stated form. Conversely each such subgroup has index p, and so is maximal. □

3 The point stabilizer

Let $F = \mathbb{F}_q$, $q = 2^n$, $n \geq 3$ odd, and put $\theta = 2^{(n+1)/2}$. We use the classical Suzuki root group

$$P = \{u(a, b) : a, b \in F\}, \qquad u(a, b)u(c, e) = u(a + c, b + e + ac^\theta).$$

Let $Z = \{u(0, b) : b \in F\}$ and $H = \{h(t) : t \in F^*\}$, with

$$u(a, b)^{h(t)} = u(t^\theta a, t^{\theta+2} b).$$

Then the point stabilizer is $B = P \rtimes H$. These formulas and their realization in the faithful Suzuki matrix group are retained from the classical construction, also used in Section 2.2 of the source and in the project's separately reviewed scalar-subspace note. We do not reconstruct the whole Suzuki group here.

Theorem 5. *The maximal subgroups of B are, up to conjugacy in B,*

$$P \rtimes C_{(q-1)/p} \quad (p \mid q - 1 \text{ prime}), \qquad Z \rtimes H.$$

In particular every maximal subgroup of B contains Z.

Proof. The multiplication law shows that Z is central in P, $P/Z \cong (F, +)$, and

$$u(a, b)^2 = u(0, a^{\theta+1}).$$

The exponent $\theta + 1$ is coprime to $q - 1$. Indeed, if an odd integer h divides both, then $\theta = -1 \pmod h$, whereas $\theta^2 = 2q = 2 \pmod h$. It follows that $1 = 2 \pmod h$, so $h = 1$. Every

element of Z is consequently a square, and $Z \leq \Phi(P)$. Conversely, the inverse images of the prime-field hyperplanes of P/Z are maximal subgroups of P whose intersection is Z. Thus $\Phi(P) = Z$.

Lemma 2 shows that every maximal subgroup of B contains Z. Modulo Z, the conjugation action of H on F is multiplication by t^θ. This ranges over all nonzero scalars, so is irreducible over $\mathbb{F}_2$. Lemma 3 classifies the maximal subgroups of B/Z. Their inverse images are exactly the groups displayed above. □

Theorem 5 gives source Lemma 5.1, including the completeness of its list. Corollary 4 gives Lemma 5.2 for the cyclic normalizers of order $4a_\epsilon(q)$, and Lemma 5.3 for the dihedral group $C_{q-1} \rtimes C_2$. Thus the three lists use one affine complement argument. Their assertions remain attributed to the source; this note supplies the complete common derivation and does not claim that these elementary tools are new.

4 Excluding a point stabilizer in every rank

We first isolate a general geometric obstruction. In a flag-transitive coset geometry

$$\Gamma(G; (G_i)_{i \in I}),$$

the residue of the base element of type 0 has acting group G_0 and stabilizers $G_0 \cap G_i$, $i \neq 0$. A transitive coset action is primitive if and only if its point stabilizer is maximal: blocks containing the base coset correspond to intermediate subgroups.

Lemma 6. *Let G be finite and $B < G$ be maximal. Suppose that every maximal subgroup J of B is contained in no maximal subgroup of G other than B. Then no firm flag-transitive residually weakly primitive geometry of rank greater than one has B as one of its parabolic subgroups.*

Proof. Suppose $G_0 = B$. Weak primitivity of the residue at the base element of type 0 gives an $i \neq 0$ such that $J = B \cap G_i$ is maximal in B. Every proper subgroup of the finite group G containing J lies in a maximal subgroup, hence lies in B. Thus $G_i \leq B$, so $G_i = J$. Set $G_{I\setminus\{0\}} = \bigcap_{j \neq 0} G_j$ and $G_I = \bigcap_j G_j$. Since $G_i \leq B$, these two intersections are equal. The base flag of all types except 0 therefore extends to exactly

$$[G_{I\setminus\{0\}} : G_I] = 1$$

chamber. This contradicts firmness. The same proof applies when the geometry has rank two; no truncation assumption is needed. □

Theorem 7. *For $G = \mathrm{Sz}(q)$, $q = 2^n$, $n \geq 3$ odd, the point stabilizer B cannot occur as a parabolic subgroup in a firm flag-transitive residually weakly primitive geometry of rank greater than one.*

Proof. We use Suzuki's classical list of maximal subgroups, quoted as Table 1 of the source: point stabilizers, $D_{2(q-1)}$, $C_{a_+(q)} \rtimes C_4$, $C_{a_-(q)} \rtimes C_4$, and maximal simple subfield groups $\mathrm{Sz}(s)$ with $q = s^r$, r an odd prime. Let J be maximal in B. Theorem 5 gives $Z \leq J$, where Z is elementary abelian of order $q \geq 8$.

The dihedral maximal subgroup has Sylow two-subgroups of order two. The two cyclic normalizers have cyclic Sylow two-subgroups of order four. None can contain Z. A simple proper subfield group has Sylow two-subgroups of order s^2, whereas $q = s^r \geq s^3 > s^2$. It too cannot contain Z.

Finally, a nonidentity element of the root group P fixes exactly one point in the classical ovoid action. In the affine coordinates on the complement of the distinguished point, its action is

$$(x, y) \longmapsto (x + a, y + b + a^{\theta} x).$$

A fixed affine point forces $a = 0$, then $b = 0$, so no nonidentity element fixes such a point. Consequently every point stabilizer containing a fixed nonidentity element of Z is B itself. The only maximal subgroup of G containing J is therefore B, and Lemma 6 applies. □

This proves the full rank range of source Lemma 6.1. The geometric step is written out because merely excluding rank-two configurations does not, by itself, justify an assertion in every rank.

5 Attribution and the remaining source obligations

The source is D. Leemans, *The Rank 2 Geometries of the Simple Suzuki Groups* Sz(q), Beiträge zur Algebra und Geometrie 39(1) (1998), 97–120, in the fixed twenty-four-page edition. The assertions treated here are its seven arithmetic lemmas, three maximal-subgroup lemmas, and Lemma 6.1. We retain Suzuki's maximal-subgroup theorem and the standard faithful root-group and ovoid constructions as explicit premises. The source credits Suzuki's *On a class of doubly transitive groups*, Annals of Mathematics 75 (1962), 105–145, and the treatments by Tits and Lüneburg for this structure. No fresh proof of that classification or new historical priority is asserted.

The common proofs here use elementary finite group theory: the Frattini generating property, irreducibility of the full scalar action, affine averaging, and the correspondence between blocks and intermediate subgroups. The distinct scalar-subspace correction to the source's mixed-parabolic family remains unchanged. The remaining assertions in Section 6, the corrected global family list, Tables 2–12 and their counts still require their own source review. This component supplies neither a complete treatment of the original nor an open-conjecture resolution. No Lean formalization or compression ratio is claimed for it.

Normalizer fibres in rank-two Suzuki geometries

Human Compression project

Abstract

The families with a dihedral or a nonsplit torus normalizer as one parabolic can be recovered from two elementary constructions. A normalizer calculation gives uniqueness of the relevant subfield overgroups. The remaining configurations are fibres of an incidence relation between conjugacy classes of subgroups; these fibres are small quotients of a Suzuki root group. Their field automorphisms and inversion give the counts in the prime cases. This treats source Lemmas 6.2–6.6 and 6.8, with a correction to the prime-divisor condition in Table 6. The classical subgroup classification quoted as source Lemma 6.7 is retained as an input. The families with a subfield group as the only maximal parabolic, and the global classification, are not concluded here.

1 Scope and structural inputs

Write $q = 2^n$, where $n \geq 3$ is odd, and put

$$\theta = 2^{(n+1)/2}, \qquad a_\epsilon(q) = q + \epsilon\theta + 1 \quad (\epsilon \in \{+1, -1\}).$$

Let $G = \mathrm{Sz}(q)$. A configuration in this note is an unordered pair of proper subgroups $\{M, L\}$, with intersection $J = M \cap L$. Equivalence means the action of $\mathrm{Aut}(G)$, allowing interchange of the two subgroups. This is the specified-group convention of the source [1]; we do not identify geometries by arbitrary permutations which need not preserve their specified G action.

The following classical structure is retained, with its attribution to Suzuki, Tits and Lüneburg as in Sections 2.2 and 4 of the source. The nonidentity elements of G are partitioned by its Sylow two-subgroups and its cyclic tori of orders $q - 1, a_+(q), a_-(q)$. The normalizers of these parts are respectively

$$B = P \rtimes \mathbb{F}_q^*, \quad D = C_{q-1} \rtimes C_2, \quad N_\epsilon = C_{a_\epsilon(q)} \rtimes C_4. \tag{1}$$

These groups are maximal in G. Each indicated type of part forms one conjugacy class. Every subgroup of G either lies in one of these normalizers or is a conjugate of a standard subfield group $S = \mathrm{Sz}(s)$, where $s = 2^d \geq 8$ and $d \mid n$. In the latter case the conjugacy class for each s is unique. The same description holds inside S; its maximal proper simple subfield groups have prime field index. This is precisely the classical subgroup input quoted from Suzuki [2] as source Lemma 6.7, together with the maximal-subgroup list in Section 4. It is not proved anew here.

We also retain the standard action on the Suzuki ovoid: B fixes one point, a nonidentity involution fixes just one point, and a split torus fixes two points whose stabilizers are interchanged by D. The centralizer of an involution is its Sylow two-subgroup. Finally $\mathrm{Aut}(G) = G \rtimes \langle\phi\rangle$, where ϕ is coordinate Frobenius of order n.

We use the previously reviewed preliminary note for the pairwise coprimality of the three odd orders, their proper-subfield arithmetic, and the maximal subgroups of the solvable groups in (1). Explicitly, in $C_t \rtimes C_4$ these maximal subgroups are $C_t \rtimes C_2$ and $C_{t/p} \rtimes C_4$, with $p \mid t$ prime; in D they are C_{q-1} and $D_{2(q-1)/p}$. In a point stabilizer they are $P \rtimes C_{(q-1)/p}$ and $Z \rtimes C_{q-1}$, where $Z = Z(P)$. We write D_{2m} for a dihedral group of order $2m$. The arithmetic

also shows that if any one of $q-1, a_+(q), a_-(q)$ is prime, then n is prime: if n were composite, a proper subfield of order at least eight would make all three composite.

For rank two, the elementary coset criterion we use is as follows. If M is maximal in G, $L \not\leq M$, and $J = M \cap L$ is maximal in both M and L, then $\Gamma(G; M, L)$ is firm, connected, flag transitive and residually weakly primitive. Indeed, the local degrees are $[M : J]$ and $[L : J]$, both at least two; rank-two coset geometries are flag transitive; $\langle M, L\rangle = G$ gives connectivity; and the three required primitive coset actions are those with stabilizers $M < G$, $J < M$, and $J < L$. Conversely, these maximality conditions on J are necessary when M is a maximal parabolic.

2 Two root-group quotients

Use the standard root coordinates

$$P = \{u(a, b) : a, b \in F\}, \qquad F = \mathbb{F}_q, \qquad u(a, b)u(c, e) = u(a + c, b + e + ac^\theta). \tag{2}$$

Put $z = u(0, 1)$, $w = u(1, 0)$, $J_2 = \langle z\rangle$ and $J_4 = \langle w\rangle$. Then $w^2 = z$ and both J_2, J_4 are fixed pointwise by ϕ. We retain the classical realization of (2) in the faithful Suzuki matrices, rather than reconstructing the ambient simple group.

Lemma 1. *The normalizers of these two subgroups are*

$$\mathrm{N}_G(J_2) = P, \qquad \mathrm{N}_G(J_4) = R := \{u(a, b) : a \in \mathbb{F}_2,\ b \in F\}. \tag{3}$$

The group R is abelian of order $2q$ and is isomorphic to $C_4 \times C_2^{n-1}$. The quotient R/J_4 is naturally the additive group $F/\mathbb{F}_2$. The cyclic subgroups of order four containing J_2 are parametrized by $F/\mathbb{F}_2$, the four-groups containing J_2 by $(F/\mathbb{F}_2) \setminus \{0\}$, and the subgroups of order eight containing J_4 by the same nonzero quotient. These parametrizations commute with Frobenius. Also

$$P^\phi = R^\phi = J_4. \tag{4}$$

Proof. The first equality in (3) is the retained involution-centralizer fact, since an automorphism of a group of order two is trivial. Every element normalizing J_4 fixes $w^2 = z$, so lies in P. The multiplication law gives

$$u(a, b)^2 = u(0, a^{\theta+1}), \qquad [u(a, b), w] = u(0, a + a^\theta), \tag{5}$$

where the choice of commutator convention does not change the second formula because its value has order at most two. Conjugation preserves the first coordinate of w, so normalizing J_4 is equivalent to $a + a^\theta \in \mathbb{F}_2$. The absolute trace of $a + a^\theta$ is zero, whereas the trace of 1 is one because n is odd. Thus $a = a^\theta$. The fixed field of $x \mapsto x^\theta$ is $\mathbb{F}_2$, since $\gcd(n, (n+1)/2) = 1$. This proves (3). Formula (2) on first coordinates in $\mathbb{F}_2$ proves that R is abelian, has exponent four, and has squares exactly J_2. Its stated abstract structure and $R/J_4 \cong F/\mathbb{F}_2$ follow, the latter map being represented by $u(0, b) \mapsto b + \mathbb{F}_2$.

The exponent $\theta + 1$ is relatively prime to $q - 1$: a common odd divisor would give $\theta = -1$ and $\theta^2 = 2$ modulo that divisor, hence $1 = 2$. Thus (5) shows that an element whose square is z has first coordinate one. The groups $\langle u(1, b)\rangle$ are therefore all cyclic groups of order four in P containing J_2, and two values of b give the same group precisely when they differ by one. A cyclic group of order four containing J_2 centralizes its square z, so is in P. Likewise, a four-group containing J_2 centralizes z and lies in P. All involutions of P belong to $Z = \{u(0, b) : b \in F\}$, so four-groups containing J_2 are the two-dimensional binary subspaces of F containing one. These are parametrized by the nonzero elements of $F/\mathbb{F}_2$.

If a group of order eight contains J_4, that index-two subgroup is normal; the group is therefore in R. Conversely, inverse images of order-two subgroups of R/J_4 have order eight. They are $C_4 \times C_2$ since R is abelian of exponent four. All formulas commute with squaring of field coordinates. Formula (4) follows because the only fixed coordinates are zero and one. □

Lemma 2. *Suppose n is prime. Frobenius has the following numbers of fixed cosets:*

$$\begin{array}{c|ccc} X & P/J_2 & P/J_4 & R/J_4 \\ \hline |X| & q^2/2 & q^2/4 & q/2 \\ |X^\phi| & 2 & 1 & 1. \end{array} \tag{6}$$

On $(F/\mathbb{F}_2) \setminus \{0\}$ it has no fixed point. On $(P/J_2) \setminus \{1\}$ modulo inversion, it has exactly one fixed orbit. The number of inversion orbits in this last set is

$$q^2/4 + q/2 - 1. \tag{7}$$

Proof. For either pointwise fixed subgroup $J = J_2, J_4$, a fixed right coset gJ gives $\phi(g) = gh$ for some $h \in J$. Iterating n times gives $h^n = 1$, so $h = 1$ because n is odd and h has two-power order. Hence fixed cosets are exactly the cosets represented by fixed elements. Equation (4) proves (6). The statement for $F/\mathbb{F}_2$ follows from R/J_4.

Let $H = P/J_2$. It has q elements with square one: by (5), there are $2q$ elements of P with square in J_2, with first coordinate zero or one, and each coset has two elements. Inversion on the set H fixes exactly these q elements. Pairing the others and removing the identity gives $(|H| + q)/2 - 1$, which is (7). An inversion orbit fixed by ϕ has size at most two. The permutation of it induced by an element of odd prime order cannot exchange its two members; each member is therefore fixed. By (6), H^ϕ has order two, and its sole nonidentity element gives the one fixed orbit. □

3 Uniqueness of three subfield overgroups

Lemma 3. *Let a finite group K have one conjugacy class of subgroups S of the specified type. Suppose $H < S$, and all the K-conjugates of H lying in S form one S-conjugacy class. Then the number of conjugates of S containing a fixed conjugate of H is*

$$\frac{|S|\,|\,\mathrm{N}_K(H)|}{|\,\mathrm{N}_K(S)|\,|\,\mathrm{N}_S(H)|}. \tag{8}$$

In particular it is one if S and H are self-normalizing in K.

Proof. Count incident pairs of conjugates (S', H') with $H' \leq S'$. There are $|K|/|\,\mathrm{N}_K(S)|$ first entries, each incident with $|S|/|\,\mathrm{N}_S(H)|$ second entries. Divide by the total number $|K|/|\,\mathrm{N}_K(H)|$ of second entries. □

Lemma 4. *For each subfield order $s \geq 8$, every subgroup abstractly isomorphic to one of the three groups in (9) below is conjugate to its standard copy in* $\mathrm{Sz}(s)$.

Proof. For a dihedral group or a group $C_{a_\delta(s)} \rtimes C_4$, the odd cyclic subgroup is characteristic. Its nonidentity elements determine the unique ambient torus. The order of that torus is the same as for the standard copy: the three ambient odd orders are pairwise coprime. All tori of a given type are conjugate, and each has a unique subgroup of each dividing order. The complements are conjugate by the torus. In the dihedral case this is the invertibility of squaring on an odd cyclic group; in the order-four case it follows from the invertibility of $q-1$ on the nonsplit torus. This proves the assertion for the two cyclic normalizers.

Now let H be abstractly isomorphic to B_s. Its normal Sylow two-subgroup V has order s^2, with exactly $s-1$ involutions, and its odd complement A has order $s-1$. The centre of V contains an involution z', so $V \leq \mathrm{C}_G(z')$, a Sylow two-subgroup. Any nonidentity element of V is in a unique part of the classical partition. Consequently A normalizes this Sylow subgroup, and after conjugation $H \leq B = P \rtimes T$, where T denotes the diagonal torus. The image of A in $B/P = T$ is its unique subgroup K^*, where $K = \mathbb{F}_s$. Conjugating by an element of P makes $A = K^*$: first apply the affine complement lemma to $(P/Z) \rtimes K^*$, and then to $Z \rtimes K^*$. The lemma permits a nonfaithful action, so applies to both quotients.

All involutions of P are in Z. Hence $|V \cap Z| = s$ and the image of V in P/Z also has order s. The action of K^* on each coordinate is the full scalar group of K. Thus these two additive subgroups are lines bK and aK in F. Squares in (5) give $bK = a^{\theta+1}K$. A diagonal conjugation makes $aK = K$ and consequently $bK = K$. The possible subgroup V is now described by a function $f : K \longrightarrow F/K$:

$$V = \{u(c, e) : c \in K,\ e + K = f(c)\}.$$

The multiplication law shows that f is additive, since $cd^\theta \in K$ for $c, d \in K$. Invariance under K^* gives $f(k^\theta c) = k^{\theta+2} f(c)$. Since $\theta^2 = 2$ as an exponent on K^*, this implies $f(c) = c^{\theta+1} f(1)$ for every $c \in K$. Choose $c \in K \setminus \mathbb{F}_2$. Additivity at $c+1$ gives $(c^\theta + c)f(1) = 0$ in the K-vector space F/K. The coefficient is nonzero, because the fixed field of $x \mapsto x^\theta$ on K is $\mathbb{F}_2$. Therefore $f(1) = 0$, so $f = 0$ and $V = U(K, K)$. This proves $H = B_s$ after conjugation. □

Theorem 5. *Let $K = \mathbb{F}_s \subseteq F$, with $s \geq 8$. Each subgroup of one of the following types is contained in exactly one subgroup isomorphic to* $\mathrm{Sz}(s)$*:*

$$B_s = U(K, K) \rtimes K^*, \qquad D_{2(s-1)}, \qquad C_{a_\delta(s)} \rtimes C_4 \quad (\delta = +1, -1). \tag{9}$$

Proof. By Lemma 4, it suffices to use the groups in (9) inside the standard $S = \mathrm{Sz}(s)$. Their respective conjugacy classes in S are unique by the retained structure. We prove that each is self-normalizing in G. The subfield group S is also self-normalizing. Indeed, its normalizer contains the nonsolvable simple group S, so cannot be contained in a solvable normalizer in (1). The subgroup classification makes it a simple subfield group; since it normalizes its nontrivial normal subgroup S, it equals S.

For B_s, its characteristic Sylow two-subgroup determines the unique fixed ovoid point, so any normalizer lies in $B = PH$. Write its elements using $u(a, b)$ and $h(t)$, where

$$u(a, b)^{h(t)} = u(t^\theta a, t^{\theta+2} b). \tag{10}$$

The exponent $\theta + 2$ is coprime to $q - 1$: a common odd divisor would give $\theta = -2$ and $\theta^2 = 2$, hence $4 = 2$. Preserving the centre $U(0, K)$ of the Sylow subgroup in B_s therefore forces $t^{\theta+2} \in K^*$, and hence $t \in K^*$. Remove this factor. For $k \in K^* \setminus \{1\}$, the first coordinate of the commutator of $u(a, b)$ with $h(k)$ is a nonzero K-multiple of a. Membership in B_s forces $a \in K$. Remove the factor $u(a, 0) \in B_s$ as well. The remaining central factor $u(0, b)$ has commutator coordinate $(k^{\theta+2} - 1)b$, up to multiplication by a nonzero element of K. This forces $b \in K$. Thus $\mathrm{N}_G(B_s) = B_s$.

For either odd normalizer in (9), its cyclic odd subgroup is characteristic. The partition into tori forces its normalizer to preserve the unique ambient torus containing that subgroup. In the split case this gives a subgroup of $C_{q-1} \rtimes C_2$. A torus element x normalizes $D_{2(s-1)}$ precisely when $x^2 \in C_{s-1}$; since the torus has odd order this is equivalent to $x \in C_{s-1}$.

In the nonsplit case the ambient torus normalizer is $C_t \rtimes \langle v \rangle$, with $t = a_\epsilon(q)$ and $|v| = 4$. The standard action on the cyclic torus is the power q or its inverse. Both $q - 1$ and $q^{-1} - 1$ are coprime to t, because $\gcd(q - 1, t) = 1$. For $m = a_\delta(s)$, a torus element x normalizes $C_m \rtimes \langle v \rangle$ precisely when the commutator of x with v lies in C_m. This is equivalent to $x \in C_m$.

All complements of order four are torus conjugate, so this calculation applies to the subgroup in (9). It is self-normalizing. Equation (8) now proves all three uniqueness assertions. □

These are source Lemmas 6.5, 6.6 and 6.8, including the version of 6.8 for the other nonsplit torus. The common step is incidence counting after self-normalization. No new conjugacy classification of subfield groups is being asserted.

4 Fibres of subgroup incidence

Fix $J = J_2$ or J_4. If the copies of J in a subgroup M form one conjugacy class in M, then the conjugates of M containing this fixed J form one $\mathrm{N}_G(J)$-orbit. To prove this, first conjugate one copy of M to another; conjugate again within the second copy to send the image of J to J. The stabilizer is $\mathrm{N}_G(J) \cap \mathrm{N}_G(M)$. Thus the orbit is an actual coset space, not merely a set of the same cardinality.

Applied to (1) and to its prime subgroups, this gives

J	M	$\mathrm{N}_G(J) \cap \mathrm{N}_G(M)$	fibre
J_2	$D_{2(q-1)}$	J_2	P/J_2
J_2	$D_{2p},\ p \mid a_\epsilon(q)$ prime	C_4	P/C_4
J_4	$C_t \rtimes C_4$	J_4	R/J_4.

(11)

In the last row t may be the whole nonsplit torus order or a prime divisor of it. For the second row, the full normalizer of D_{2p} is $C_p \rtimes C_4$. This follows by preserving its characteristic odd subgroup, then using the odd-order squaring argument from Theorem 5. Its stabilizer of its given involution is a cyclic group of order four. For the last row, the commutator argument in the same theorem shows that $C_t \rtimes C_4$ is self-normalizing, even when t is a proper prime divisor of the ambient torus order. The required conjugacy of involutions or complements inside these subgroups follows from the cyclic semidirect products of odd order by C_2 or C_4. This also follows from the affine complement argument in the preliminary note.

Suppose now that n is prime. Every fibre in (11) has a Frobenius-fixed member: its cardinality is respectively $q^2/2$, $q^2/4$ or $q/2$, congruent to $2, 1, 1$ modulo n by Fermat's congruence. The orbits of a cyclic group of prime order have size one or n, so a fixed member exists. Choose that member as base. In the second row its stabilizer C_4 is Frobenius invariant, and Frobenius acts trivially on it because $\mathrm{Aut}(C_4)$ has order two. By (4), this stabilizer is the particular J_4 used above. Thus (11) is Frobenius equivariant with exactly the quotients in (6); no assumption about a preferred complement is needed.

For two distinct dihedral parabolics with common J_2, the first fibre in (11) is a regular set for $H = P/J_2$. Simultaneous translation sends an ordered pair to $(1, h)$ with $h \neq 1$; interchanging its entries sends h to h^{-1}. Hence the unordered configurations are the inversion classes in $H \setminus \{1\}$, followed by Frobenius. For two equal types of nonsplit normalizer, the same argument uses R/J_4, an elementary abelian group, so inversion does nothing.

For distinct parabolic types there is no interchange. Once the first parabolic containing J_2 is fixed, its stabilizer in P is J_2, which acts trivially on P/J_4 and on the order-four groups in Lemma 1. The nonsplit calculation has the analogous trivial action of J_4 on R/J_4. These observations explain why the small fibres give the complete configuration counts, rather than just numbers of possible second subgroups.

5 The split normalizer

Theorem 6. *Take $M = D_{2(q-1)}$. If $q - 1$ is prime, the possible second parabolics and their numbers of equivalence classes are precisely the rows of the following table. Each prime p in*

the third row gives a separate family.

L	J	*number*
$Z \rtimes C_{q-1}$	C_{q-1}	1
$D_{2(q-1)}$	C_2	$1 + \dfrac{(q-2)(q+4)}{4n}$
$D_{2p},\ p \mid a_+(q)a_-(q)$	C_2	$1 + \dfrac{(q-2)(q+2)}{4n}$
C_2^2	C_2	$\dfrac{q-2}{2n}$
C_4	C_2	$1 + \dfrac{q-2}{2n}.$

(12)

If $q-1$ is composite, the first row still occurs once. The only other possibilities are

$$L = \mathrm{Sz}(s), \qquad J = D_{2(s-1)}, \qquad (q-1)/(s-1) \text{ prime}, \tag{13}$$

with $s \geq 8$ a proper subfield order. Each occurs once.

Proof. First consider $J = C_{q-1} = A$, which is always maximal in M. A proper overgroup L of A not in M must lie in a point stabilizer, by the subgroup classification and coprimality of the three odd factors. A proper subfield group cannot contain A: if $q \geq s^3$, every cyclic torus in $\mathrm{Sz}(s)$ has order less than $q-1$. The torus A fixes two ovoid points, giving the two possible point stabilizers.

In either one write $L = VA$, with $V = L \cap P$ invariant under the full scalar torus. If $V \leq Z$, nonzero invariance forces $V = Z$, since the action is all nonzero scalars. If $V \not\leq Z$, the same argument on P/Z shows $VZ = P$. The Frattini property $Z = \Phi(P)$ from the preliminary note then gives $V = P$. But $A < ZA < PA$, so A would not be maximal in L. Consequently $L = ZA$. The two choices are interchanged by M, and their intersection with M is exactly A. This proves the first row for every q.

Suppose $q-1$ is prime. The other maximal intersection in M is J_2. We classify proper overgroups L in which J_2 is maximal. If L is contained in a point stabilizer, its normal two-subgroup $L \cap P$ contains J_2. By maximality it either equals J_2 or equals L. In the first case L normalizes J_2, hence lies in its two-group centralizer P as well. In both cases L is a two-group, and maximality makes its order four. It is therefore C_4 or C_2^2. The cyclic normalizers give D_{2p}, with p prime, or C_4. The split case has $p = q-1$; the other two odd orders give the third row. No simple subfield group has a maximal subgroup of order two, by its retained maximal-subgroup list. This proves exhaustion. The intersections in all distinct-type cases are exactly J_2 by their orders. Two distinct split normalizers containing J_2 cannot have a common odd element, since such an element lies in a unique torus; their intersection is also exactly J_2. All the groups are outside M, so the coset criterion applies.

Here n is prime. The inversion quotient for the second row has size (7) and one fixed point by Lemma 2. An action of order n on a finite set X has $(|X| + (n-1)|X^\phi|)/n$ orbits. This gives $1 + (q-2)(q+4)/(4n)$. The third row uses P/J_4, of size $q^2/4$ with one fixed point, giving $1 + (q-2)(q+2)/(4n)$. The last two rows use respectively the nonzero quotient $F/\mathbb{F}_2$ and the whole quotient, of size $q/2 - 1$ and $q/2$, with zero and one fixed points. This proves all counts in (12).

Finally suppose $q-1$ is composite. Apart from A, a maximal intersection has form $J = D_{2m}$, with $m = (q-1)/p > 1$ and p prime. Any overgroup of J lying in a solvable normalizer must lie in M: the characteristic odd rotation subgroup determines its ambient split torus; a point stabilizer cannot contain its inverting involution, since its quotient by the normal two-group is abelian of odd order. Coprimality excludes nonsplit tori. Thus an overgroup outside M is $L = \mathrm{Sz}(s)$. Maximality of J in L forces $m = s-1$ by its classical list.

Conversely the standard such J occurs when the condition in (13) holds. It lies in a unique L by Theorem 5. Its conjugacy class in M is unique, so this produces exactly one equivalence class. The maximality conditions and exact intersection follow as before. □

Theorem 6 recovers source Lemma 6.2 and Tables 2 and 3. The common proof does not separately enumerate the subgroups of each ambient Suzuki group.

6 The nonsplit normalizers

Theorem 7. *Let $t = a_\epsilon(q)$ and $M = C_t \rtimes C_4$. If t is prime, the complete list is*

L	J	*number*
$C_t \rtimes C_4$	C_4	$\dfrac{q-2}{2n}$
$C_p \rtimes C_4,\ p \mid a_{-\epsilon}(q)$ *prime*	C_4	$1 + \dfrac{q-2}{2n}$
$C_4 \times C_2$	C_4	$\dfrac{q-2}{2n}$.

(14)

If t is composite, the only possibilities are

$$L = \mathrm{Sz}(s), \qquad J = C_{a_\delta(s)} \rtimes C_4, \qquad t/a_\delta(s) \text{ prime}, \tag{15}$$

where $s \geq 8$ is a proper subfield order and $a_\delta(s) \mid t$. Each has one equivalence class.

Proof. One maximal subgroup of M is $C_t \rtimes C_2$. Its nontrivial characteristic odd subgroup determines the ambient nonsplit torus. Any overgroup outside M must therefore be a simple subfield group, by the same partition and coprimality argument as above. For $C_t \rtimes C_2$ to be maximal in $\mathrm{Sz}(s)$ the classical list would require $t = s - 1$. This is impossible: $s - 1 \mid q - 1$, whereas $\gcd(t, q-1) = 1$ and $t > 1$.

If t is prime, the remaining maximal intersection is J_4. A proper overgroup L in which J_4 is maximal, if contained in a point stabilizer, must be a two-group. Indeed its normal two-subgroup contains J_4, and if it equals J_4 the whole group lies in $\mathrm{N}_G(J_4) = R$, still a two-group. Maximality makes $|L| = 8$, whence Lemma 1 gives $C_4 \times C_2$. A split torus normalizer has no element of order four. A nonsplit one gives $C_p \rtimes C_4$, with p prime. The prime $p = t$ gives another conjugate of M; all other primes must divide the other nonsplit torus order. The classical maximal list excludes a simple subfield group with maximal subgroup C_4. This proves exhaustion. Distinct conjugates of M containing J_4 intersect exactly in J_4, since any common odd element would determine the same torus and normalizer. Different types have intersection J_4 by their orders.

Since n is prime, the first row is the nonzero elementary abelian quotient R/J_4 modulo Frobenius, giving $(q-2)/(2n)$. The second is the whole quotient, with one fixed point, giving $1 + (q-2)/(2n)$. The third is the nonzero quotient from Lemma 1. These are actual incidence fibres by (11), and the interchange argument has already been included.

If t is composite, the remaining maximal intersections are $J = C_m \rtimes C_4$, where $m = t/p > 1$. Their nontrivial odd subgroups again exclude any solvable overgroup outside M. Thus $L = \mathrm{Sz}(s)$, and maximality in its classical list forces $m = a_\delta(s)$. Conversely each condition (15) gives the standard subgroup and its unique subfield overgroup by Theorem 5. Conjugacy within M and this uniqueness give one class. The coset criterion proves the claimed geometric properties in every listed case. □

For $\epsilon = +1$, this gives source Lemma 6.3 and Tables 4–5. For $\epsilon = -1$, it gives Lemma 6.4 and Tables 6–7, subject to the following explicit correction. The second row of printed

Table 6, page 110, requires its prime p to divide β_q, the source's notation for $a_-(q)$. That table already assumes β_q prime, so this repeats the same-normalizer family in its first row and omits the opposite torus family. The condition must be $p \mid \alpha_q = a_+(q)$, as (14) proves. At $q = 8$ the missing family has parabolics $C_5 \rtimes C_4$ and $C_{13} \rtimes C_4$. It is not a new conjecture or a global classification correction; it is a corrected condition in this particular table.

7 Attribution and remaining obligations

The recovered assertions are source Lemmas 6.2–6.6 and 6.8, and Tables 2–7, with the Table 6 condition just specified. The source's Lemma 6.7 remains an explicitly attributed classical input from Suzuki. Our contribution in this component is the common derivation through normalizer incidence, the two root-group quotients and their Frobenius actions. We make no priority claim for incidence counting, Burnside's orbit formula, or these elementary group-theoretic tools. The source's original case proofs and numerical assertions remain attributed to [1].

No part of this note supplies the remaining global classification. In particular, configurations whose only maximal parabolic is a subfield group require separate treatment of all its maximal intersections. The already reviewed scalar-subspace correction addresses one such family and shows that the printed global classification cannot be retained unchanged. Neither the present note nor that correction alone settles all remaining cases or the source's global numerical tables. The attached finite controls at $q = 8, 32, 128$ form the literal root groups, their indicated subgroups and coset partitions, and the Frobenius and inversion orbits. All seven count formulas agree. These checks concern the quotient algebra and do not test the borrowed classification of ambient subgroups or replace the quantified arguments. This component has no Lean certificate and carries no whole-original completion credit by itself.

References

[1] D. Leemans, *The rank 2 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Beiträge zur Algebra und Geometrie 39 (1998), no. 1, 97–120.

[2] M. Suzuki, *On a class of doubly transitive groups*, Annals of Mathematics 75 (1962), 105–145.

Subfield parabolics and the corrected Suzuki classification

Human Compression project

Abstract

We complete the subgroup argument for configurations with a maximal subfield parabolic. A normalizer condition in a finite two-group forces every cover of a subfield point stabilizer into the central overgroup already treated by the correspondence with scalar subspaces. Thus that correspondence supplies the entire corrected family. We count its field orbits by an explicit formula for fixed projective lines under Frobenius. Together with the separately established normalizer families, this gives a corrected classification and a finite arithmetic formula for the total number of configurations. The numerical source tables and diagrams still require their separate complete audit; this draft does not by itself claim whole-source completion.

1 Acquired results and notation

Let $q = 2^n$, where $n \geq 3$ is odd, and let $G = \mathrm{Sz}(q)$. Write $a_\pm(q) = q \pm 2^{(n+1)/2} + 1$. We use the classical Suzuki partition, subgroup classification, ovoid action and automorphism group specified in the accepted note *Normalizer fibres in rank-two Suzuki geometries*. In particular every proper nonsolvable subgroup is a simple subfield group, all subfield groups of a prescribed order are conjugate, and their proper maximal simple subfield groups have prime relative field degree. The solvable maximal subgroups have already been treated, and a point stabilizer cannot be a parabolic in the geometries under consideration.

Fix a prime r dividing n, put $d = n/r \geq 3$, $s = 2^d$, $F = \mathbb{F}_q$, $K = \mathbb{F}_s$, and let $S = \mathrm{Sz}(s)$ be the standard maximal subfield subgroup. Its point stabilizer is $B_s = U_K H_K$, with $U_K = \{u(a,b) : a,b \in K\}$ and $H_K = \{h(k) : k \in K^*\}$. We retain the same faithful root-group coordinates as in the preceding notes:

$$u(a,b)u(c,e) = u(a+c, b+e+ac^\theta), \quad u(a,b)^{h(k)} = u(k^\theta a, k^{\theta+2}b), \qquad \theta = 2^{(n+1)/2}. \tag{1}$$

Put $P = U(F,F)$ and $Z_F = U(0,F)$. The two coordinate actions of K^* are the full scalar group of K. The automorphism $x \mapsto x^\theta$ on K has fixed field $\mathbb{F}_2$.

The accepted scalar-subspace note proves that the overgroups of B_s in $Z_F B_s$ are exactly

$$Q_T = Z_T B_s, \qquad K \leq T \leq F \text{ a } K\text{-subspace}, \quad Z_T = \{u(0,t) : t \in T\}. \tag{2}$$

A group Q_T covers B_s precisely when $\dim_K T = 2$; it then has order $s^3(s-1)$. The geometries $\Gamma(G; S, Q_T)$ are firm, connected, flag transitive, residually weakly primitive and locally doubly transitive. Their equivalence classes, in the source's convention of automorphisms of the specified group with type interchange allowed, are exactly the Frobenius orbits on the projective space $\mathbb{P}_K(F/K)$. These statements are reused, not proved twice. The task here is to prove exhaustion and compute those orbits.

2 Every point-stabilizer cover is central

Lemma 1. *Inside the point stabilizer $B_q = P \rtimes F^*$,*

$$\mathrm{N}_P(U_K) = Z_F U_K, \qquad \mathrm{N}_{B_q}(U_K) = Z_F U_K H_K. \tag{3}$$

Proof. An element $u(a,b) \in P$ normalizes U_K precisely when $ac^\theta + ca^\theta \in K$ for every $c \in K$, by (1). Taking $c = 1$ gives $a + a^\theta \in K$. Subtracting c times this relation from the general one gives $a(c^\theta + c) \in K$. Choose $c \in K \setminus \mathbb{F}_2$; its coefficient is nonzero, so $a \in K$. There is no restriction on b. This proves the first equality.

Conjugation by P does not change first coordinates. Therefore, if $u(a,b)h(t)$ normalizes U_K, its action on the first-coordinate image K gives $t^\theta K = K$, hence $t \in K^*$. Removing this diagonal factor reduces to the first equality. Conversely, all displayed elements normalize U_K. □

Theorem 2. *If $B_s < L < B_q$ and B_s is maximal in L, then $L = Q_T$ for a two-dimensional K-subspace T containing K. These are all proper subgroups outside S in which B_s is maximal.*

Proof. Put $V = L \cap P$, a normal two-subgroup of L containing U_K. If $V = U_K$, then $L \le \mathrm{N}_{B_q}(U_K)$. Equation (3) makes its image in B_q/P equal to H_K. Because $H_K \le L$, we obtain $L = VH_K = B_s$, a contradiction. Thus $V > U_K$.

The normalizer condition in a finite p-group gives $\mathrm{N}_V(U_K) > U_K$. For completeness, a proper subgroup H of a finite p-group acts on its cosets; the fixed cosets are $\mathrm{N}(H)/H$, all other orbits have length divisible by p, and the total index is divisible by p. There is a fixed coset, so there are at least p of them. This proves the condition. Now H_K normalizes both V and U_K, hence normalizes $\mathrm{N}_V(U_K)$. Therefore

$$B_s < \mathrm{N}_V(U_K)H_K \le L.$$

Maximality forces equality with L. Consequently U_K is normal in L, and (3) gives $L \le Z_F B_s$. The acquired correspondence (2) proves the claimed form and dimension.

Finally consider any proper overgroup L of B_s outside S. By the classical subgroup classification it either lies in a solvable normalizer or is a simple subfield group. Its large two-subgroup U_K excludes the cyclic torus normalizers. If it lies in a point stabilizer, a nonidentity involution of U_K forces that stabilizer to be the particular B_q. If $L = \mathrm{Sz}(t)$ and B_s is maximal in L, the classical maximal list forces B_s to be a point stabilizer of L, hence $t = s$. The accepted uniqueness theorem for a subfield overgroup of B_s would give $L = S$. Thus this case is excluded and the list already obtained is exhaustive. □

This strengthens the accepted correction: the groups Q_T are the entire replacement for the first row of source Table 8, not only a family contradicting that row. The printed group of order $qs(s-1)$ does not cover B_s when $r \ge 3$; the correct groups have order $s^3(s-1)$ and are parametrized as in (2).

3 The other subfield intersections

We shall use a further consequence of the normalizer incidence formula. If $\mathbb{F}_t \subseteq \mathbb{F}_s \subseteq \mathbb{F}_q$, every fixed $\mathrm{Sz}(t)$ is in exactly one $\mathrm{Sz}(s)$ of the prescribed order. Here and below both smaller field orders are at least eight. Indeed, all the subfield groups of each order are conjugate, both are self-normalizing by simplicity and the classical subgroup classification, and the subgroups of order $|\mathrm{Sz}(t)|$ inside $\mathrm{Sz}(s)$ form one conjugacy class. The incidence formula therefore gives one overgroup. This is a uniqueness assertion for each prescribed field order, not a claim that the entire interval has only one intermediate subgroup.

Theorem 3. *Besides the central covers in Theorem 2, all configurations with maximal parabolic $S = \mathrm{Sz}(s)$ are as follows:*

$J = S \cap L$	L	*condition*
$D_{2(s-1)}$	D_{2m}	$m/(s-1)$ *prime,* $m \mid q-1$
$C_{a_\delta(s)} \rtimes C_4$	$C_m \rtimes C_4$	$m/a_\delta(s)$ *prime,* $m \mid a_\epsilon(q)$
$\mathrm{Sz}(2^e)$	$\mathrm{Sz}(2^{er})$	$n = erp$, r, p *distinct primes,* $e \geq 3$.

(4)

In the second row $\delta \in \{+1, -1\}$ and the ambient sign ϵ is the unique one for which $a_\delta(s) \mid a_\epsilon(q)$. Each permitted numerical choice in the first two rows gives one equivalence class. Each unordered pair of distinct prime indices in the last row gives one class.

Proof. The maximal-subgroup list in S leaves precisely the three intersection types in (4), after its point stabilizer. For $J = D_{2(s-1)}$, its odd rotation subgroup determines the unique split torus of G. An overgroup in a point stabilizer cannot contain the involution inverting that nontrivial odd subgroup: projection to the point stabilizer's abelian odd quotient would make inversion trivial. Coprimality excludes the nonsplit normalizers. Thus a solvable overgroup is in the unique split dihedral normalizer, and the cyclic maximal-subgroup calculation gives exactly the first row. A simple subfield overgroup in which J is maximal must have field order s; the already accepted uniqueness theorem would make it S. Within the ambient torus normalizer the group of each displayed order containing this fixed J is unique, so each choice gives one configuration class.

For $J = C_{a_\delta(s)} \rtimes C_4$, its characteristic odd subgroup similarly determines its unique ambient nonsplit torus. The point stabilizer and split normalizer are excluded by their odd orders. Its solvable covers are therefore exactly the prime torus extensions in the second row. A simple overgroup in which J is maximal must again have field order s: the two functions $a_+(2^d)$ and $a_-(2^d)$ strictly increase through nonoverlapping intervals as the odd exponent $d \geq 3$ increases, and the two signs at a fixed exponent are distinct. Hence its odd torus order determines s and δ. The uniqueness theorem then excludes it. For a fixed J each allowed m gives one group, proving the count one.

If $J = \mathrm{Sz}(2^e)$ is a simple subfield group, then $d/e = p$ is prime, by maximality in S. An overgroup L is nonsolvable, so is a subfield group $\mathrm{Sz}(2^{e\ell})$, with ℓ prime by maximality of J in L. Since $n/e = rp$, necessarily $\ell = r$ or p. If $\ell = p$, uniqueness gives $L = S$. Thus $\ell = r \neq p$. Conversely distinct r, p give two different maximal subfield groups over the fixed J. Their intersection is exactly J: it contains J, maximal in each, and neither contains the other. Their conjugacy class is unique by the same incidence argument, and type interchange identifies the two orders of the prime pair.

For the first two rows, the proposed L lies outside S and J is maximal in both; hence the intersection is exactly J. In every row the maximality of S gives generation of G and the rank-two coset criterion supplies firmness, connectivity, flag transitivity and residual weak primitivity. □

The nonoverlap used in the proof is elementary. For $s \geq 8$, $a_+(s) \leq 3s/2 + 1$, whereas $a_-(4s) = 4s - 2\sqrt{2s} + 1 > 3s/2 + 1$. Adjacent odd exponents multiply the field order by four, proving the assertion. The case $2^e = 2$ is not included in the simple-subfield row: $\mathrm{Sz}(2)$ is the solvable group $C_5 \rtimes C_4$, and any such maximal intersection is already governed by the nonsplit row.

4 Frobenius orbits of the central covers

The following argument is finite field algebra; it is independent of the subgroup classification. It applies to every odd $d \geq 1$ and odd prime r, with $n = dr$, even when r divides d.

Lemma 4. *For a cyclic finite-field extension E/D with generator τ, the elements of norm one are exactly $\tau(c)/c$ for $c \in E^*$. If the extension degree is odd in characteristic two, invariants are exact on every sequence of vector spaces with compatible actions: an invariant vector in a quotient has an invariant lift.*

Proof. The map $c \mapsto \tau(c)/c$ has kernel D^*, and its image has size $(|E|-1)/(|D|-1)$. The norm map on the cyclic group E^* is the power $(|E|-1)/(|D|-1)$, is onto D^*, and has a kernel of that same size. The image lies in the kernel by telescoping, proving equality. For the second assertion, average any lift over the acting cyclic group. Its odd order is invertible in characteristic two, and averaging preserves the specified quotient vector. □

Theorem 5. *For $0 \le j < n$, put $h = \gcd(d, j)$ and $g = \gcd(n, j)$, with the usual convention $\gcd(a, 0) = a$. The number b_j of K-projective lines in F/K fixed by ϕ^j is*

$$b_j = \begin{cases} \dfrac{2^{h(r-1)} - 1}{2^h - 1}, & g = rh, \\ \gcd(r, 2^h - 1) - 1, & g = h. \end{cases} \tag{5}$$

These are the only cases. Consequently the number of central cover classes is

$$C(n, d) = \frac{1}{n} \sum_{j=0}^{n-1} b_j. \tag{6}$$

Proof. Set $\sigma = \phi^j$ and $m = d/h$, the order of its restriction to K. Since $\gcd(d/h, j/h) = 1$,

$$g = h \gcd(r, j/h) \in \{h, rh\}.$$

This identity does not require r to be coprime to d. The order of σ on F is n/g.

Suppose $g = rh$. Then σ has order m on both fields. The invariant fields are $F^\sigma = \mathbb{F}_{2^{rh}}$ and $K^\sigma = \mathbb{F}_{2^h}$. Exactness by averaging gives

$$|(F/K)^\sigma| = 2^{h(r-1)}.$$

On a fixed K-line choose $v \neq 0$ with $\sigma(v) = \lambda v$. Applying $\sigma^m = 1$ gives $N_{K/\mathbb{F}_{2^h}}(\lambda) = 1$. Lemma 4 permits a scalar change making v fixed. Each fixed line then has exactly $2^h - 1$ nonzero fixed vectors. Dividing gives the first expression in (5).

Suppose $g = h$. The automorphism $\rho = \sigma^m = \phi^{jd/h}$ has order r on F and fixes K. Its one-eigenspace in F is exactly K. Since r is odd, trace identifies F/K with the trace-zero complement $V = \ker(\mathrm{Tr}_{F/K})$; this complement is also σ-invariant. The one-eigenspace is removed, so $V^\rho = 0$. On a σ-fixed K-line, σ acts semilinearly and ρ acts as a scalar $\mu = N_{K/\mathbb{F}_{2^h}}(\lambda) \in \mathbb{F}_{2^h}^*$. Necessarily $\mu^r = 1$ and $\mu \neq 1$.

Conversely, for every such μ, Lemma 4 applied to F/K gives a nonzero solution of $\rho(v) = \mu v$. Its solutions form one K-line: the ratio of two nonzero solutions is fixed by ρ. For $\mu \neq 1$ this line has trace zero, so lies in V. Commutation of σ and ρ sends the μ-eigenline to the $\sigma(\mu)$-eigenline. Thus the line is σ-stable exactly when $\mu \in \mathbb{F}_{2^h}$. There are $\gcd(r, 2^h - 1) - 1$ such nontrivial roots. This proves the second expression. Finally, counting the pairs consisting of an orbit and an automorphism fixing one of its points gives Burnside's formula (6). □

For example,

$$C(9, 3) = 1, \quad C(15, 3) = 41, \quad C(15, 5) = 3, \quad C(21, 3) = 1791, \quad C(21, 7) = 7.$$

The first two agree with the independently accepted literal projective-orbit certificates in the scalar-subspace note. The formula also covers the repeated prime index cases, such as $C(27, 9) = 19$; no coprimality of the absolute and relative field degrees has been imposed.

5 A corrected global list and total

Let $\omega(a)$ denote the number of distinct prime divisors of the positive integer a, with $\omega(1) = 0$. Each simple subfield case above has maximal absolute field exponent $d = n/r \geq 3$, for a prime divisor r of n. For each sign δ, let $t_\delta(q, s)$ be the unique ambient nonsplit torus order divisible by $a_\delta(s)$.

Theorem 6. *The complete classification consists of the solvable maximal parabolic families in the accepted normalizer note, together with (2) and (4), identifying a configuration when both descriptions give the same unordered pair. If n is composite, its total is*

$$T(n) = 1 + E(n) + \sum_{\substack{r \mid n \ \text{prime} \\ d = n/r \geq 3}} [C(n,d) + A(n,d)], \tag{7}$$

where

$$A(n,d) = \omega\left(\frac{q-1}{2^d - 1}\right) + \sum_{\delta = \pm 1} \omega\left(\frac{t_\delta(q, 2^d)}{a_\delta(2^d)}\right),$$
$$E(n) = \#\{\{r,p\} : r \neq p \text{ prime},\ rp \mid n,\ n/(rp) \geq 3\}.$$

If n is prime, put

$$v = \frac{q-2}{2n}, \qquad I = 1 + \frac{(q-2)(q+4)}{4n}, \qquad H = 1 + \frac{(q-2)(q+2)}{4n}.$$

Let η_0, η_+, η_- be the indicators that respectively $q-1, a_+(q), a_-(q)$ are prime. Then

$$T(n) = 1 + \eta_0\big[I + 2v + 1 + H(\omega(a_+(q)) + \omega(a_-(q)))\big] + \sum_{\epsilon = \pm 1} \eta_\epsilon\big[2v + (v+1)\omega(a_{-\epsilon}(q))\big] - \eta_+\eta_-(v+1). \tag{8}$$

Proof. Residual weak primitivity gives at least one maximal parabolic in G. The accepted preliminary obstruction excludes a point stabilizer. The accepted normalizer classification treats the three torus normalizers; Theorems 2 and 3 treat every maximal intersection in a subfield parabolic. These exhaust the classical maximal-subgroup list, proving the classification.

If n is composite, all three ambient odd factors are composite. The universal pair $(D_{2(q-1)}, q{:}(q-1))$ contributes one. All other families have a subfield parabolic. For a fixed s, the first two rows of (4) are counted by the distinct prime divisors of the indicated quotients, and the central family by $C(n,d)$. The last row has two maximal subfield parabolics and must be counted once for each unordered pair of distinct prime indices. Their simple intersection must have field order at least eight. Distinct field orders or distinct torus orders do not identify these families. Including the full ambient torus normalizer among the possible L in (4) already includes the corresponding family from the normalizer note, so it is not added again. This proves (7).

If n is prime there are no proper simple subfield groups. The split prime case adds the same-type contribution I, the four-group and cyclic-four contributions v and $v+1$, and one H for each prime divisor of either nonsplit order. For each prime nonsplit order there are two contributions v and one $v+1$ per prime divisor of the opposite order. When both nonsplit orders are prime, their common mixed family has been counted twice; subtract $v+1$. No other families overlap. This is (8). □

6 Local double transitivity and source disposition

The universal pair is locally doubly transitive: its local degrees are two and q, and the latter action is the affine action of $\mathbb{F}_q \rtimes \mathbb{F}_q^*$. Every corrected central cover is also locally doubly transitive by the acquired scalar-subspace theorem. Thus the source's claim that only the universal family survives this condition for $n > 5$ requires correction when n is composite.

There are no further locally doubly transitive families except, at $q = 8$, the pairs $(C_5 : C_4, C_5 : C_4)$ and $(C_5 : C_4, C_4 \times C_2)$. Here a cyclic complement of order four acts regularly on the four nonzero elements of C_5, and the other local action in the second pair has degree two. For the prime split families, an intersection of order two cannot be transitive on the $q-2 \geq 6$ other cosets. For a prime nonsplit normalizer with intersection C_4, double transitivity requires its prime torus order to satisfy $p-1 \leq 4$, so $p = 5$; the opposite torus cannot also have order five. For the remaining subfield families the local action of S already fails double transitivity. If J is a torus normalizer of S, then $|J| \leq 8s$ and $|S| > |J|(|J| + 1)$ for $s \geq 8$, so $[S : J] - 1 > |J|$. If $J = \mathrm{Sz}(t)$ is a proper simple subfield group, then $s \geq t^3$, $|J| < t^5$ and $|S| > s^5/2 > t^{10} + t^5$; the same inequality applies. A point stabilizer of size $|J|$ cannot act transitively on more than $|J|$ points.

This supplies a corrected form of source Lemma 6.9 and its Table 8, the complete family statement of Theorem 7.1, and the corresponding local double-transitivity column in Tables 9–11. The source's numerical Table 12 and its 29 small-group diagrams need explicit numerical and invariant joins. They are not silently certified by a general formula. The original source is D. Leemans, *The rank 2 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Beiträge zur Algebra und Geometrie 39 (1998), no. 1, 97–120. Its retained structural inputs remain attributed to Suzuki, Tits and Lüneburg. The earlier component's phrase "right coset gJ" should read simply "coset gJ"; its equations are unaffected. No Lean or whole-original completion claim is made by this draft.